Energy and Particle Technology Laboratory

Research Group at Mechanical Engineering Department
Carleton University

Designing a Common Platform for Modelling Soot in Flames and Reactors

Author:
Mohammad Adib
moadib@cmail.carleton.ca

Supervisor: Dr. Reza Kholghy

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Nomenclature

Acronyms

CFD Computational Fluid Dynamics
DEM Discrete Element Modelling
HACA H-abstraction-C_2H_2/Carbon-addition
MPBM Monodisperse Population Balance Model
SPBM Sectional Population Balance Model

Constants

Av	Avogadro's number	$6.02214076\ \times 10^{23}$ 1/mol
k_B	Boltzmann constant	$1.3806488 \times 10^{-23}$ J/K
R	Global gas constant	8.314 J/mol

English symbols

A	Surface area	m^2
c_p	Heat capacity at constant pressure	J/kgK
c_v	Heat capacity at constant volume	J/kgK
c_{soot}	Heat capacity of soot	J/kgK
C_{tot}	Total carbon content of soot particles (per section)	mol/kg
D	Diffusion coefficient of particles	m^2/s
d	Diameter	m
d_c	Collision diameter	m
d_g	Gyration diameter	m
D_H	Duct hydraulic diameter	m
d_m	Mobility diameter	m
d_p	Primary particle diameter	m
e	Internal energy	J/kg
$E(m)$	Absorption function of soot	–
f	Friction factor	–
f_v	Soot volume fraction based on gas volume	m^3/m^3
h	Entalpy	J/kg
H_{tot}	Total hydrogen content of soot particles (per section)	mol/kg
I	Partial source terms for soot variables	$mol/(kg \cdot s)$
k	Reaction rate constant	$m^3/(mol \cdot s)$
k_{dep}	Deposition velocity	m/s
Kn	Knudsen number	–
m	Mass	kg
n_p	Number of primary particles of per agglomerate	
N_{agg}	Number density of agglomerates	mol/kg
N_{pri}	Number density of primary particles	mol/kg
p	Pressure	Pa
P_c	Wetted perimeter of the flow reactor	m

R_H	Duct hydraulic radius	m
Re	Reynolds number	–
S	Source terms for soot variables	$\mathrm{mol/(kg \cdot s)}$
Sc	Schmidt number	–
SF	Sectional spacing factor	–
Sh	Sherwood number	–
T	Temperature	K
u	Velocity	m/s
V	Volume	$\mathrm{m^3}$
W	Molecular weight	kg/mol
Y	Mass fraction of gaseous species	–

Greek symbols

β	Collision frequency	$\mathrm{m^3/s}$
δ_a	Mean distance of particles	m
ϵ	Wall roughness	m
η_{ads}	PAH adsorption adjustment factor	–
η_{inc}	Inception adjustment factor	–
λ	Mean free path	m
λ_a	Mean stopping distance of particles	m
μ	Gas dynamic viscosity	$\mathrm{Pa \cdot s}$
ω	The rate of production of gaseous species	$\mathrm{mol/m^3 s}$
ρ	Density	$\mathrm{kg/m^3}$
τ_w	Wall shear stress	Pa
τ_{psr}	Nominal residence time of perfectly stirred reactor	Pa
φ	Soot volume fraction based on reactor volume	$\mathrm{kg/m^3}$
ζ	Collision efficiency	–

Subscripts

ads	Adsorption
agg	Agglomerate
coag	Coagulation
cont	Continuum
f	Forward
fm	Free molecular
grow	Surface growth
inc	Inception
ox	Oxidation
pri	Primary particle
r	Reverse
reac	Reactive

Chapter 1

Introduction

1.1 Motivation

Carbonaceous nanoparticles, such as Carbon Black (CB) and soot, are widely encountered in nature and engineering. Every year, nearly 9.5 megatons of soot (black carbon) is emitted into the atmosphere from anthropogenic activities and natrual sources such as wildfires and volcanoes [12]. The wide light absorption range of soot reduces the albedo of snow-covered area and alters the radiative forcing balance in the atmosphere making soot the third strongest contributor to climate change after methane and carbon dioxide [12]. Also, exposure to combustion-generated soot could promote respiratory and cardiovascular diseases [13]. So, strict regulations are targeting combustion engines to limit environmental and health risks from soot formation [14]. Accurate and affordable models are essential for the prediction of soot composition and morphology in combustion devices and the reduction of emissions in engines. They can also provide better understanding of soot optical properties that are major indicator of soot environmental effects.

On the other hand, CB with a similar synthesis process and structure to soot but higher elemental carbon to hydrogen ratios (>97%) [15] is commercially produced and sold in large scales. In fact, CB is the largest industrially produced nanomaterial by value and volume (∼15 megatons per year with a value of $17B) with applications as a reinforcing agent in rubber and tire industries [16], and conductive additive in lithium-ion batteries [17]. CB is primarily manufactured by the so-called furnace process in which about 50% of heavy fuel oil is partially combusted to convert the rest of it into CB [18]. This process suffers from low mass yield and excessive emission, generating 4 tons of CO_2 per each ton of product on average [19]. Plasma reactor is an emerging alternative production method with distinct advantages over flame-based methods: They can achieve 100% carbon yields with no direct CO_2 emission or other pollutants [20], and the energy required for pyrolysis is supplied by an electric arc that does not depend on the feedstock composition. Controlling CB properties such as its specific surface area (or primary particle diameter), hard agglomerate size (or gyration diameter of agglomerates with primary particles connected to each other by strong chemical bonds), and composition (or particle carbon to hydrogen ratio) is important to make the process economical and to achieve specific grades of CB for different target applications. However, this is a challenging task because of the complexity of CB formation and mass growth processes and its coupling with gas phase chemistry, dependence on local temperature and pressure. This requires accurate process design and optimization tools built on physics of CB formation and evolution to inform manufacturers' decisions for production of CB with desired grades [21].

The term "*soot*" usually refers to the unwanted particulate matter formed during incomplete combustion of any carbon-containing material from jet and diesel fuel to wood, heavy oil, and plastics with variable organic content and a large variation in C/H ratio [15], but this research focuses on soot particles generated under controlled laboratory conditions from fuels with known compositions. The mature soot formed in methane and ethylene premixed flame can reach 95% elemental C/H ratio [22], which is close to CB composition. The comparison of transmission electron microscopy (TEM) images of industrially produced CB [2] with soot sampled from diesel fuel [3, 4] shown in

Fig.1.1 indicates similarity of their morphology and structure. Hereafter, soot will be used to collectively refer to carbonaceous nanoparticles produced in flame/reactor during combustion/pyrolysis processes.

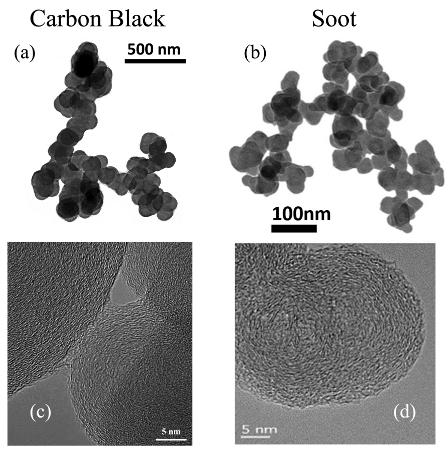

Figure 1.1: The TEM images of Carbon black (a & c) [2] and soot (b & d) [3, 4] that shows soot and CB has similar morphology and structure

1.2 Background

1.2.1 Soot inception and surface growth

Soot formation involves concurrent processes with different time and length scales starting from the breakdown of hydrocarbon molecules to formation of intermediate species and radicals that transition into soot particles and grow via heterogeneous reactions on the surface as well as coagulation [23]. The TEM analysis of soot sampled from flames [4], reactors [24], and engines [25] with different fuels and process conditions revealed a common fractal-like morphology, often characterized as agglomerates of primary particles. High resolution transmission electron microscopy (HRTEM) of soot primary particles showed clusters of precondensed Polycyclic Aromatic Hydrocarbons (PAHs) [26] pointing to PAHs as main soot precursors. This hypothesis was also supported by thermodynamic stability of PAHs enabling them to resist dissociation at high flame temperatures [27].

However, the transition of PAHs (soot precursors) to incipient soot, known as *soot inception* has not been well understood at the level of pathways and elementary reactions [5] primarily due to uncertainties in PAH (precursor) chemistry. The formation of carbonaceous particles is kinetically controlled and known to be an entropy-driven process [5]. Wang [5] uses the pyrolysis of propane at 1600 K to highlight how thermodynamics of soot formation is distinct from synthesis of inorganic nanomaterials. As shown in Fig. 1.2, acetylene is highly endothermic, but it is accompanied with drastic increase in entropy due to hydrogen release resulting in overall decrease of Gibbs free energy, which explains acetylene (C_2H_2) being the dominant hydrocarbon species. The growth of PAHs from acetylene does not undergo a significant enthalpy release or entropy increase, so the path is driven by a gradual reduction in Gibbs free energy. As a result, the kinetics of PAH growth into subsequent soot particles can be highly reversible hence sensitive to local temperature, pressure and intermediate species concentration.

The growth of PAHs beyond first ring (benzene) is predominantly driven by so-called hydrogen abstraction carbon (acetylene) addition (HACA) mechanism [28] where a hydrogen radical abstracts an hydrogen atom at the edge of PAH, providing a reactive site for acetylene addition. The kinetic reversibility of HACA opens the door for competing pathways such as chain reactions of resonance-stabilized radical (RSR). Propargyl is a prominent example of these radicals, whose combination is known as a major contributor to benzene formation [29]. Built on this hypothesis, Johansson et al. [30] proposed a radical-driven growth mechanism by addition of vinyl (C_2H_3) starting from cyclopentadienyl (C_5H_5) to larger hydrocarbon radicals that can survive long enough in high temperatures to react with other radicals, PAHs and unsaturated aliphatic species, through radical chain reactions. However, low concentrations of these radicals limit the growth rate through RSR pathways [31]. Moreover, some of the intermediate steps for radical regeneration such as formation of vinylcyclopentadienyl were shown to be kinetically unfavorable [31] compared to HACA. Regardless of their mechanistics, these sequential growth mechanisms, termed as *chemical growth* cannot account for rapid soot formation [28] and its nanostructure [32].

There is abundant but mostly indirect experimental and computational evidence pointing to an alternative collision-based mechanism for soot inception. HRTEM images of nascent and mature soot shows disordered PAH clusters highlighting the role of PAH collisions. The bimodality of particle size distribution (PSD) of nascent soot particles in premixed flames [33] indicates that the kinetics of inception is second order in precursor concentration [34]. The time-of-flight mass spectrometry (TOFMS) experiments in a 13 kPa acetylene-oxygen flame showed a series of peaks with a periodicity of 500 amu [35].

It is not clear how PAH clusters form and what forces allow the binding occur and resist dissociation at flames temperatures (T≥1600 K). Frenklach [28] described clustering of PAHs as an irreversible process where two PAHs collide and stick upon collision forming dimers held together by Van der Waals (vdW). Herdman and Miller [36] found that the binding energy of PAHs dimers due to dispersive and electorstatic forces increases linearly with molecular mass and reaches the limit of exfoliation energy for graphite. However, the entropy barrier of dimerization increases with PAH size making them unfavorable under equilibrium conditions. So, PAHs as large as circumcoronene ($C_{54}H_{18}$) can only form dimer to survive flame temperatures [5]. However, the concentration of large PAHs are too low to account for the observed number density of soot particles in flames [37].

There is also the possibility of PAH clustering governed by non-equilibrium kinetics. PAH molecules can form rovibrationally excited dimer with long enough life-time [38] to react with H atoms forming covalent bonds. There is another possibility for these clusters to be joined by aliphatic linkages. Micro-FT-IR spectroscopy analysis by Cain et al. [39] showed the ratio of aliphatic-to-aromatic C–H bonds can exceed unity at the flame temperature. Using these findings, they suggested that the aliphatic components in the form of alkyl and alkenyl can covalently bound to aromatic units in soot particles. Although such a mechanism is viable near the flame region, it cannot explain persistent soot inception in the post-flame zone [40] where the H atom concentration is too low to initiate H abstraction reactions that produce those radicals.

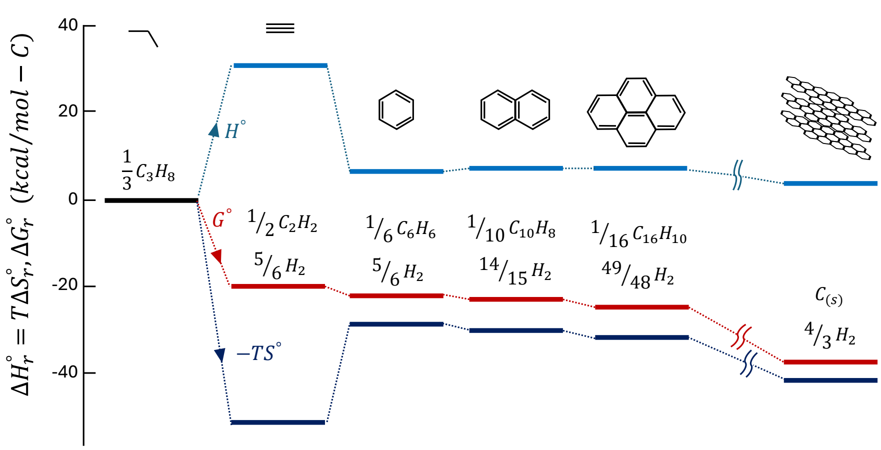

Figure 1.2: Standard enthalpy (ΔH) and entropy ($T\Delta S$) contributions to Gibbs function of reaction ΔG at 1600 K for carbon formation from propane(reprinted from Ref. [5])

There are other complicating factors that hinders the fundamental understanding of soot inception such as lack of a decisive criterion to distinguish gaseous molecules from particles [23], and the overlap of inception with surface growth and agglomeration [41]. Measurement techniques have limited capability in soot detection due to short time and length scales of soot inception and surface growth ranging from pico- to milliseconds and micro- to millimeter [42]. Fig. 1.3 demonstrates the length and time scales relevant to different stages of soot formation from PAH precursors to incipient, nascent and mature soot in flames. Moreover, the collected data from measurements might not be true representatives of soot formed at the probed location of the studied process. For example, intrusive diagnostic methods based on thermophoresis or dilution have a sampling probe that can perturb flow dynamics or alter structure and composition of soot [43]. Non-intrusive techniques such as optical methods also rely on assumptions about light absorption and scattering of soot [44] and its morphology [45].

Despite the gaps in fundamental understanding of soot formation and limitations of diagnostics methods, models have been developed describe the soot inception and growth. These models have been formulated as a set of clear pathways that explain soot inception based on collisions of PAH molecules. They have to be consistent with current knowledge of soot inception, feasible to be coupled with chemistry and particle dynamics models, and able to predict soot mass, PSD and morphology observed in flames and reactors.

The classic description of soot inception relies on PAH dimerization where collision of two PAH molecules (monomers in this context) forms a dimer held together by Van der Wales forces [46]. The dimerization is a irreversible process with an efficiency that accounts for the reversibility or dissociation of dimers. The theory postulates that PAH growth continues by sequential addition of a monomer (PAH molecule) forming stacks of dimers, trimers, tetramers and so on to reach a certain mass threshold that marks the emergence of incipient soot [46], but for practical purposes, a dimer is usually considered as an incipient soot. Here, we call this model *Irreversible Dimerization*. Irreversible Dimerization has been used to predict soot formation in burner-stabilized premixed [47, 48], counterflow diffusion flames [49, 50], coflow diffusion flames [51, 52]. A collision efficiency factor ranging between 10^{-6} to 1 is also employed to adjust the inception flux and PAH adsorption rates to achieve desired soot mass and size distribution. PAHs of moderate sizes such as pyrene (4 rings) to coronene (7 rings) have been considered as the starting point of inception due to their thermodynamic instability that justifies the irreversibility at high temperatures [46]. However, the theoretical calculations [53] and experiments [54] indicated that PAH dimerization is highly reversible in flame conditions.

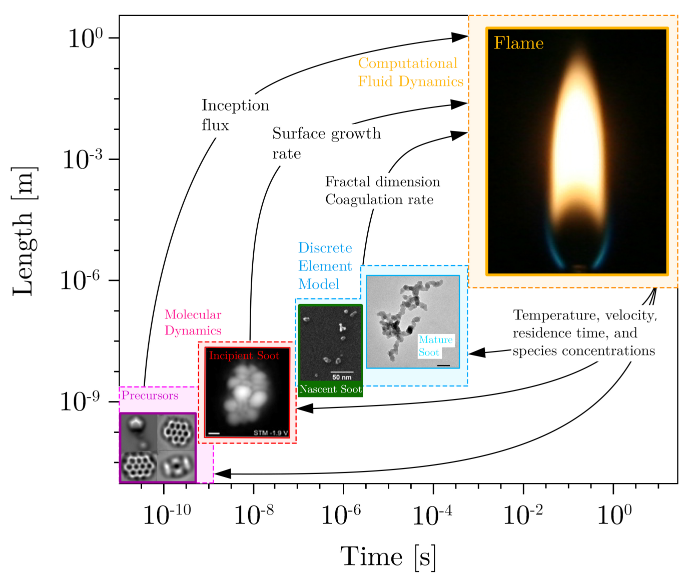

Figure 1.3: The range of time and length scales of the processes involved in soot formation from molecular reactions to particle-fluid interaction in flames

The inception flux of irreversible dimerization is determined from collision of PAHs, which is proportional to $\sqrt{T}$, so new particles can be formed at low temperatures (even less than 500 K) [55] despite experimental evidence for termination of inception below 1200 K [56, 57]. Moreover, the arbitrary selection of efficiency factors alters the distribution of mass between inception of surface growth could significantly change soot mass, PSD, and morphology [58]. Miller [59] used equilibrium constant for PAH dimerization to calculate the net dimerization rate and demonstrated that the collision of PAHs larger than circumovalene ($\sim$800 amu) could last long enough grow into incipient soot. However, the concentration of PAHs drops rapidly with size [5]. The entropy barrier of dimerization is significant for larger PAHs [60].

Eaves et al. [61] relaxed the irreversibility assumption, and developed a reversible clustering model to simulate inception using an array of PAHs from naphthalene to benzo-pyrene. Building on that work, Kholghy et al. [62] emphasized on the necessity of chemical bond formation after physical PAH clustering for accurate prediction of volume fraction, primary particle diameter and PSD in ethylene coflow diffusion flames. Later, Kholghy et al. [63] proposed the *"Reactive Dimerization"* model which starts with reversible collision of PAHs leading to physical dimers held with vdW forces that are graphitized and form chemically-bonded dimers that serve as soot nuclei grow via surface reactions. They also performed a systematic analysis on the contribution of different PAHs, and concluded that one- and two-ring aromatics account for almost all of inception flux in the so-called *"sooting flame"* [48]. However, Frenklach and Mebel [31] pointed out that an inception model that initiated with a highly reversible step similar to Reactive Dimerization [63] cannot produce sufficient flux of particles to match measurements of the benchmark burner-stabilized stagnation flame [34]. Instead, they proposed a HACA-driven mechanism where addition of monomer molecule to its radical activated by hydrogen abstraction for a stable dimer via an E-Bridge bond formation, and this sequential process continues to form trimers, tetramers, and larger PAH clusters.

The gas-phase chemistry of aromatics can be extended to account for chemical growth of incipient soot via surface reactions [28]. This hypothesis, known as "chemical similarity" postulates that the reactions occurring on the soot surface are similar to those involving large molecules of PAHs in the gas phase. It also provides means to describe the rates of surface growth and particle oxidation in terms of elementary chemical reactions. In other words, it is assumed that the surface of soot particles is made up of lateral faces of larges PAHs covered with C-H bonds. This is the basis for HACA mechanism [46, 64] that assumes the soot surface to consist of hydrogenated sites with a predefined density. Mass growth on soot surface requires H-abstraction to form a radical site, followed by acetylene attack similar to growth of PAH molecules in the gas-phase. The reactivity of these sites changes with time and temperature [65, 66], described as soot aging. For modelling purposes, a temperature-dependent multiplier, usually represented by α, was introduced to account for these effects. Appel et al. [64] showed α changes with temperature and particle size.

Adsorption of PAHs on the surface of soot particles is also a viable growth mechanism [46], more specifically called physiorption or chemisorption depending on the mechanisms driving the adsorption process [67]. There is still debate over the stability of adsorbed PAH molecules on soot surface [68]. Following the hypothesis that PAHs are building blocks of soot particles, a mechanism similar to inception is often used to describe PAH-soot growth.

1.3 Coagulation and agglomeration

In typical soot formation processes such as those occurring in flames and reactor, soot particles are formed at high number concentrations (10^{12} 1/cm^3), and inception and surface growth are relatively short compared to the total residence time of soot particles. As a result, coagulation becomes dominant rapidly attaining both [69] self-preserving size distribution (SPSD) [70] and asymptotic fractal-like structure [71]. The evolving fractal-like structure of agglomerates quantified by their mobility diameter normalized by primary particle, d_m/d_p, and gyration, d_m/d_g, diameters can be described with power laws derived from mesoscale simulations [72]. The collision frequency of agglomerates depends on their evolving fractal-like morphology. Also, polydisperse agglomerates collide more frequently than monodisperse ones. The enhancement in their collision frequency reaches an asymptotic value of 35% [69] or 82% [73] in the free molecular or transition regimes, respectively at SPSD regardless of the polydispersity in their constituent primary particles. Particle morphology formed by inception, surface growth and agglomeration can be tracked precisely by mesoscale simulations, such as Discrete Element Modeling (DEM) [74] where each agglomerate is tacked separately. However, they are computationally expensive and interfacing them with chemical kinetics in computational fluid dynamics (CFD) is not trivial [75]. This limits their application in simulation of realistic flames and reactors. So, sectional population balance models (SPBM) are often used to track agglomerate and primary particle size distribution [76], morphology [77], and composition [51] in complex laminar [51] and turbulent flows [78]. Using the SPBMs coupled with relations for agglomerate fractal-like structure [79] and collision frequency [36], particle size distribution, morphology and composition can be tracked accurately. However, the computational cost of SPBMs increases exponentially with the number of sections [76] and particle properties [51] tracked. Thus, one property (e.g. agglomerate mass) is typically tracked with SPBMs to reduce computational cost. This does not allow to account for agglomerate fractal-like structure [80, 81] which limits SPBM accuracy in predicting surface growth and coagulation rate of agglomerates and their size distribution.

Alternatively, particle dynamics can be tracked by the method of moments (MOM) [82] or monodisperse population balance models (MPBM) [83]. Such models only track average particle properties (e.g. moment ratios) and their accuracy could be limited if unrealistic assumptions (e.g. approximating agglomerates as monodisperse and perfect spheres) are used. However, when inception and surface growth are short [84] and high particle (number) concentrations are formed [72], they lead to rapid attainment of self-preserving size distributions (SPSD) and agglomerates having asymptotic structure [69]. In this case a MPBM or MOM can be assembled on a firm scientific basis with accuracy on par with DEM [74], SPBM [85] and experimental data [86, 87, 88]. Such models can be readily interfaced with CFD simulations [89] without significant computational cost, making them ideal for three-dimensional and even turbulent flame simulations.

The MOM tracks moments of the PSD and estimates average particle properties such as mass [90], surface area [91], the number of constituent primary particles per agglomerate, $\mathrm{n_p}$ [82], or even particle composition [92] using the ratio of the moments. The MOM with four equations was used to describe synthesis of optical fibers by simultaneous reaction, diffusion, coagulation and thermophoresis of SiO_2 in laminar flow reactors assuming a lognormal PSD [93]. The MOM with interpolative closure (MOMIC) was developed to predict simultaneous nucleation, surface growth and coagulation of soot agglomerates and estimate its PSD with six equations [82]. To calculate source terms of the transported moments, additional moments that are not tracked are needed preventing the closure of the system of differential equations with the MOM [90, 94]. Thus, often the PSD shape is assumed a priori [90] or extra equations are solved to estimate it [83].

The MPBMs do not have the closure problem and calculate average particle properties by tracking their total concentration, mass [83] and area [95, 96]. Leung et al. [97] used a 2-equation MPBM to track soot concentration and mass in (non-premixed) flames assuming spherical particles. Good agreement was achieved for measured soot mass. However, the specific surface area [96] and coagulation frequency of spheres are significantly smaller compared to that of agglomerates with the same mass underestimating their oxidation rate [85] and overestimating their concentration [83]. Kruis et al. [83] proposed a 3-equation MPBM to account for the fractal-like structure of nanoparticle agglomerates during coagulation and sintering. Agglomerate volume and area were used to obtain their equivalent primary particle diameter, $\mathrm{d_p}$, and $\mathrm{n_p}$. Then, agglomerate collision diameter, i.e. $\mathrm{d_g}$, was calculated by $\mathrm{D_f}$, $\mathrm{d_p}$ and $\mathrm{n_p}$ to account for their fractal-like structure that affects their collision frequency. Tsantilis and Pratsinis [95] extended the MPBM to predict hard-(chemically-bonded) and soft- (physically-bonded) agglomerates during synthesis of SiO_2 and TiO_2 [98] nanoparticles with simultaneous reaction, surface growth, coagulation and sintering. Such a MPBM applies best at high concentrations when inception and surface growth are short [84] resulting in the dominance of coagulation where particles rapidly reach their SPSD and asymptotic fractal-like structure. This is often the case for soot emitted from a variety of combustion devices or CB reactors where inception and surface growth are limited to only a few milliseconds when temperature is very high (i.e. $\mathrm{T} \geq 1500\mathrm{K}$) [63].

1.4 Soot maturity and its optical properties

The maturity level of soot is described as the evolution of physical and chemical properties from incipient to graphite-like mature soot [99]. It involves the growth of the graphitic crystallite fine structure of soot within, and perpendicular to, the aromatic layers. [100, 101], known as graphitization, followed by increase in the size of the crystallite-layer planes and decrease of the interlayer spacing [26]. This process is accompanied by the pyrolytic conversion of hydrocarbon species and substituted hydrocarbons toward elemental carbon and increase in the carbon-to-hydrogen (C/H) ratio, known as carbonization. Soot maturity is closely associated with reactivity of surface sites [102]. Fig. 1.4 compares nanostructure of nascent soot composed of disoriented PAH clusters with that of mature soot with a core-shell pattern where disordered core is surrounded by concentrically oriented graphitic layers [35]. The core-shell nanostructure of mature soot strongly depends on process conditions such as pressure, fuel identity, temperature and residence time of the particles.

Evolution of soot maturity and morphology impacts its optical properties. Incipient and nascent soot absorb shorter wave lengths (λ <600 μm) as opposed to mature soot particles that are broadband light absorbers [103]. Non-intrusive optical diagnostic methods such as light extinction (absorption and scattering) [104] and Laser Induced Incandescence (LII) [44] are widely utilized to measure soot volume fraction, f_v using extinction coefficient and the absorption function, E(m) that depends on soot refractive index, m, soot composition and morphology [105].

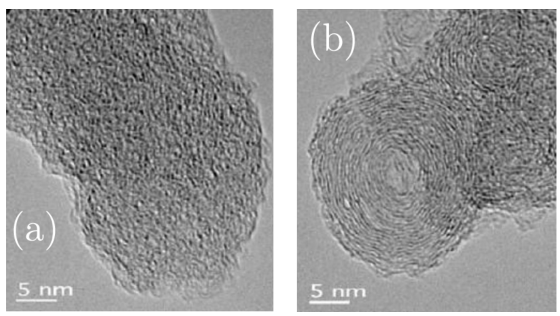

Figure 1.4: The HRTEM image of (a) a nascent soot primary particle with disordered internal nanostructure juxtaposed with that of (b) a mature soot primary particle with well-organized cluster near the shell sampled from soot generated by pyrolysis of ethanol at 1250 and 1650, respectively. Reprinted from Ref.[6]

However, the effects of soot composition, and morphology on its refractive index, and absorption function are not fully understood yet. E(m) of soot at λ =1064 nm measured by two-color LII measurements increases from 0.193 to 0.349, and 0.226 to 0.340 for ethylene premixed flames with equivalence ratio of ϕ=2.1 and ϕ=2.3, respectively when height above burner (HAB) changes from 8 to 14 mm [106]. During acetylene pyrolysis in a shock tube, E(m) of soot particles increases from 0.05 to 0.25 as their primary particle diameter, d_p, grows to 20 nm within 1.6 ms [107]. Using m and E(m) of mature spherical soot [108] at different HABs neglecting the impact of soot morphology and composition on m [109] can overestimate soot volume fraction by 100% [110]. So, accurate estimation of the evolving optical properties of soot is essential to close the carbon mass balance in the measurements [110], and analyze reaction kinetics for soot inception [63] surface growth [64] and oxidation [111] that are essential for development and validation of aerosol dynamics models coupled with chemistry to predict soot formation.

The dependence of soot m on its morphology and composition can be quantified by soot optical band gap, E_g [112]. The optical band gap concept was originally proposed by Tauc et al. [113] for semi-conductors and later developed for amorphous carbon [114], and it can be used to describe the crystalline character of PAH clusters in soot [114]. Incipient flame-made soot with diameters less than 20 nm exhibit quantum dot behavior [115] with 0.7< E_g <2 eV that is larger than E_g of graphitized soot, 0.12 eV [115]. Optical band gap of organic carbon coated on soot remains nearly constant (1.8-1.9 eV) over soot evolution [116]. In contrast, Russo et al. [117] measured soot optical band gap using ex-situ and in-situ methods and showed that E_g drops from 0.7 eV at HAB= 8 mm to 0.2 eV at HAB=14 mm in an ethylene premixed flame with ϕ=2.3 as particles grow and become more mature by carbonization. Kelesidis and Pratsinis [118] correlated the evolving refractive index of soot with its E_g obtained from quantum confinement theory (QCT) [115] for wavelengths of λ= 532 and 1064 nm [118] by linear interpolations between that of nascent and mature soot. The proposed relations were then used with soot morphologies obtained from Discrete Element Modeling (DEM) simulations to obtain the evolving Mass Absorption Cross-section (MAC), E(m), and the absorption function ratio of soot (E(λ=532nm)/ E(λ=1024nm)) and compared with measurements [119, 120, 121]. Employing the relation in laser diagnostics to reprocess light extinction measurement data resulted in accurate prediction of f_v in moderate (ϕ=2.34) [110] and rich (ϕ >3) ethylene premixed flames [122], and fv along the centerline of a coflow ethylene diffusion flame [123] which is necessary to close the carbon mass balance in soot generating processes.

1.5 Oxidation

Soot oxidation is described as the removal of soot mass by reaction of molecular oxygen (O_2), oxygen radical (O), and hydroxyl radical (OH) from soot surface. Because of its heterogeneous reaction kinetics and mechanisms, oxidation is expected to be sensitive to surface structure and composition. So, oxidation mechanisms depend on soot maturity and temperature-time history. In near stoichiometric and fuel-rich conditions, the contribution of OH radical to soot oxidation is predominant [124] compared to that of O radicals [125]. Fenimore and Jones [126] investigated soot oxidation rate for low oxygen partial pressures and temperatures from 1530 to 1890 K, highlighted the importance of OH as a major oxidation agent, and attributed the faster rates compared to those predicted by Lee et al. [127], to OH oxidation. One approach for describing OH oxidation is by a introducing collision efficiency representing the fraction of collisions of OH with soot particles that resulted in the removal of a carbon atom [124]. Scanning Mobility Particle Sizer (SMPS) with a two-stage burner also showed a collision efficiency of 0.13 [128].

The empirical relation of Nagle and Strickland-Constable (NSC) [129] originally developed for oxidation of pyrolytic graphite have been widely used to describe soot oxidation by O_2. It describes O_2 oxidation rates based on partial pressure of O_2 and the fraction of reactive edge sites to less reactive basal planes using the graphite analogy for soot. The dominance of edge sites in typical combustion conditions accounts for the relatively small reactivity of soot and basal planes become reactive at high temperature (>2500 K) [125]. O_2 oxidation kinetics was also described with a power-law kinetics [127], and shown to be first order in oxygen concentration [130]. Alternatively, soot oxidation can be explained based on chemical similarity using HACA mechanism assuming that active site are attacked by O_2 and OH leading to loss ot carbon and release of CO.

A different oxidation regime has been identified for soot at low temperatures (T<1000 K) where O_2 diffuses and reacts with bulk soot accounting for most of soot mass consumption [87]. Internal oxidation compacts the pore network of soot resulting in hollow CB [131], diesel [132] and biodiesel [133] soot particles and increases their SSA up to a factor of four [132]. Oxidation can cause fragmentation of soot particles affecting their morphology. The increase in number concentration and the change of soot morphology in lean premixed flames [134] and in the oxidation region of diffusion flames [135] was attributed to fragmentation. Such a change in agglomerate morphology has not been observed in fuel-rich conditions, so fragmentation was linked to O_2 oxidation.

1.6 Research objectives

The purpose of this project is developing a process design tool to predict the yield, morphology, composition and size distribution of soot particles in industrial flames and reactors under different temperatures, pressures and residence times. Integrating gas chemistry, all existing inception and surface growth models and particle dynamics in a single package provides a fully coupled soot description from breakdown of fuel to agglomeration and oxidation of particles that enables sensitivity analysis to shed light on key pathways for soot inception and surface growth in various targets. This

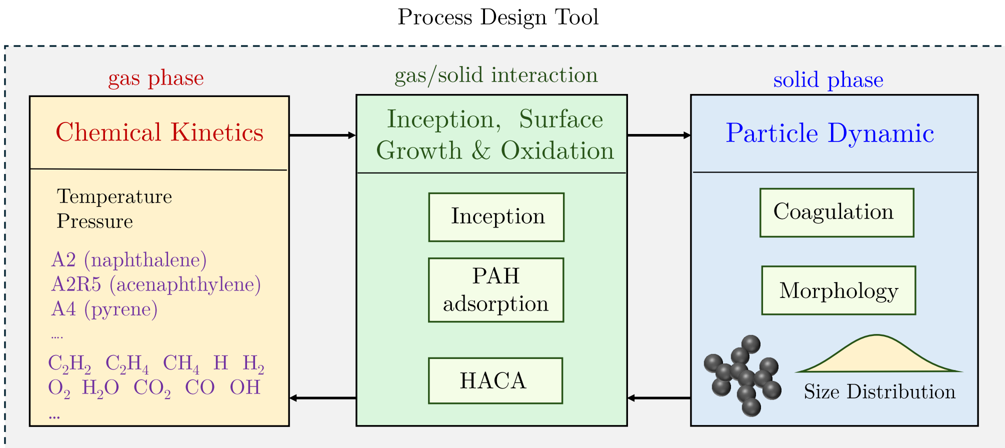

Figure 1.5: The conceptual structure of developed process design tool that account for gas mixture properties, and soot chemistry involving inception, surface growth and oxidation as well as its coagulation leading to the fractal-like morphology of soot particles

tool can also be used for rate constant optimization and mechanism reduction under soot forming conditions. The conceptual structure of this tool is shown in Fig 1.5 that includes a chemical kinetics unit coupled with Cantera [136] to compute thermal and physical properties of gas mixture such as temperature, pressure, density, and enthalpy based on ideal equation of state for the gas mixture. The chemistry of gas mixture is quantified using detail reaction mechanisms that enables tracking concentration of fuel, intermediate species such as C_2H_2 and H_2 essential to surface growth, and PAHs such as benzene (A1), naphthalene (A2) and pyrene (A4) known as building blocks of soot. The second unit accounts for soot inception and surface growth via PAH adsorption and HACA as well as surface oxidation. It accommodates the four inception models widely used in literature. The third unit deals with particle dynamics using a MPBM and SPBM to describe the coagulation of soot particles that results in the fractal-like structure of particle with an evolving size distribution. A multi-step validation procedure is followed to assess reliability of each sub-model that entails the comparison of collision frequency kernels and agglomerate morphology of different particle dynamics models with those of detailed DEM simulations and establishing energy and elemental carbon, and hydrogen balances for all combinations of particle dynamics and inception models with reactors and flames. This tool can be applied to a variety of experimental targets with different temperature- and pressure-time-histories, residence times, and fuel compositions.

As the first step, methane pyrolysis in shock-tube will be simulated using a constant volume reactor. The species measurements will be used to assess the reaction mechanism and examine carbon conversion flux from fuel to smaller intermediates and larger hydrocarbons. The inception flux and surface growth rates of the soot model strongly depend on the reaction mechanism employed to track the concentration of soot precursors. The performance of inception models will be analyzed by comparing the predicted soot volume fraction, f_v and primary particle diameter, d_p, with data collected from extinction and TEM measurements. A sensitivity analysis will be conducted on model parameters to identify the determining factors that control both f_v and d_p in the framework of shock-tube simulations with short residence time (t$\approx$ 2 ms) and high temperatures ($\geq$ 2000K). Then, the optimization of inception models will be performed by adjusting the rate constants within their physical limit (maximum collision rate of molecules) to minimize the prediction error for yield and morphology. This shows the range of inception flux that ensures the accurate prediction of soot yield and morphology represented by f_v and d_p, respectively in good agreement with measurements. Then, the optimized rates will be applied to shock-tube in a wide temperature range, and the f_v predictions will be compared with data. The measurement on shock-tube pryolysis of methane with different H_2 concentrations will be used to investigate the effect of hydrogen on pyrolysis chemistry as well as soot yield and morphology. A main focus will be finding the reaction mechanism that describes methane decomposition rate and acetylene and PAH concentrations in good agreement with the data from absorbance measurements. The inception flux and surface growth rate that results in accurate prediction of soot yield and morphology will be estimated and compared with cases without hydrogen. A similar investigation will be conducted for flow reactors using plug flow reactor model of omnisoot at lower temperatures and longer residence times and then the range of expected inception flux with be compared with those of shock-tubes.

In the next step, a reactor network model based on omnisoot will be developed to enable building a series of plug flow reactors (PFR) and perfectly stirred reactor (PSR) models with multiple gas sources to more accurately represent complex industrial reactors. A set of tests will be developed to ensure gas and solid mass and energy balance during the development process. The connections between elements will be developed in a way that recognizes upstream and downstream flows. Moreover, safeguards will be put in place for the network to avoid faulty designs that can result in cyclical flows and detached networks. The performance of the reactor network will be evaluated using benchmark data from literature [137].

The flame solver of omnisoot will be developed from its current experimental stage to a functional level. After validation of the coagulation and surface growth sub-models and ensuring the conservation of carbon and hydrogen mass, and energy balances, the flame solver will be applied to a variety of counterflow diffusion [138, 139] and burner-stabilized premixed flames[34] with stagnation plate [88] at atmospheric and elevated pressures. The reaction mechanism study will be performed to reduce the uncertainty in tracking gas chemistry by comparing the model predictions of species concentration with measurements. The performance of the soot model will be evaluated by comparing the predicted soot volume fraction, agglomerate number concentration, PSD, mobility and primary particle diameters with those obtained from light extinction measurement, scanning mobility particle sizer (SMPS), and TEM samples, and the comparisons will be use to estimated the range of inception flux and surface growth rates that can predict soot mass, morphology and size distribution with minimum error compared to data.

Chapter 2

Theoretical Model

The mathematical basis for *Omnisoot* is explained in the top-to-bottom hierarchical order. First, the governing equations for constant volume, constant pressure, perfectly stirred and plug flow reactor models are reviewed in Sections 2.2.1-2.2.4. These equations differ from pure gas-phase transport formulation as they consider the solid phase. The transport equations of "soot variables" include the source terms that account for change in each soot variable due to inception, surface growth, oxidation and coagulation. Then, particle dynamics models are explained in Sections 2.3 that entails description of size distribution and morphology of soot particles and their collision rate, which is used to determine the coagulation source term. Section 2.4 focuses on the *"PAH growth model"*s that take care of inception and adsorption from designated precursors and calculates the corresponding source terms. Similarly, the *"surface reactions"* model are detailed in Section 2.5 that elaborates on the surface growth and oxidation rates based on HACA mechanism. Figure 2.1 illustrates the general structure of Omnisoot and the sub-models.

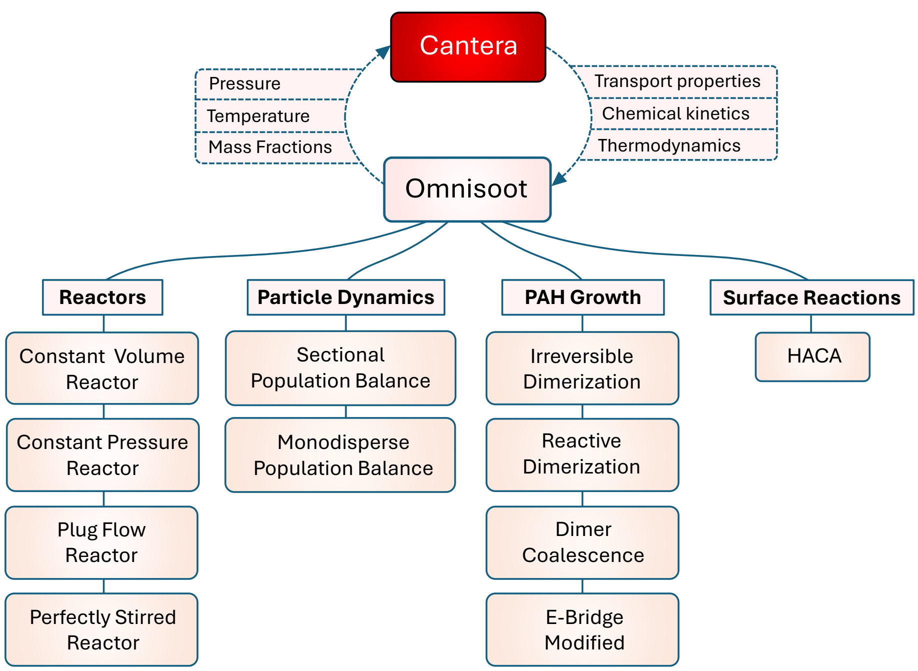

Figure 2.1: The structure of Omnisoot that illustrates the coupling with Cantera and sub-models including reactors, particle dynamic models, PAH growth models and surface reactions models

2.1 Assumptions and conventions

Here, the main conventions and assumptions used in the derivation of the mathematical models are listed below.

1. f_v and φ denote the volume of soot particles normalized by the gas volume and reactor volume, respectively. Their relationship can be expressed as:

$$\begin{aligned} f_v &= \frac{V_{soot}}{V_{gas}}, \\ \varphi &= \frac{V_{soot}}{V_{gas} + V_{soot}}, \\ \varphi &= \frac{f_v}{1 + f_v}. \end{aligned} \tag{2.1}$$

2. $\dot{s}_k$ denotes the rate production/consumption of k_{th} gaseous species (i.e., gas scrubbing) due to soot inception, surface growth and oxidation. It is positive when the species is released to the gas mixture.

3. Each soot agglomerate consists of monodisperse spherical primary particles, which are in point contact. So, the model does not consider formation of necking (or sintering) in soot agglomerates by surface growth.

4. The word *"particle"* refers to soot both in spherical and agglomerate shape.

5. The density of soot is assumed constant at the value of 1800 kg/m^3. Soot density changes with its maturity level, which is often linked to the elemental C/H ratio of soot particles [140]. Here, the considered value represents an average between density of mature soot with high C/H ratios ($\rho = 2000$ kg/m^3) and that of nascent soot with low C/H ratios ($\rho = 1600$ kg/m^3) [141, 140].

6. The incipient soot particles are 2 nm in diameter, so the model does not allow particles with a primary particle diameter smaller than 2 nm. The number of carbon atoms in the incipient soot particle, $n_{c,min}$, is calculated from the mass of a sphere with a diameter of $d_{p,min} = 2$ nm assuming pure carbon content, which results in:

$$n_{c,min} = \frac{\pi}{6} \rho_{soot} d_{p,min}^3 \frac{1}{W_{carbon}} \approx 378. \tag{2.2}$$

7. The specific heat, internal energy and enthalpy of soot are approximated by those of pure graphite, and are employed to close the energy balance in the system [142].

8. Soot particles and gas are in thermal equilibrium during soot formation processes, and there is no temperature gradient within each agglomerate.

9. ψ denotes a *soot variable* that represents a mean property of soot particles in each section tracked in Omnisoot by solving transport equations for the total concentration of agglomerates, N_{agg} and primary particles, N_{pri}, and the total carbon, C_{tot} and hydrogen content, H_{tot} of soot. S_ψ is the source term of the soot variable, ψ, which appears in soot equations.

10. *PAH growth* is a sub-model of Omnisoot with a set of pathways that determine the rate of inception and adsorption from PAHs (designated as soot precursors) in the gas mixture.

11. *Surface reactions* is a sub-model of Omnisoot that describes the addition of acetylene to soot surface, and removal of carbon via oxidation by OH and O_2 following the HACA scheme. The model does not consider soot oxidation with CO_2, H_2O and NO_x.

Table 2.1: The names, symbols, chemical formula and molecular weight of the soot precursors used by Omnisoot.

Species name	Symbol	Chemical formula	W [kg/mol]
Naphthalene	A2	$C_{10}H_8$	0.128
Phenanthrene	A3	$C_{14}H_{10}$	0.178
Pyrene	A4	$C_{16}H_{10}$	0.202
Acenaphthylene	A2R5	$C_{12}H_8$	0.152
Acephenanthrylene	A3R5	$C_{16}H_{10}$	0.202
Cyclopentapyrene	A4R5	$C_{18}H_{10}$	0.226

12. The single superscript i denotes the section number of a soot variable or a derived property. For example, d_p^i represents the primary particle diameter of section i. The double superscript ij indicates a property related to two sections; for example, β^{ij} denotes the collision frequency between sections i and j. In the case of the monodisperse model, the section number can be omitted because it is equivalent to the sectional model with only one section.

13. The computation of morphological parameters is done similarly in both particle dynamics models, but they are explained separately in Section 2.3.1.1.

14. *precursors* refer to the PAHs larger than naphthalene used for inception and PAH adsorption. The list of precursors with their chemical formula and molecular mass is provided in Table 2.1. It should be noted that the precursors can be dynamically changed by Omnisoot's user interface.

2.2 Reactors

The governing equations of reactor models implemented in Omnisoot are briefly presented in the following sections. The control volume encompasses the gas mixture and soot particles, as illustrated in Figure 2.2. The equations ensure conservation of total mass and energy of the gas and particle system, which can also receive or lose heat through the reactor walls.

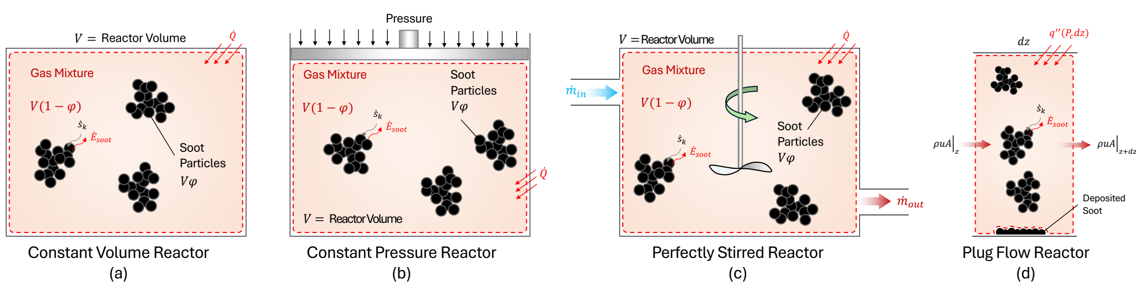

Figure 2.2: The schematics of constant volume reactor (a), constant pressure reactor (b) perfectly stirred reactor (c), and plug flow reactor (d)

2.2.1 Constant Volume Reactor (CVR)

As shown in Figure 2.2a, CVR represents a closed system constant volume where chemical reactions converts part of the gas mixture to soot particles. The mass balance equation is written as:

$$\frac{\mathrm{d}}{\mathrm{d}t}(m) = (1-\varphi)V\sum_i \dot{s}_i W_i, \tag{2.3}$$

where m is the mass of the gas mixture. The rate of change of m is equal to the rate of production of soot mass. Similarly, the species equation for species k is expressed as:

$$\frac{\mathrm{d}Y_k}{\mathrm{d}t} = \frac{1}{\rho}\left(\dot{\omega}_k + \dot{s}_k\right)W_k - \frac{1}{\rho}Y_k\sum_i \dot{s}_i W_i. \tag{2.4}$$

The transport equation for a generic soot variable, ψ, can be written as:

$$\frac{\mathrm{d}\psi}{\mathrm{d}t} = S_\psi - \frac{\psi}{\rho}\sum_i \dot{s}_i W_i. \tag{2.5}$$

The second term on RHS of Equations (2.4) and (2.5) denotes the change in Y_k and ψ, respectively, due to the removal or addition of gas mass by soot-related processes. The energy conservation for the gas mixture is written in terms of the rate of change of temperature. An external heat source of $\dot{\mathrm{Q}}$ is considered to account for possible heat loss/gain of the reactor.

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\rho c_v + \rho_{soot} f_v c_{soot}} \Bigg[&-\sum_k e_k\left(\dot{\omega}_k + \dot{s}_k\right)W_k \\ &+ e_{soot}\sum_k \dot{s}_k W_k + \frac{\dot{Q}}{V(1-\varphi)}\Bigg],\end{aligned} \tag{2.6}$$

where $\rho_{soot} f_v c_{soot}$ and $e_{soot}\sum_k \dot{s}_k W_k$ represent the formation and sensible energy of soot, respectively. We investigated the effect of considering soot formation and sensible energy on gas and soot properties by simulating the pyrolysis of 30% CH_4-Ar with and without considering the above mentioned term. As shown in Figure 2.3a, neglecting soot sensible energy results in the overprediction of temperature by nearly 150 K and mobility diameter by a factor of 3 during the 80 ms of the simulation. The overprediction of temperature changes gas chemistry, leading to a noticeable decrease in the residual methane and benzene (Figure 2.3b).

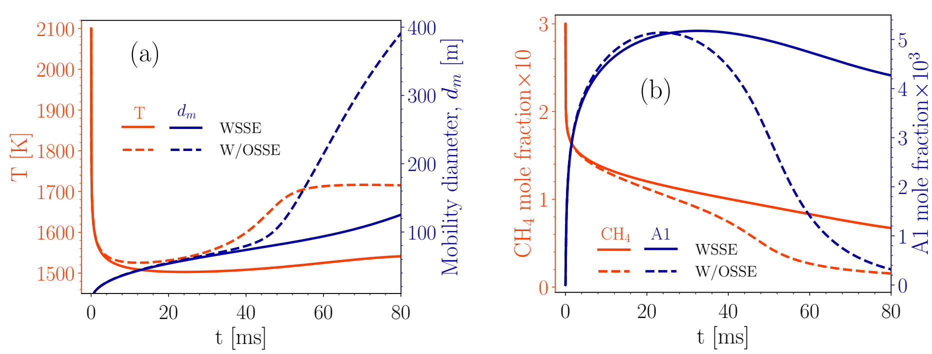

Figure 2.3: The comparison of temperature and soot mobility diameter, d_m (a) and the mole fraction of methane, CH_4, and benzene, A1 in the simulation of the pyrolysis of 30%CH_4-Ar when soot sensible energy is considered (labeled as "WSSE") and neglected (labeled as "W/OSSE"). The CVR is used along with Caltech mechanism [7], Reactive Dimerization and MPBM

2.2.2 Imposed Pressure Reactor

As show in Figure 2.2b, this reactor is a closed system similar to CVR with a main difference that the boundary can move and change the volume of the system due to an external pressure that can stay constant or vary with time. In either case, the pressure is known a priori in this model. The heat transfer can occur through reactor walls, which changes the internal energy of the gas-particle system.

The equations of mass, species, and soot variables in CPR are the same as those in CVR, but the energy equation is written based on enthalpy, h, instead of internal energy, e, as:

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\rho c_p + \rho_{soot} f_v c_{soot}} \Bigg[&- \sum_k h_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k \\ &+ h_{soot} \sum_k \dot{s}_k W_k + \frac{\mathrm{d}P}{\mathrm{d}t} + \frac{\dot{Q}}{V(1-\varphi)} \Bigg].\end{aligned} \tag{2.7}$$

where $\mathrm{d}P/\mathrm{d}t$ represents the effect of variable pressure on the gas temperature, which is zero when the pressure is constant throughout the process.

2.2.3 Perfectly Stirred Reactor (PSR)

In this reactor, gas enters with a mass flow rate, composition, and temperature of $\dot{m}_{in}$, Y_{in}, and T_{in}, respectively, and homogeneously reacts with the mixture inside the reactor. The reacting gas reaches a spatially uniform temperature and composition described by T and Y. It is assumed that temperature, composition and soot properties of the outflow are the same as the mixture inside reactor. Without soot formation, the inlet and outlet mass flow rates are equal (i.e. $\dot{m}_{in} = \dot{m}_{out}$), but when soot is formed, $\dot{m}_{out}$ is slightly less than $\dot{m}_{in}$. The nominal residence time of PSR is calculated as:

$$\tau_{psr} = \frac{\rho V}{\dot{m}_{in}}. \tag{2.8}$$

The conservation of mass of PSR can be written by considering the mass flux of in- and outflow, and the removal of mass due to soot formation as:

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \dot{m}_{in} - \dot{m}_{out} + V(1-\varphi) \sum_i \dot{s}_i W_i. \tag{2.9}$$

Gas composition is obtained by solving the species transport equations as:

$$\frac{\mathrm{d}Y_k}{\mathrm{d}t} = \frac{\dot{m}_{in}}{\rho V\left(1-\varphi\right)} \left(Y_{k,in} - Y_k\right) + \frac{1}{\rho} \left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i \right]. \tag{2.10}$$

The soot transport equations can also be expressed as:

$$\frac{\mathrm{d}\psi}{\mathrm{d}t} = \frac{\dot{m}_{in}}{\rho V\left(1-\varphi\right)} \left(\psi_{in} - \psi\right) + S_\psi - \frac{1}{\rho} \psi \sum_i \dot{s}_i W_i. \tag{2.11}$$

The energy equation for this reactor is written as:

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\rho c_p + \rho_{soot} c_{p,soot} f_v} \Bigg[&\frac{\dot{m}_{in}}{V(1-\varphi)} \left(h_{in} - h\right) - \frac{\dot{m}_{in}}{V(1-\varphi)} \sum_k \left(Y_{k,in} - Y_k\right) h_k \\ &- \sum_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k h_k + \sum_i \dot{s}_i W_i h_{soot} + \frac{\dot{Q}}{V(1-\varphi)} \Bigg].\end{aligned} \tag{2.12}$$

2.2.4 Plug Flow Reactor (PFR)

PFR is an ideal representation of a channel or duct where the temperature, composition, and soot properties of a steady-state one-dimensional flow evolve along the reactor. There is no spatial gradient over cross-section due to strong mixing. The diffusion along reactor is negligible.

The continuity equation for PFR is written as:

$$\frac{\mathrm{d}\dot{m}}{\mathrm{d}z} = (1-\varphi)A\sum_i \dot{s}_i W_i. \tag{2.13}$$

The momentum equation can also be established as:

$$u(1-f_v)\sum_i \dot{s}_i W_i + \rho u(1-\varphi)\frac{\mathrm{d}u}{\mathrm{d}z} = -\frac{\mathrm{d}}{\mathrm{d}z}(p(1-\varphi)) - \frac{\tau_w}{R_H}, \tag{2.14}$$

where τ_w, and R_H are the wall shear stress and hydraulic radius of the reactor. τ_w can be determined from the friction factor, f, as:

$$\tau_w = \frac{1}{2}\rho u^2 f, \tag{2.15}$$

where f can be accurately calculated over the entire range of Reynolds numbers, from laminar to turbulent flow, using the explicit formula provided by Haaland [143]:

$$\frac{1}{f^{1/2}} = -1.8\log\left(\frac{6.9}{Re} + \left[\frac{\epsilon/D_H}{3.7}\right]^{1.11}\right), \tag{2.16}$$

where ϵ and D_H are the wall roughness and the hydraulic radius of the reactor. The species equation can be expressed as:

$$\frac{\mathrm{d}Y_k}{\mathrm{d}z} = \frac{1}{\rho u}\left[\left(\dot{\omega}_k + \dot{s}_k\right)W_k - Y_k\sum_i \dot{s}_i W_i\right]. \tag{2.17}$$

The soot transport equations can also be written as:

$$\frac{\mathrm{d}\psi}{\mathrm{d}z} = \frac{S_\psi}{u} - \frac{\psi}{\rho u}\sum_i \dot{s}_i W_i - \frac{4}{D_H}\frac{k^i_{dep}\psi}{u}, \tag{2.18}$$

where k^i_{dep} is the deposition velocity of soot particles of section i, which is calculated as:

$$k_{dep} = \frac{Sh\cdot D^i}{D_H}, \tag{2.19}$$

where $Sh = 3.66$ for laminar flows, and it calculated using the Berger and Hau correlation [144] for the turbulent flow as:

$$Sh = 0.0165Re^{0.86}Sc^{1/3}. \tag{2.20}$$

The energy equation can be expressed as:

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{1}{\rho u c_p + \rho_{soot} u f_v c_{p,soot}} \Bigg[&-\sum_k h_k\left(\dot{\omega}_k + \dot{s}_k\right)W_k \\ &+ h_{soot}\sum_k \dot{s}_k W_k + q''\frac{P_c}{A}\Bigg],\end{aligned} \tag{2.21}$$

where q'' is the wall heat flux of the reactor.

2.3 Particle Dynamics

Population balance models rely on the Eulerian description of particles where average properties of particle population such as number density, mass or surface area are treated as continuous quantities and tracked by solving scalar transport equations. These methods are computationally cheaper compared with mesoscale models such as DEM, and can be easily interfaced with chemical kinetics in CFD solvers to simulate soot formation in laminar and turbulent configurations. Here, we use two particle dynamics models: a monodisperse population balance model (MPBM), which tracks four variables and results in four transport equations, and a fixed sectional population balance model (SPBM), which tracks three variables per section. In the SPBM approach, the total number of transport equations equals the number of tracked variables multiplied by the number of sections.

Having the total number concentration of agglomerates and primary particles and total carbon content of soot enables the model to describe evolving fractal-like morphology and surface area of soot agglomerates, which is essential to compute their collision frequency [145] as well as oxidation and surface growth rates [85]. Tracking hydrogen content of soot allows capturing the soot composition, thereby its maturity [51], and surface reactivity [92].

The common features of implemented particle dynamics models are reviewed in Section 2.3.1. The composition (H/C), diffusion coefficient, and morphology of soot particles are calculated in the same manner for MPBM and each section in SPBM. However, particle dynamics models differ in terms of the calculation of coagulation, and the processing of the contributions from inception, surface growth, and coagulation to generate the source terms of soot variables, S_φ, incorporated into the transport equations.

2.3.1 Common Features

2.3.1.1 Soot Morphology

The evolving fractal-like structure of agglomerates is quantified by their mobility diameter normalized by primary particle diameter, d_m/d_p, and gyration diameter, d_m/d_g, that can be described with power-laws derived from mesoscale simulations. Incipient soot is initially a sphere formed of PAHs that grows in size by surface reactions and forms agglomerates by coagulation. The collision frequency of particles depends on their evolving fractal-like structure [145]. Mobility and gyration diameters are calculated using power-laws developed to describe the morphology of soot from premixed [86], diffusion [146] flames, and diesel engines [147]. Figure 2.4 shows the schematic of a soot agglomerate composed of 12 primary particles, with d_p, d_m, and d_g annotated.

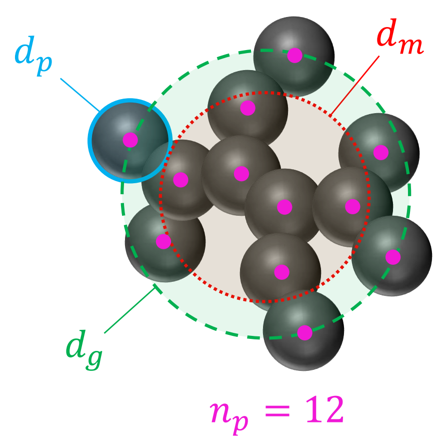

Figure 2.4: The schematics of a soot agglomerates with 12 primary particles ($n_p = 12$). Primary particle, d_p, mobility d_m, and gyration d_g, diameters are shown.

n_p^i is the number of primary particles per agglomerate of the i^{th} section that can be obtained by dividing the number concentration of primary particles in the i^{th} section by that of agglomerates in that section as:

$$n_p^i = \frac{N_{pri}^i}{N_{agg}^i}. \tag{2.22}$$

The primary particle diameter, d_p^i, is determined from the total carbon content and the number density of primary particles using:

$$d_p^i = \left(\frac{6}{\pi} \frac{C_{tot}^i \cdot W_{carbon}}{\rho_{soot}} \frac{1}{N_{pri}^i \cdot Av} \right)^{1/3}. \tag{2.23}$$

The DEM-derived power-laws [72] relate d_m^i and d_g^i to d_p^i and n_p^i as:

$$d_m^i = d_p^i \cdot {n_p^i}^{0.45}, \tag{2.24}$$

$$d_g^i = \begin{cases} d_m^i/({n_p^i}^{-0.2} + 0.4), & \text{if } n_p^i > 1.5 \\ d_m^i/1.29 & \text{if } n_p^i \leq 1.5 \end{cases} \tag{2.25}$$

The collision diameter, d_c^i, is the maximum of d_m^i and d_g^i:

$$d_c^i = \max\left(d_m^i, d_g^i\right). \tag{2.26}$$

The volume equivalent diameter, d_v^i, is the diameter of the sphere with the same mass as the agglomerate, and it is obtained as:

$$d_v^i = d_p^i \cdot {n_p^i}^{1/3}. \tag{2.27}$$

The primary particle surface area is calculated from d_p^i assuming spherical primary particles as:

$$A_p^i = \pi {d_p^i}^2. \tag{2.28}$$

A_{tot}^i (for each section) is defined as the total surface area of soot particles per unit mass of gas mixture obtained as:

$$A_{tot}^i = N_{pri}^i \cdot Av \cdot A_p^i. \tag{2.29}$$

Mass of each agglomerate, m_{agg}^i is expressed as:

$$m_{agg}^i = \frac{C_{tot}^i \cdot W_{carbon}}{N_{agg}^i \cdot Av}. \tag{2.30}$$

2.3.1.2 Soot Composition

The composition of soot characterized by their elemental carbon to hydrogen ratio (C/H) is a measure of soot maturity and increases from C/H < 2 for incipient soot [148] to $2 <$ C/H < 10 for nascent soot [149] and C/H > 20 for mature soot [99]. The soot agglomerates are assumed to have pure carbon graphitic core [51] with all hydrogen atoms on the surface [92]. C/H ratio can be obtained from the total carbon and the hydrogen content as:

$$\left(\frac{C}{H}\right)^i = \frac{C_{tot}^i}{H_{tot}^i}. \tag{2.31}$$

The carbon content of each agglomerate is a predefined parameter in the SPBM (depending on the section the agglomerate is placed), but it can be calculated from dividing C_{tot} by N_{agg} for the MPBM. The hydrogen content of each agglomerate is calculated for both particle dynamics models as:

$$H_{agg}^i = \frac{H_{tot}^i}{N_{agg}^i}. \tag{2.32}$$

2.3.1.3 Diffusion of soot particles

The diffusion coefficient of soot particle, D^i, is calculated as:

$$D^i = \frac{k_B T}{f^i}, \tag{2.33}$$

where f^i is the friction factor of particles in the gas mixture, and it is calculated as:

$$f^i = \frac{3\pi\mu d_m^i}{C^i(d_m^i)}, \tag{2.34}$$

where C^i is the Cunningham correction factor for the particle friction factor that accounts for non-continuum effects in the transition and free molecular regimes. C^i is calculated for a given diameter, d, as:

$$C^i(d) = 1 + \frac{2\lambda}{d}\left(1.21 + 0.4\exp(\frac{-0.78d}{\lambda})\right), \tag{2.35}$$

where λ is the mean free path of gas given as:

$$\lambda = \frac{\mu}{\rho}\sqrt{\frac{\pi W_{gas}}{2k_B Av T}}. \tag{2.36}$$

Note that λ is a property of the gas mixture that does not depend on particle mass and morphology.

2.3.2 Sectional Population Balance Model

A SPBM with fixed pivots is used to describe particle dynamics [150]. The mass range of particles is divided into discrete sections each of which includes agglomerates of the same mass. Figure 2.5 illustrates the interaction between the gas phase and mass sections, as well as the mechanisms by which particles move between sections. Inception introduces new particles to the first section with the mass corresponding to the incipient particle. The particles of each section can migrate to upper sections by gaining mass via surface growth and coagulation, and return to lower sections when they lose mass through oxidation without breaking into smaller particles (i.e., fragmentation is not considered). Particles attach at point contact without coalescence. The mass of sections is determined by a geometric progression that has an initial value equal to the mass of incipient soot particle, and a common ratio of SF, known as sectional spacing factor. The mass of each section is approximated by the carbon content of agglomerates in moles as:

$$C_{agg}^i = \frac{n_{c,min}}{Av} \cdot SF^{(i-1)}, \tag{2.37}$$

where $(i-1)$ represents the exponent of SF. The mass of hydrogen is ignored in the placement of agglomerates in the sections. The total number density of agglomerates, N_{agg}^i, and primary particles, N_{pri}^i, and the hydrogen content, H_{tot}, are tracked for each section. The mass of each section is fixed, so C_{tot}, for each section can be easily calculated by knowing the number of agglomerates; that is, $C_{tot} = N_{agg}^i \cdot Av \cdot C_{agg}^i$. Morphological parameters for each section are determined according to the equations provided in Section 2.3.1.1.

New particles formed by coagulation are assigned to an upper section, with a total mass equal to the sum of the masses of the colliding particles. If the mass of resulting particle falls between two consecutive sections, the mass is distributed between them proportionally. In some cases, the mass of the newly formed particle may exceed the upper bound of the last section, causing it to fall outside the tracked mass range. This leads to potential mass loss, which is a known limitation of the fixed pivot sectional model [151]. However, this issue can be mitigated by choosing an appropriate number of sections and a suitable spacing factor to ensure the uppermost sections remain unoccupied during the simulation. In the SPBM, the source terms of tracked soot variables (which appear in Equations 2.5, 2.11, and 2.18) are split into four parts showing the contribution of inception, surface growth, oxidation and coagulation. The effect of HACA and PAH adsorption are combined (denoted

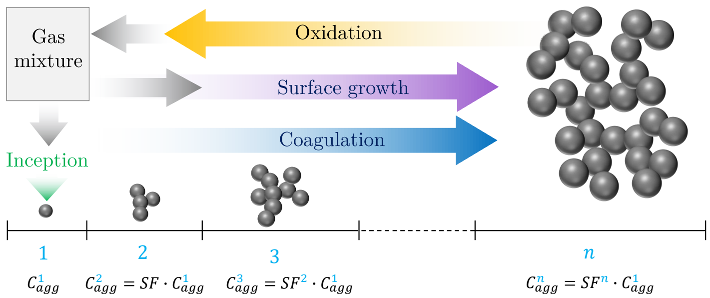

Figure 2.5: The illustration of sections in SPBM. The mass of sections grows progressively by the scale factor of SF. Inception introduces new particles to the first section that propagate to the upper section via coagulation and surface growth and return to lower sections by oxidation. Carbon and hydrogen pass from gas to solid phase through inception and surface growth and goes back via oxidation.

by the subscript *haca,ads*) because they are similar mass-gaining mechanisms. The source terms for section i are given as:

$$S^i_{N_{agg}} = \left(S^i_{N_{agg}}\right)_{inc} + \left(S^i_{N_{agg}}\right)_{haca,ads} + \left(S^i_{N_{agg}}\right)_{ox} + \left(S^i_{N_{agg}}\right)_{coag}, \tag{2.38}$$

$$S^i_{N_{pri}} = \left(S^i_{N_{pri}}\right)_{inc} + \left(S^i_{N_{pri}}\right)_{haca,ads} + \left(S^i_{N_{pri}}\right)_{ox} + \left(S^i_{N_{pri}}\right)_{coag}, \tag{2.39}$$

$$S^i_{H_{tot}} = (S^i_{H_{tot}})_{inc} + (S^i_{H_{tot}})_{haca,ads} + (S^i_{H_{tot}})_{ox} + (S^i_{H_{tot}})_{coag}. \tag{2.40}$$

2.3.2.1 Coagulation Source Term

Coagulation redistributes the total number of agglomerates and primary particles as well as hydrogen atoms among the sections. The partial coagulation source terms for N^i_{agg}, N^i_{pri} and H^i_{tot} can be calculated as:

$$\left(S^i_{N_{agg}}\right)_{coag} = \sum_{k=1}^{n_{sec}} \sum_{j=k}^{n_{sec}} \left(1 - \frac{\delta_{jk}}{2}\right) \eta_{ijk} \zeta^{jk} \beta^{jk} N^j_{agg} N^k_{agg} - N^i_{agg} \sum_{m=1}^{n_{sec}} \zeta^{im} \beta^{im} N^m_{agg}, \tag{2.41}$$

$$\left(S^i_{N_{pri}}\right)_{coag} = \sum_{k=1}^{n_{sec}} \sum_{j=k}^{n_{sec}} \left(1 - \frac{\delta_{jk}}{2}\right) \eta_{p,ijk} \eta_{ijk} \zeta^{jk} \beta^{jk} N^j_{agg} N^k_{agg} - N^i_{pri} \sum_{m=1}^{n_{sec}} \zeta^{im} \beta^{im} N^m_{agg}, \tag{2.42}$$

$$(S^i_{H_{tot}})_{coag} = \sum_{k=1}^{n_{sec}} \sum_{j=k}^{n_{sec}} \left(1 - \frac{\delta_{jk}}{2}\right) \eta_{h,ijk} \eta_{ijk} \zeta^{jk} \beta^{jk} N^j_{agg} N^k_{agg} - H^i_{tot} \sum_{m=1}^{n_{sec}} \zeta^{im} \beta^{im} N^m_{agg}. \tag{2.43}$$

where δ_{jk} is the Kronecker delta defined as:

$$\delta_{jk} = \begin{cases} 1, & \text{if } j = k \\ 0. & \text{if } j \neq k \end{cases} \tag{2.44}$$

The collision frequency between sections j and k (β^{jk}) can be obtained from the harmonic mean of the values in the continuum (β^{jk}_{cont}) and free molecular (β^{jk}_{fm}) regimes as:

$$\beta^{jk} = \frac{\beta^{jk}_{fm} \cdot \beta^{jk}_{cont}}{\beta^{jk}_{fm} + \beta^{jk}_{cont}}, \tag{2.45}$$

$$\beta_{fm}^{jk} = \sqrt{\frac{\pi k_B T}{2}\left(\frac{1}{m_{agg}^{j}} + \frac{1}{m_{agg}^{k}}\right)}\left(d_c^j + d_c^k\right)^2, \tag{2.46}$$

$$\beta_{cont}^{ij} = \frac{2k_B T}{3\mu}\left(\frac{C^j}{d_m^j} + \frac{C^k}{d_m^k}\right)\left(d_c^j + d_c^k\right)^2. \tag{2.47}$$

The collision frequency can also be determined from the Fuchs interpolation as:

$$\beta^{jk} = \beta_{cont}^{ij}\left[\frac{d_c^j + d_c^k}{d_c^j + d_c^k + 2 + \delta_r^{jk}} + \frac{8\left(D^j + D^k\right)}{\bar{c}_r^{jk}\left(d_c^j + d_c^k\right)}\right]^{-1}, \tag{2.48}$$

where δ_r^{jk} and $\bar{c}_r^{jk}$ are the mean square root of mean distance and velocity of particles, respectively, and they are obtained as:

$$\delta_r^{jk} = \sqrt{{\delta_a^j}^2 + {\delta_a^k}^2}, \tag{2.49}$$

$$\bar{c}_r^{jk} = \sqrt{{c^j}^2 + {c^k}^2}. \tag{2.50}$$

The mean velocity, c^i, and mean stop distance of particles, λ_a^i, can be calculated as:

$$c^i = \sqrt{\frac{8k_B T}{\pi m_{agg}^i}}, \tag{2.51}$$

$$\delta_a^i = \frac{1}{d_c^i \lambda_a^i}\left[\left(d_c^i + \lambda_a^i\right)^3 - \left({d_c^i}^2 + {\lambda_a^i}^2\right)^{3/2}\right] - d_c^i, \tag{2.52}$$

$\lambda_a^i$is the agglomerate stopping distance defined as:

$$\lambda_a^i = \frac{8D^i}{\pi c^i}. \tag{2.53}$$

In Equation (2.41), η_{ijk} assigns newly formed agglomerates to the two consecutive sections in order to conserve mass during coagulation [77].

$$\eta_{ijk} = \begin{cases} \dfrac{C_{agg}^{i+1} - C_{agg}^{jk}}{C_{agg}^{i+1} - C_{agg}^{i}}, & \text{if } C_{agg}^{i} \le C_{agg}^{jk} < C_{agg}^{i+1} \\ \dfrac{C_{agg}^{i} - C_{agg}^{jk}}{C_{agg}^{i} - C_{agg}^{i-1}}, & \text{if } C_{agg}^{i-1} \le C_{agg}^{jk} < C_{agg}^{i} \\ 0, & \text{else} \end{cases} \tag{2.54}$$

where $C_{agg}^{jk} = C_{agg}^{j} + C_{agg}^{k}$. Similarly, $\eta_{p,ijk}$ in Equation (2.42) and $\eta_{h,ijk}$ in Equation (2.43) adjust the number of primary particles and hydrogen atoms added to consecutive sections based on their mass, respectively.

$$\eta_{p,ijk} = \frac{C_{agg}^{i}}{C_{agg}^{jk}}\left(n_p^j + n_p^k\right), \tag{2.55}$$

$$\eta_{h,ijk} = \frac{C_{agg}^{i}}{C_{agg}^{jk}}\left(H_{agg}^{j} + H_{agg}^{k}\right). \tag{2.56}$$

In Equation (2.41), ζ^{jk} is the coagulation efficiency of soot particles in sections j and k. A value of $\zeta^{jk} = 1$ indicates that every collision between two soot particles successfully results in the formation of a new agglomerate. However, numerical models [152] and experimental evidence [153] have shown that coagulation efficiency drastically decreases for particles smaller than 10 nm in the free-molecular regime (Kn $\gg$ 10), due to their high kinetic energy exceeding the magnitude of attractive forces [154]. The coagulation efficiency between two colliding particles can be described by [152]:

$$\zeta^{ij} = 1 - \left(1 + \frac{\Phi_0^{ij}}{k_B T}\right) \exp\left(-\frac{\Phi_0^{ij}}{k_B T}\right), \tag{2.57}$$

where Φ_0 is the potential well depth, i.e., the minimum interaction energy between two colliding particles. Hou et al. [155] calculated Φ_0 for soot particles ranging from 1 to 15 nm by considering the attractive and repulsive interactions between constituent carbon and hydrogen atoms, and proposed an equation based on the reduced diameter, d_r^{jk}, of colliding particles as:

$$\Phi_0^{ij} = -6.6891 \times 10^{-23}(d_r^{jk})^3 + 1.1244 \times 10^{-21}(d_r^{jk})^2 + 1.1394 \times 10^{-20} d_r^{jk} - 5.5373 \times 10^{-21}, \tag{2.58}$$

$$d_r^{jk} = \frac{d_c^i \cdot d_c^j}{d_c^i + d_c^j}. \tag{2.59}$$

Equation (2.58) is valid for d_r^{jk} between 1 and 7 nm, and ζ^{ij} is assumed as unity for particles with a reduced diameter larger than 7 nm.

2.3.2.2 Other Source terms

Inception introduces equal number of agglomerates and primary particles to the first section.

$$\left(S_{N_{agg}}^i\right)_{inc} = \begin{cases} \dfrac{1}{Av}\dfrac{I_{N,inc}}{C_{agg}^i}, & \text{if } i = 1 \\ 0.0, & \text{else} \end{cases} \tag{2.60}$$

$$\left(S_{N_{pri}}^i\right)_{inc} = \begin{cases} \dfrac{1}{Av}\dfrac{I_{N,inc}}{C_{agg}^i}, & \text{if } i = 1 \\ 0.0, & \text{else} \end{cases} \tag{2.61}$$

$$\left(S_{H_{tot}}^i\right)_{inc} = \begin{cases} I_{H,inc}, & \text{if } i = 1 \\ 0.0. & \text{else} \end{cases} \tag{2.62}$$

Surface growth and PAH adsorption increase both the carbon mass and hydrogen content of agglomerates, transferring them to higher sections. The rate at which agglomerates are removed from the original section and added to the target section is calculated to ensure mass conservation. Specifically, it is determined by dividing the mass growth rate by the difference in mass of adjacent sections as:

$$\left(S_{N_{agg}}^i\right)_{haca,ads} = \frac{1}{Av}\begin{cases} -\dfrac{I_{C_{tot},haca}^i + I_{C_{tot},ads}^i}{C_{agg}^{i+1} - C_{agg}^i}, & \text{if } i = 1 \\ \dfrac{I_{C_{tot},haca}^{i-1} + I_{C_{tot},haca}^{i-1}}{C_{agg}^i - C_{agg}^{i-1}} - \dfrac{I_{C_{tot},haca}^i + I_{C_{tot},ads}^i}{C_{agg}^{i+1} - C_{agg}^i}, & \text{if } 1 < i < n_{sec} \\ \dfrac{I_{C_{tot},haca}^{i-1} + I_{C_{tot},ads}^{i-1}}{C_{agg}^i - C_{agg}^{i-1}}. & \text{if } i = n_{sec} \end{cases} \tag{2.63}$$

As agglomerates move up/down through sections, they carry the number of primary particles as well as hydrogen atoms, so the transfer rate of agglomerates is multiplied by n_p^i and H_{agg}^i, respectively.

$$\left(S^i_{N_{pri}}\right)_{haca,ads} = \frac{1}{Av}\begin{cases} -\dfrac{I^i_{C_{tot},haca}+I^i_{C_{tot},ads}}{C^{i+1}_{agg}-C^i_{agg}}, & \text{if } i=1 \\ \dfrac{I^{i-1}_{C_{tot},haca}+I^{i-1}_{C_{tot},ads}}{C^i_{agg}-C^{i-1}_{agg}}n_p^{i-1} - \dfrac{I^i_{C_{tot},haca}+I^i_{C_{tot},ads}}{C^{i+1}_{agg}-C^i_{agg}}n_p^i, & \text{if } 1<i<n_{sec} \\ \dfrac{I^{i-1}_{C_{tot},haca}+I^{i-1}_{C_{tot},ads}}{C^i_{agg}-C^{i-1}_{agg}}n_p^{i-1}, & \text{if } i=n_{sec} \end{cases} \tag{2.64}$$

$$\left(S^i_{H_{tot}}\right)_{haca,ads} = \frac{1}{Av}\begin{cases} -\dfrac{I^i_{C_{tot},haca}+I^i_{C_{tot},ads}}{C^{i+1}_{agg}-C^i_{agg}}H^i_{agg}+I^i_{H_{tot},haca}+I^i_{H_{tot},ads}, & \text{if } i=1 \\ \dfrac{I^{i-1}_{C_{tot},haca}+I^{i-1}_{C_{tot},ads}}{C^i_{agg}-C^{i-1}_{agg}}H^{i-1}_{agg} - \dfrac{I^i_{C_{tot},haca}+I^i_{C_{tot},ads}}{C^{i+1}_{agg}-C^i_{agg}}H^i_{agg}+I^i_{H_{tot},haca}+I^i_{H_{tot},ads}, & \text{if } 1<i<n_{sec} \\ \dfrac{I^{i-1}_{C_{tot},haca}+I^{i-1}_{C_{tot},ads}}{C^i_{agg}-C^{i-1}_{agg}}H^{i-1}_{agg}+I^i_{H_{tot},haca}+I^i_{H_{tot},ads}. & \text{if } i=n_{sec} \end{cases} \tag{2.65}$$

Similarly, the agglomerates lose carbon mass by oxidation, and descend to the lower sections carrying primary particle and hydrogen.

$$\left(S^i_{N_{agg}}\right)_{ox} = \frac{1}{Av}\begin{cases} \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}} - \dfrac{I^i_{C_{tot},ox}}{C^i_{agg}}, & \text{if } i=1 \\ \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}} - \dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}, & \text{if } 1<i<n_{sec} \\ -\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}, & \text{if } i=n_{sec} \end{cases} \tag{2.66}$$

$$\left(S^i_{N_{pri}}\right)_{ox} = \frac{1}{Av}\begin{cases} \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}n_p^{i+1} - \dfrac{I^i_{C_{tot},ox}}{C^i_{agg}}, & \text{if } i=1 \\ \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}n_p^{i+1} - \dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}n_p^i, & \text{if } 1<i<n_{sec} \\ -\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}n_p^i, & \text{if } i=n_{sec} \end{cases} \tag{2.67}$$

$$\left(S^i_{H_{tot}}\right)_{ox} = \frac{1}{Av}\begin{cases} \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}H^{i+1}_{agg} - \dfrac{I^i_{C_{tot},ox}}{C^i_{agg}}H^i_{agg}+I^i_{H_{tot},ox}, & \text{if } i=1 \\ \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}H^{i+1}_{agg} - \dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}H^i_{agg}+I^i_{H_{tot},ox}, & \text{if } 1<i<n_{sec} \\ -\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}H^i_{agg}+I^i_{H_{tot},ox}. & \text{if } i=n_{sec} \end{cases} \tag{2.68}$$

2.3.3 Monodisperse Population Balance Model

The MPBM used in this study is based on the monodisperse model presented in [9], and it tracks N_{pri}, N_{agg}, C_{tot} and H_{tot}. The coagulation model follows the same principles as the SPBM, but the calculations are greatly simplified due to the monodispersity assumption. This approach has been shown to achieve accuracy comparable to that of DEM and SPBM when applied with well-justified assumptions [74, 85]. The rate of decay of agglomerates is simply described as:

$$I_{coag} = -\frac{1}{2}\zeta\beta N_{agg}^2, \tag{2.69}$$

where ζ is the collision efficiency of agglomerates calculated using Equation (2.57), and β is the collision frequency of agglomerates for the free-molecular (Kn > 10) to continuum regimes ($Kn < 0.1$). The value of β in the transition regime ($0.1 < Kn < 10$) can be calculated from the harmonic mean of the continuum (β_{cont}) and free-molecular (β_{fm}) regime values. Additionally, an enhancement factor of %82 is applied to take into account the effect of polydispersity [73] as:

$$\beta = 1.82\frac{\beta_{fm}\cdot\beta_{cont}}{\beta_{fm}+\beta_{cont}}, \tag{2.70}$$

$$\beta_{fm} = 4\sqrt{\frac{\pi k_B T}{m_{agg}}}d_c^2, \tag{2.71}$$

$$\beta_{cont} = 8\pi d_m D, \tag{2.72}$$

where d_c, D, and m_{agg} are calculated using Equations (2.26), (2.33), and (2.30), respectively. Alternatively, β can be obtained using Fuchs interpolation [156] as:

$$\beta = \beta_{cont}\left(\frac{d_c}{d_c + 2\sqrt{2}\delta} + \frac{8D}{\sqrt{2}cd_c}\right)^{-1}, \tag{2.73}$$

where c (mean velocity of particles) and δ (mean stop distance of particles) are calculated using Equations (2.51) and (2.52). The source terms of tracked variables combines the effect of the inception, surface growth, oxidation and coagulation.

$$S_{N_{agg}} = \frac{I_{N,inc}}{n_{c,min}} + I_{coag}, \tag{2.74}$$

$$S_{N_{pri}} = \frac{I_{N,inc}}{n_{c,min}}, \tag{2.75}$$

$$S_{C_{tot}} = I_{C_{tot},inc} + I_{C_{tot},haca} + I_{C_{tot},ads} - I_{C_{tot},ox}, \tag{2.76}$$

$$S_{H_{tot}} = I_{H_{tot},inc} + I_{H_{tot},haca} + I_{H_{tot},ads} - I_{H_{tot},ox}. \tag{2.77}$$

2.4 PAH growth models

Four different PAH growth models are implemented in Omnisoot to describe the conversion of PAHs into incipient particles and their adsorption onto existing agglomerates. In other words, these models explain the calculation of $I_{\varphi,inc}$ and $I^i_{\varphi,ads}$. The implemented models are: Irreversible Dimerization [46], Reactive Dimerization [63], Dimer Coalescence [91], and E-Bridge Modified [31]. All models are based on PAH collision, as supported by ample evidence in the literature [33, 34, 35], but they differ in terms of reversibility, temperature dependence, and the number of steps involved. The collision frequency of gaseous species including PAH molecules and polymers depends on their mass and diameter, and it is obtained as:

The collision frequency of gaseous species, including PAH molecules and polymers, depends on their mass and diameter and is calculated as:

$$\beta_{dim_{jk}} = 2.2\cdot d_r^2\sqrt{\frac{8\pi k_B T}{m_r}}, \tag{2.78}$$

where 2.2 is the vdW enhancement factor [63]. d_r and m_r are reduced diameter and mass for two PAH molecules/dimers, respectively, calculated as:

$$d_r = \frac{d_{PAH_k} \cdot d_{PAH_j}}{d_{PAH_k} + d_{PAH_j}}, \tag{2.79}$$

$$m_r = \frac{m_{PAH_k} \cdot m_{PAH_j}}{m_{PAH_k} + m_{PAH_j}}, \tag{2.80}$$

where m_{PAH_j} and d_{PAH_j} are the mass and equivalent diameter of the j^{th} PAH molecule, respectively, and are obtained as:

$$m_{PAH_j} = \frac{W_{PAH_j}}{Av}, \tag{2.81}$$

$$d_{PAH_j} = \left(\frac{6 \cdot m_{PAH_j}}{\pi \cdot \rho_{PAH_j}}\right)^{1/3}, \tag{2.82}$$

where ρ_{PAH_j} is the equivalent density of PAH estimated using the relation proposed by Johansson et al. [157] as:

$$\rho_{PAH_j} = 171943.5197 \frac{W_{carbon} \cdot n_{C,PAH_j} + W_{hydrogen} \cdot n_{H,PAH_j}}{n_{C,PAH_j} + n_{H,PAH_j}}, \tag{2.83}$$

where n_{C,PAH_j} and n_{H,PAH_j} denote the number of carbon and hydrogen atoms in the j^{th} PAH molecule, respectively. The collision frequency of PAH_j and soot agglomerates in each section can be determined for the entire regime by harmonic mean of the collision frequencies in the free-molecular and continuum regimes as:

$$\beta^i_{ads_j} = \frac{\beta^i_{fm,ads} \cdot \beta^i_{cont,ads}}{\beta^i_{fm,ads} + \beta^i_{cont,ads}}, \tag{2.84}$$

$$\beta^i_{fm,ads_j} = 2.2\sqrt{\frac{\pi k_B T}{2}\left(\frac{1}{m^i_{agg}} + \frac{1}{m_{PAH_j}}\right)}\left(d^i_g + d_{PAH}\right)^2, \tag{2.85}$$

$$\beta^i_{cont,ads_j} = \frac{2k_B T}{3\mu}\left[\frac{C^i\left(d_m\right)}{d^i_g} + \frac{C^i\left(d_{PAH_j}\right)}{d_{PAH_j}}\right]\left(d_g + d_{PAH_j}\right), \tag{2.86}$$

where C^i is the Cunningham correction factor calculated using Equation (2.35).

2.4.1 Irreversible Dimerization

The Irreversible Dimerization is based on the collision of a pair of PAH molecules forming a dimer. The sequential growth continues leading formation of trimers and tetramers until the PAH cluster mass reaches a threshold that can be considered a solid particle. For practical purposes, a dimer is usually considered as an incipient particle that grows by surface growth and coagulation. A single-step collision of two similar PAHs forms a new dimer as:

$$\mathrm{PAH}_j + \mathrm{PAH}_j \xrightarrow{k_{f,dim_j}} \mathrm{Dimer}_j \tag{R 2.1}$$

Similarly, the adsorption of each PAH molecule on soot particles is described by the irreversible collision of soot and PAH_j as:

$$\mathrm{PAH}_j + \mathrm{Soot} \xrightarrow{k_{f,ads_j}} \mathrm{Soot{-}PAH}_j \tag{R 2.2}$$

The forward rate of dimerization, k_{f,dim_j}, and adsorption, k_{f,ads_j}, in Reactions (R 2.1) and (R 2.2) are calculated as:

$$k_{f,dim_j} = \gamma_{inc} \cdot \beta_{jj,PAH} \cdot Av, \tag{2.87}$$

$$k^i_{f,ads_j} = \gamma_{ads_j} \cdot \beta^i_{j,ads} \cdot Av, \tag{2.88}$$

where $\beta_{jk,\mathrm{PAH}}$ and $\beta^i_{j,\mathrm{ads}}$ are computed using Equations (2.78) and (2.84), respectively, and γ_{inc} and γ_{ads} are the collision efficiencies for dimerization and adsorption, respectively. Their values range from 10^{-7} to 1 and are typically chosen to match the predicted soot properties with experimental data. The dimerization rate of PAH_j is then calculated as:

$$\omega_{dim_j} = \eta_{inc} k_{f,dim_j} [\mathrm{PAH}_j][\mathrm{PAH}_j]. \tag{2.89}$$

The partial source terms of inception are calculated as:

$$I_{N,inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 2\omega_{dim_j} n_{PAH_j,C}, \tag{2.90}$$

$$I_{C_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 2\omega_{dim_j} n_{PAH_j,C}. \tag{2.91}$$

$$I_{H_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 2\omega_{dim_j} n_{PAH_j,H}. \tag{2.92}$$

The rate of PAH adsorption for each section is obtained as:

$$\omega^i_{ads_j} = \eta_{ads} k^i_{f,ads_j} [\mathrm{soot}^i][\mathrm{PAH}_j]. \tag{2.93}$$

The contribution of PAH adsorption to the source terms are expressed as:

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH_j}} \omega^i_{ads_j} n_{PAH_j,C}, \tag{2.94}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH_j}} \omega^i_{ads_j} (n_{PAH_j,H} - 2). \tag{2.95}$$

Note that PAH adsorption is a mass growth phenomenon that only changes C_{tot} and H_{tot} but does not affect number of N_{agg} and N_{pri}. Each PAH molecule loses one H atom becoming a radical that forms bonds with a dehydrogenated site on soot surface, so two H atoms are released during adsorption that is taken into account in Equation (2.95).

The formation of a dimer consumes two PAH molecules, and during adsorption one PAH molecule is removed from the gas mixture, so the total rate of removal of PAH_j by Irreversible Dimerization is obtained as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -2\omega_{dim_j}, \tag{2.96}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -\sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.97}$$

Moreover, one H_2 is released to the gas mixture as a result of the adsorption process.

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{ads} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.98}$$

2.4.2 Reactive Dimerization

This model was built on Irreversible Dimerization with a main difference: The first step of dimerization and adsorption is reversible forming physically bonded dimers followed by a irreversible carbonization that leads to chemical bond formation in dimers [63]. This approach allows the formation of homo- and hetero-dimers. The dimerization of PAH_j and PAH_k is described as:

$$\mathrm{PAH}_j + \mathrm{PAH}_k \underset{k_{r,dim_{jk}}}{\overset{k_{f,dim_{jk}}}{\rightleftarrows}} \mathrm{Dimer}^*_{ij} \tag{R 2.3}$$

$$\mathrm{Dimer}^*_{jk} \xrightarrow{k_{reac}} \mathrm{Dimer}_{jk} \tag{R 2.4}$$

where $\mathrm{Dimer}^*{}_{jk}$ and Dimer_{jk} are physically and chemically bonded dimers, respectively, from PAH_j and PAH_k. The forward rate of physical dimerization, $k_{f,dim_{jk}}$, is calculated as:

$$k_{f,dim_{jk}} = p^{''} \cdot \beta_{jk,PAH} \cdot Av, \tag{2.99}$$

where $\beta_{jk,\mathrm{PAH}}$ is calculated using Equation (2.78), and $p'' = 0.1$ accounts for the probability of PAH–PAH collisions occurring in the "FACE" configuration that result in successful van der Waals (vdW) bond formation [158]. The reverse rate of physical dimerization, $k_{r,\mathrm{dim}_{jk}}$, is obtained from the dimerization equilibrium constant [59] as:

$$k_{r,dim_{jk}} = \frac{k_{f,dim_{jk}}}{K_{eq}}, \tag{2.100}$$

$$\log_{10} K_{eq} = a\frac{\epsilon_{jk}}{RT} + b, \tag{2.101}$$

$$\epsilon_{jk} = cW_{jk} - d, \tag{2.102}$$

$$W_{jk} = \frac{W_j \cdot W_k}{W_j + W_k}, \tag{2.103}$$

where $a = 0.115$ (obtained from pyrere dimerization data [54]) and $b = 1.8$ [63], $c = 933420$ J/kg, and $d = 34053$ J/mol [63].

The rate constant of chemical bond formation, k_{reac}, is defined in the Arrhenius form [55] as

$$k_{reac} = 5 \times 10^6 \cdot e^{(-96232/RT)}. \tag{2.104}$$

Assuming a steady state condition for physical dimers, i.e., $\partial[\mathrm{Dimer}^*_{\mathrm{jk}}]/\partial \mathrm{t} = 0$, the rate of formation of chemically-bonded dimers can be obtained as:

$$\omega_{dim_{jk}} = k_{reac}\frac{k_{f,dim_{jk}}[\mathrm{PAH}_j][\mathrm{PAH}_k]}{k_{r,dim_{jk}} + k_{reac}}. \tag{2.105}$$

The contribution of dimer formation to the partial source terms is evaluated by looping over all combinations of PAH precursors as:

$$I_{N,inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}}\sum_{k=j}^{n_{PAH}} \omega_{dim_{kj}} \left(n_{PAH_j,C} + n_{PAH_k,C}\right), \tag{2.106}$$

$$I_{C_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}}\sum_{k=j}^{n_{PAH}} \omega_{dim_{kj}} \left(n_{PAH_j,C} + n_{PAH_k,C}\right), \tag{2.107}$$

$$I_{H_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}}\sum_{k=j}^{n_{PAH}} \omega_{dim_{kj}} \left(n_{PAH_j,H} + n_{PAH_k,H}\right). \tag{2.108}$$

Similarly, PAH adsorption is described by a two-step process where the collision of PAH_j with soot agglomerates leads to a physically bonded Soot−PAH_j^* that is carbonized and forms a chemically bonded Soot−PAH_j on soot surface.

$$\mathrm{PAH}_j + \mathrm{Soot} \underset{k_{r,ads_j}}{\overset{k_{f,ads_j}}{\rightleftarrows}} \mathrm{Soot{-}PAH}_j^* \tag{R 2.5}$$

$$\mathrm{Soot{-}PAH}_j^* \xrightarrow{k_{c,ads}} \mathrm{Soot{-}PAH}_j \tag{R 2.6}$$

The forward and reverse rate of PAH-soot collision are calculated as:

$$k^i_{f,ads_j} = \beta^i_{j,ads} \cdot Av, \tag{2.109}$$

$$k^i_{r,ads_j} = k^i_{f,ads_j} \cdot 10^{-b} e^{-a\epsilon^i_{soot,j} \ln(10)/(RT)}, \tag{2.110}$$

$$\epsilon^i_{soot,j} = cW^i_{soot,j} - d, \tag{2.111}$$

where $\beta^i_{j,ads}$ is calculated using Equation (2.84). The values of a, b, c, and d are the same as those explained in the inception part. Computing $\epsilon^i_{soot,j}$ also requires the equivalent soot molecular weight, W^i_{soot}, which is estimated from carbon mass of each agglomerate as:

$$W^i_{soot} = \frac{C^i_{tot} W_{carbon}}{N^i_{agg}}. \tag{2.112}$$

The rate constant of carbonization of Soot−PAH_j^* is defined as in an Arrhenius form similar to the inception formulation (Equation (2.104)). The pre-exponential factor is adjusted by matching the numercial PSD [55] with measurements in the ethylene pyrolysis in a flow reactor [159].

$$k_{c,ads} = 2 \times 10^{10} \cdot e^{(-96232/RT)}. \tag{2.113}$$

The total adsorption rate can be calculated assuming a steady-state concentration for the physically adsorbed PAH on soot, i.e., $\partial[\mathrm{Soot{-}PAH}_j^*]/\partial t = 0$, calculated in a similar way to the inception flux (Equation (2.105)) as:

$$\omega^i_{ads_j} = k_{c,ads} \frac{k_{f,ads_j} [\mathrm{soot}^i][\mathrm{PAH}_j]}{k_{r,ads_j} + k_{c,ads_j}}, \tag{2.114}$$

The contribution of PAH adsorption rate to the partial source terms can be expressed as:

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} n_{C,PAH_j}, \tag{2.115}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} \left(n_{H,PAH_j} - 2\right). \tag{2.116}$$

The rate of removal of PAH from the gas mixture due to inception and PAH adsorption is given as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -\sum_{k=1}^{n_{PAH}} \omega_{dim_{jk}}, \tag{2.117}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{ads} = -\sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.118}$$

Additionally, during the PAH adsorption process one H_2 is released to the gas mixture changing the concentration of H_2 as:

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{ads} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.119}$$

2.4.3 Dimer Coalescence

The Dimer Coalescence model is a multi-step irreversible model proposed by Blanquart and Pitsch [91] where self-collision of PAH molecules forms dimers, which are an intermediate state between gaseous PAH molecules and solid soot particles. The dimers can either form incipient soot particles through self-coalescence or adsorb on the surface of existing soot particles and contribute to their surface growth. The following equations describing the inception and surface growth in Dimer Coalescence adopted from the work of Sun et al. [160].

$$\mathrm{PAH}_j + \mathrm{PAH}_j \xrightarrow{k_{dim_j}} \mathrm{Dimer}_j \tag{R 2.7}$$

$$\mathrm{Dimer}_j + \mathrm{Dimer}_j \xrightarrow{k_{inc_j}} \mathrm{Tetramer}_j \tag{R 2.8}$$

$$\mathrm{Dimer}_j + \mathrm{Soot} \xrightarrow{k_{ads_j}} \mathrm{Soot{-}PAH}_j \tag{R 2.9}$$

The rate constant of dimerization, k_{dim_j}, and inception, k_{inc_j}, are calculated from the collision rate of PAHs as:

$$k_{dim_j} = \gamma_{dim_j} \cdot \beta_{jj,PAH} \cdot Av, \tag{2.120}$$

$$k_{inc_j} = \beta_{jj,dimer} \cdot Av, \tag{2.121}$$

where $\beta_{jj,PAH}$, $\beta_{jj,dimer}$ are both calculated using Equation (2.78). γ_{dim_j} is the dimerization efficiency assumed to scale with the fourth power of the PAH molecular weight [92] as:

$$\gamma_{dim_j} = C_{N,j} \cdot W^4_{PAH_j}. \tag{2.122}$$

Blanquart and Pitsch [91] estimated the constant $C_{N,j}$ in Equation (2.122) by comparing the profiles of several PAH species with the experimental measurements in a single premixed benzene flame [161], and provide $C_{N,j}$ values for various PAHs, which are listed in Table 1 in [92]. The rate of dimer collision is expressed as:

$$\omega_{dim_j} = \eta_{inc} k_{inc_j} [\mathrm{Dimer}_j][\mathrm{Dimer}_j]. \tag{2.123}$$

Similarly, the rate of adsorption of dimers on soot particles is obtained as:

$$\omega^i_{ads_j} = \eta_{ads} k^i_{ads_j} [\mathrm{soot}^i][\mathrm{Dimer}_j], \tag{2.124}$$

$$k^i_{ads_j} = \beta^i_{ads_j} \cdot Av. \tag{2.125}$$

Assuming fast dimer consumption leads to the steady-state concentration of dimers, which can be determined by solving a quadratic equation [92] as:

$$a_{inc_j} [\mathrm{Dimer}_j]^2 + b_{ads_j} [\mathrm{Dimer}_j] = \omega_{dim,j}, \tag{2.126}$$

$$[\mathrm{Dimer}_j] = \begin{cases} \dfrac{-b_{ads_j} + \sqrt{\Delta_j}}{2a_{inc_j}}, & \text{if } \Delta_j \geq 0 \\ 0 & \text{if } \Delta_j < 0 \end{cases}, \tag{2.127}$$

$$\Delta_j = b^2_{ads_j} + 4a_{inc_j}\omega_{dim,j}, \tag{2.128}$$

where $a_{inc_j} = k_{inc_j}$ and b_{ads_j} is calculated by summing the adsorption rate of dimers for all sections as:

$$b_{ads_j} = \sum_{i=1}^{n_{sec}} k^i_{ads_j} [\mathrm{soot}^i] \tag{2.129}$$

After determining the concentration of each dimer, the contributions of inception and PAH adsorption to the source terms of the tracked soot variables can be calculated in a manner similar to previous inception models, by accounting for the number of carbon and hydrogen atoms involved in the process as:

$$I_{N,inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 4\omega_{inc_j} n_{PAH_j,C}, \tag{2.130}$$

$$I_{C_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 4\omega_{inc_j} n_{PAH_j,C}, \tag{2.131}$$

$$I_{H_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 4\omega_{inc_j} n_{PAH_j,H}, \tag{2.132}$$

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} 2\omega^i_{ads_j} n_{C,PAH_j}, \tag{2.133}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} 2\omega^i_{ads_j} \left(n_{H,PAH_j} - 1\right). \tag{2.134}$$

The rate of removal of PAHs and release of H_2 molecule due to PAH adsorption is calculated as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -4 \sum_{k=1}^{n_{PAH}} \omega_{inc_j}, \tag{2.135}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{ads} = -2 \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}, \tag{2.136}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{ads} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.137}$$

2.4.4 E-Bridge Modified

The E-Bridge model was originally proposed by Frenklach and Mebel [31] to describe soot inception using a HACA-like scheme that starts with the dehydrogenation of PAH monomers, often pyrene, which forms the monomer radicals and continues with the sequential addition of the radicals that form dimers, trimers and larger polymers until the PAH structure reaches a mass threshold and the clustering process becomes irreversible [31]. Here, a modified version of this model, called E-Bridge Modified, is used where dimers are considered as incipient soot, and monomer radicals are adsorbed on soot agglomerates. This PAH growth model is described using the following set of pathways:

$$\mathrm{PAH}_j + \mathrm{H} \underset{k_{r,d_j}}{\overset{k_{f,d_j}}{\rightleftarrows}} \mathrm{P\dot{A}H}_j + \mathrm{H}_2 \tag{R 2.10}$$

$$\mathrm{P\dot{A}H}_j + \mathrm{H} \xrightarrow{k_{f,h_j}} \mathrm{PAH}_j \tag{R 2.11}$$

$$\mathrm{P\dot{A}H}_j + \mathrm{P\dot{A}H}_j \xrightarrow{k_{inc_j}} \mathrm{Dimer}_j \tag{R 2.12}$$

$$\mathrm{P\dot{A}H}_j + \mathrm{Soot} \xrightarrow{k_{ads_j}} \mathrm{Soot{-}PAH}_j \tag{R 2.13}$$

The rate constants of Reactions (R 2.10) and (R 2.11) are listed in Table 2.2. k_{inc_j} and k_{ads_j} are the rate constants of dimer production and PAH adsorption, respectively, which are calculated as:

Table 2.2: Rate coefficients for the monomer de-/hydrogenation reaction of E-bridge formation in Arrhenius form $\mathrm{k = AT^n \cdot e^{-E/RT}}$ [31].

Reaction		A $\left[\mathrm{m^3/mol \cdot s}\right]$	n	E/R [K]
(R 2.10)	f	$98\times$ $\mathrm{n_{C,PAH_j}}$	1.8	7563.519
	r	1.6×10^{-2}	2.63	2145.346
(R 2.11)	f	4.8658×10^{7}	0.13	0.0

$$k_{inc_j} = \beta_{jj,PAH} \cdot Av, \tag{2.138}$$

$$k^i_{ads_j} = \beta^i_{ads_j} \cdot Av, \tag{2.139}$$

where $\beta_{jj,PAH}$ and $\beta^i_{ads_j}$ are obtained using Equations (2.78) and (2.84), respectively. The rate of dimer formation and adsorption is calculated as:

$$\omega_{dim_j} = k_{inc_j}[\cdot\mathrm{PAH}_j]^2, \tag{2.140}$$

$$\omega^i_{ads_j} = k^i_{ads_j}[\mathrm{soot}^i][\dot{\mathrm{P}}\mathrm{AH}_j]. \tag{2.141}$$

The calculation of rate of inception and PAH adsorption from PAH_j requires the concentration of $\dot{\mathrm{P}}\mathrm{AH}_j$, which can be determined by applying the steady-state assumption for $\dot{\mathrm{P}}\mathrm{AH_j}$, i.e., $d[\dot{\mathrm{P}}\mathrm{AH}_j]/dt = 0$, resulting in a quadratic equation as:

$$a_{inc_j}[\dot{\mathrm{P}}\mathrm{AH}_j]^2 + b_{ads_j}[\dot{\mathrm{P}}\mathrm{AH}_j] + c_j = 0, \tag{2.142}$$

$$a_{inc_j} = k_{inc_j} \tag{2.143}$$

$$b_{ads_j} = k_{r,d_j}[\mathrm{H_2}] + k_{f,h_j}[\mathrm{H}] + \sum_{i=1}^{n_{sec}} k^i_{ads_j}[\mathrm{Soot}]^i \tag{2.144}$$

$$c_{inc_j} = k_{f,d_j}[\mathrm{PAH_j}][\mathrm{H}] \tag{2.145}$$

Solving the quadratic equation for each PAH gives the concentration of $\dot{\mathrm{P}}\mathrm{AH}_j$ as:

$$[\dot{\mathrm{P}}\mathrm{AH}_j] = \begin{cases} \dfrac{-b_{ads_j} + \sqrt{\Delta_j}}{2a_{inc_j}}, & \text{if } \Delta_j \geq 0 \\ 0 & \text{if } \Delta_j < 0 \end{cases} \tag{2.146}$$

$$\Delta_j = b^2_{ads_j} - 4a_{inc_j}c_j \tag{2.147}$$

The partial source terms of inception, $I_{\varphi,inc}$, are calculated in the E-Bridge Modified model as:

$$I_{N,inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}} 2\omega_{inc_j} n_{PAH_j,C}, \tag{2.148}$$

$$I_{C_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}} 2\omega_{inc_j} n_{PAH_j,C}, \tag{2.149}$$

$$I_{H_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}} \omega_{inc_j}\left(n_{PAH_j,H} - 2\right), \tag{2.150}$$

The partial source terms of PAH adsorption, $I^i_{\varphi,ads}$, are calculated for each section in this PAH growth model as:

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} n_{C,PAH_j}, \tag{2.151}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} \left(n_{H,PAH_j} - 2\right). \tag{2.152}$$

The rate of removal of each PAH involved in soot inception and PAH adsorption, and release of H_2 to the gas mixture can be expressed as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -2 \sum_{k=1}^{n_{sec}} \omega_{inc_j}, \tag{2.153}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{ads} = - \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}, \tag{2.154}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{inc} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.155}$$

2.5 Surface reactions model

Heterogeneous surface reactions are described by HACA. The soot growth in HACA scheme is based on a sequential process similar to PAH growth. The hydrogenated armchair sites ($\mathrm{C_{soot}-H}$) on the edge of aromatic rings are dehydrogenated by H abstraction forming $\mathrm{C_{soot}}^\circ$ that bonds with C_2H_2 resulting in an additional aromatic ring with hydrogenated sites. These sites can also be attacked by O_2 or OH leading to removal of carbon from soot particles by oxidation. The elementary reactions that describe this sequential process are listed in Table 2.3. The rate of mass growth by HACA is obtained from the reaction of C_2H_2 with dehydrogenated sites as:

$$\omega^i_{gr} = \alpha^i k_{f4} [\mathrm{C_2H_2}][\mathrm{C}^i_{\mathrm{soot}^\circ}], \tag{2.156}$$

where k_{f4} denotes the forward rate of Reaction (R 2.17) in Table 2.3, and $[\mathrm{C}^{\mathrm{i}}_{\mathrm{soot}^\circ}]$ is obtained by multiplying the surface density of dehydrogenated sites, $\mathrm{C}^i_{\mathrm{soot}^\circ}$ by the total surface area of soot (per unit of mass of gas mixture) of each section as:

$$[\mathrm{C}^i_{\mathrm{soot}^\circ}] = \frac{\rho}{Av} A^i_{tot} \cdot \chi_{soot^\circ}, \tag{2.157}$$

where $\chi_{\mathrm{soot}^\circ}$ is the surface density of dehydrogenated sites, and is calculated by assuming the steady-state for $[\mathrm{C}_{\mathrm{soot}^\circ}]$ in the system of reactions in Table 2.3 as:

$$\chi_{soot^\circ} = \frac{k_{f1}[\mathrm{H}] + k_{f2}[\mathrm{OH}]}{k_{r1}[\mathrm{H_2}] + k_{r2}[\mathrm{H_2O}] + k_{f3}[\mathrm{H}] + k_{f4}[\mathrm{C_2H_2}] + k_{f5}[\mathrm{O_2}]} \chi_{soot-H}, \tag{2.158}$$

where χ_{soot-H} is the surface density of hydrogenated sites estimated based on the assumption that soot "surface is assumed to be composed of outwardlooking PAH edges with PAH molecular moieties assembled into turbostratic structures" [162]. Considering the layer spacing of 3.15Å and two C–H bonds per benzene ring length, χ_{soot-H} is calculated to be $0.23\,\mathrm{site/Å^2} = 2.3 \times 10^{19}\ \mathrm{site/m^2}$, which gives the maximum theoretical limit of the reaction sites.

In Equation (2.156), α^i is the surface reactivity factor between 0 and 1 that represents the decline of reaction sites from the theoretical limit due to PAH layer orientation, particle aging, growth and maturity [163, 164], and it has been observed to depend on temperature-time history [165, 66]. The value of α has been described using constant target-specific values as well as empirical equations

Table 2.3: Arrhenius rate coefficients of the surface reactions in HACA [64], $k = AT^n \cdot e^{-E/RT}$.

No.	Reaction		A $\left[\mathrm{m^3/mol \cdot s}\right]$	n	E/R [K]
(R 2.14)	$\mathrm{C_{soot-H} + H \underset{k_{r,1}}{\overset{k_{f,1}}{\rightleftarrows}} C_{soot^\circ} + H_2}$	f	4.17×10^7	0	6542.52
		r	3.9×10^6	0	5535.98
(R 2.15)	$\mathrm{C_{soot-H} + OH \underset{k_{r,2}}{\overset{k_{f,2}}{\rightleftarrows}} C_{soot^\circ} + H_2O}$	f	10^4	0.734	719.68
		r	3.68×10^2	1.139	8605.94
(R 2.16)	$\mathrm{C_{soot^\circ} + H \xrightarrow{k_{f,3}} C_{soot-H}}$	f	10^4	0.734	719.68
(R 2.17)	$\mathrm{C_{soot^\circ} + C_2H_2 \xrightarrow{k_{f,4}} C_{soot-H} + H}$	f	80	1.56	1912.43
(R 2.18)	$\mathrm{C_{soot^\circ} + O_2 \xrightarrow{k_{f,5}} 2\,CO + product}$	f	2.2×10^6	0	3774.53
(R 2.19)	$\mathrm{C_{soot}-H + OH \xrightarrow{k_{f,6}} CO + product}$	f		$\gamma_{OH} = 0.13$	

based on particle size and flame temperature. A detailed review of these can be found in Chapter 4 of [166]. Here, the empirical equation proposed by Appel et al. [64] is used to calculate α^i as:

$$\alpha^i = \tanh\left(\frac{12.56 - 0.00563T}{\log_{10}\left(\frac{\rho_{soot}\cdot Av}{W_{carbon}}\frac{\pi}{6}{d_p^i}^3\right)} - 1.38 + 0.00068T\right). \tag{2.159}$$

Alternatively, α^i can be related to the H/C ratio of soot particles by assuming that all hydrogen atoms reside on the particle surface [91] as:

$$\alpha^i = \frac{H^i_{tot}}{C^i_{tot}}. \tag{2.160}$$

The contribution of HACA to growth source terms can be computed from HACA rates considering the number of carbon atoms in C_2H_2 and number of arm-chair and zig-zag hydrogenated sites on soot particle [92] using

$$I^i_{C_{tot},haca} = 2\omega^i_{gr}/\rho, \tag{2.161}$$

$$I^i_{H_{tot},haca} = 0.25\omega^i_{gr}/\rho. \tag{2.162}$$

The rates of concentration change of C_2H_2 and H radical due to HACA are written as:

$$\left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{gr} = -\sum_{i=1}^{n_{sec}}\omega^i_{gr}, \tag{2.163}$$

$$\left(\frac{d\,[\mathrm{H}]}{dt}\right)_{gr} = 1.75\sum_{i=1}^{n_{sec}}\omega^i_{gr}. \tag{2.164}$$

The carbons on the surface of soot are oxidized via reaction with O_2 (Reaction (R 2.18)) and OH (Reaction (R 2.19)). As a result, oxidation decreases the total carbon of soot and releases CO and H_2 to the gas mixture. The O_2 and OH oxidation rates are calculated as:

$$\omega^i_{ox,O_2} = \alpha^i k_{f5}[\mathrm{O_2}][C^i_{soot^\circ}], \tag{2.165}$$

$$\omega^i_{ox,OH} = \gamma_{OH}\beta^i_{OH}Av[\mathrm{OH}][\mathrm{soot}^i], \tag{2.166}$$

where $\gamma_{OH} = 0.13$ is the reaction probability of OH radicals with soot particles [64]. β_{OH} is the collision frequency of OH and soot particles calculated based on the kinetic theory of gases as:

$$\beta_{OH}^i = \sqrt{\frac{\pi k_B T}{2}\left(\frac{1}{m_{agg}^i} + \frac{1}{m_{OH}}\right)}\left(d_c^i + d_{OH}\right)^2, \tag{2.167}$$

where $m_{OH} = 2.824 \times 10^{-26}$ kg, and $d_{OH} = 0.3$ nm [167] are the mass and equivalent diameter of a OH radical, respectively. Then, the oxidation source term is calculated considering the number of carbon atoms removed from soot through each oxidation pathway as:

$$I_{C_{tot},ox}^i = (2\omega_{ox,O_2}^i + \omega_{ox,OH}^i)/\rho. \tag{2.168}$$

The rate of change of concentration of CO, O_2 and OH by oxidation is calculates as:

$$\left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{ox} = 2\sum_{i=1}^{n_{sec}} \omega_{ox,O_2}^i, \tag{2.169}$$

$$\left(\frac{d\,[\mathrm{O_2}]}{dt}\right)_{ox} = -\sum_{i=1}^{n_{sec}} \omega_{ox,O_2}^i, \tag{2.170}$$

$$\left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{ox} = -\sum_{i=1}^{n_{sec}} \omega_{ox,OH}^i, \tag{2.171}$$

$$\left(\frac{d\,[\mathrm{H}]}{dt}\right)_{ox} = \sum_{i=1}^{n_{sec}} \omega_{ox,OH}^i. \tag{2.172}$$

2.6 Gas scrubbing rates

The rate of production/destruction of species involved in soot formation must be taken into account to preserve the mass and energy balance in reactive systems. In order to do that, the production rate of gaseous species calculated by Cantera must be corrected for the rate of release/consumption due to PAH growth and surface reaction models.

$$\left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{inc} + \left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{ads}. \tag{2.173}$$

H_2 is released to the gas mixture due to inception, PAH adsorption as well as oxidation.

$$\left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{inc} + \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{ads} + \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{ox}. \tag{2.174}$$

Surface growth consumes C_2H_2 and adds H_2 to the gas mixture.

$$\left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{gr}. \tag{2.175}$$

$$\left(\frac{d\,[\mathrm{H}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{H}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{H}]}{dt}\right)_{gr}. \tag{2.176}$$

Oxidation uses O_2 and OH to remove carbon from soot particles and generates H_2 and CO.

$$\left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{ox}. \tag{2.177}$$

$$\left(\frac{d\,[\mathrm{O2}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{O2}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{O2}]}{dt}\right)_{ox}. \tag{2.178}$$

$$\left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{ox}. \tag{2.179}$$

Chapter 3

Results

3.1 Code Validation

A set of simulations was performed to evaluate the accuracy and reliability of Omnisoot in predicting soot formation. The aerosol dynamics models were validated by comparing the results of the population balance models implemented in Omnisoot with DEM simulations reported in the literature. In addition, carbon and hydrogen mass and energy balances were rigorously assessed to ensure that residuals remain within the bounds of acceptable numerical error.

3.2 Validation of Collision Frequency

The collision frequency function determines the rate at which two particles collide, which results in the reduction of the total number of agglomerates and the increase in particle size. In the absence of strong flow shear or external forces, Brownian motion is the main driving force for particle coagulation. As explained in Sections 2.3.2.1 and 2.3.3, Omnisoot employs harmonic mean and Fuchs interpolations to calculate collision frequency of agglomerates from free-molecular ($\mathrm{Kn} \geq 10$) to continuum ($\mathrm{Kn} \leq 0.1$) regimes based on the gas mean free path and the particle size.

The test case for validation of collision frequency is based on the DEM simulation of 2000 monodisperse spherical particles with the density of 2200 $\mathrm{kg/m^3}$ in a cubic cell with the constant temperature of 298 K and pressure of 1 atm [8]. Figure 3.1 depicts the collision frequency plotted against Knudsen number ($\mathrm{Kn} = 2\lambda/d_m$) obtained by Omnisoot using harmonic mean (red solid line) and Fuchs interpolation (blue dashed line) and the DEM results of Goudeli et al. [8]. The Fuchs interpretation perfectly matches DEM data over the free-molecular regime to the continuum regime. Harmonic mean is also in good agreement with the DEM results in the free-molecular and continuum regimes, but slightly underpredicts the collision frequency in the transition regime with relative errors less than 16%.

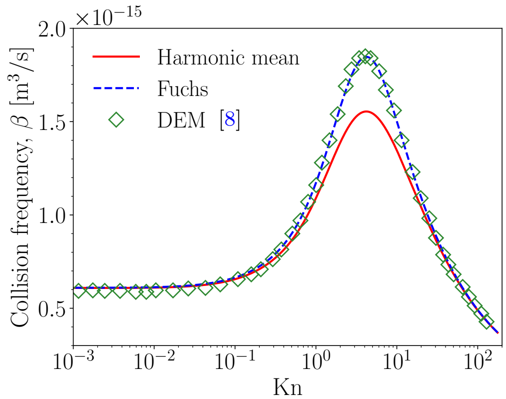

Figure 3.1: The comparison of collision frequency, β, obtained by Omnisoot using harmonic mean (red solid line) and Fuchs interpolations (blue dashed line) with DEM results (symbols) [8].

3.2.1 Coagulation

Two test cases were set up to validate the coagulation sub-model of MPBM and SPBM in the free-molecular[1] and continuum[2] regimes. An adiabatic CVR with a volume of 1 m^3 was populated with 2.6261×10^{18} spherical particles with the initial diameters of 2 nm and 668 nm for the free-molecular and continuum cases, respectively. The initial temperature in the free-molecular and continuum simulations are 1800 K and 300 K, respectively.

Figure 3.2 illustrate the schematics of the designed test in which the particles collide and grow in size through coagulation in both simulations, without inception, surface growth, or oxidation. Figure 3.3 demonstrates the evolution of N_{agg}, N_{pri}, d_m, and d_g as predicted by MPBM and SPBM in the free-molecular regime, which are in good agreement with DEM results [9]. N_{pri} is conserved during coagulation, resulting in identical flat trends for both models. In contrast, N_{agg} decreases over time, with a faster decay observed in SPBM due to its treatment of agglomerate polydispersity, which leads to a higher collision frequency compared to MPBM. Therefore, d_m and d_g predicted by SPBM, shown in Figure 3.3b, are slightly larger than those predicted by MPBM.

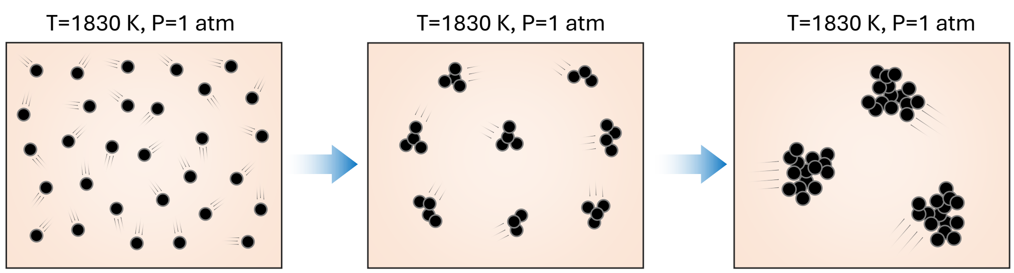

Figure 3.2: The schematic of the agglomeration process in the coagulation test case where initially spherical particles collide and form agglomerates

The MPBM model cannot resolve the PSD due to the monodispersity assumption. In contrast, the SPBM tracks the number concentration of particles in discrete sections, which allows the construction of an evolving PSD, and the assessment of the size distribution spread. Figure 3.4 shows the evolution of the non-dimensional PSD from $t = 1$ ms to 500 ms for the free-molecular and continuum simulations. The PSD is plotted as a function of the normalized concentration, $\Psi = \bar{v} n_{agg}(v,t)/N_{agg,\infty}$ and dimensionless volume, $\eta = v/\bar{v}$, where $n_{agg}(v,t)$ is the size distribution

[1] https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/validations/coagulation/free_molecular
[2] https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/validations/coagulation/continuum

function of agglomerate, v is the particle volume, $\bar{v}$ is the mean particle volume, $N_{agg,\infty}$ is the total number concentration of agglomerates. After the initial transient ($t > 22$ ms), the PSD rapidly evolves into a full bell curve and remains unchanged, indicating the attainment of a SPSD. This behavior is in good agreement with DEM results and confirms the capability of the SPBM implemented in Omnisoot to capture the SPSD for soot agglomerates in the free-molecular and continuum regimes, a characteristic outcome of Brownian-driven particle coagulation.

Figure 3.4a shows the standard deviation of the mobility diameter, σ_g, predicted by SPBM in the free-molecular regime, which is in close agreement with DEM results. Initially, $\sigma_g = 1$, indicating a monodisperse population, and it gradually increases, reaching a final value of 2.03, which corresponds to the characteristic standard deviation for the free-molecular regime [168].

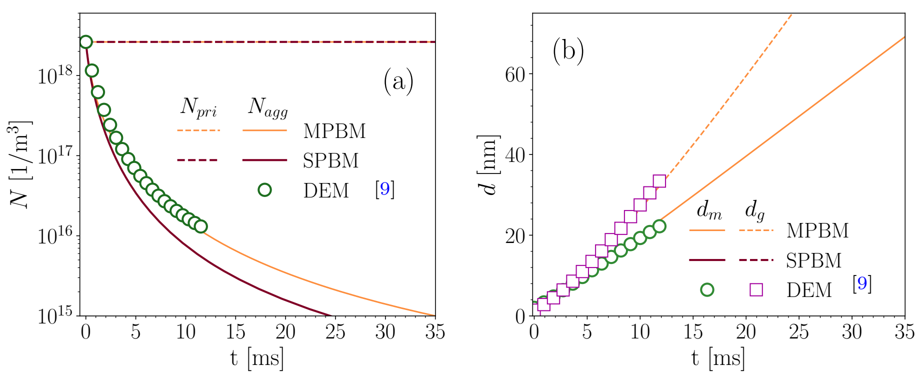

Figure 3.3: The number concentration of agglomerates and primary particles (a), and mobility and gyration diameters (b) obtained by Omnisoot using MPBM and SPBM in the free-molecular regime, which are in close agreement with the DEM results [9] indicating the validity of the coagulation submodels.

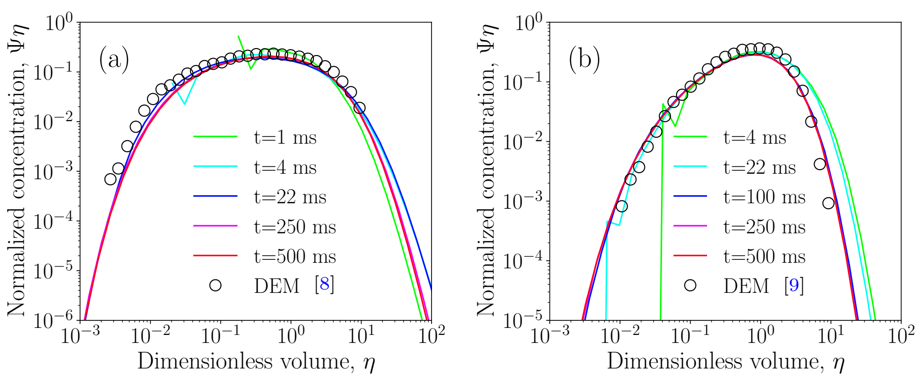

Figure 3.4: The non-dimensional particle size distribution at different residence times in the free-molecular (a) and continuum regimes (b) overlaps after the initial transient phase, indicating the attainment of a self-preserving size distribution, which is also in good agreement with DEM results [8].

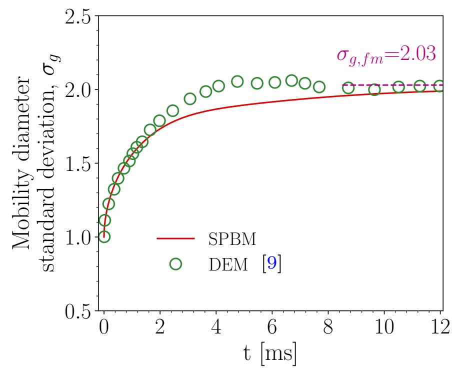

Figure 3.5: Standard deviation of the mobility diameter, σ_g, in the free-molecular regime obtained by SPBM is in close agreement with DEM results [9].

3.2.2 Elemental and Energy Balance

To ensure the accuracy and reliability of the simulations, the conservation of elemental mass and energy was assessed across all reactor models implemented in Omnisoot. The elemental balances of carbon, hydrogen, and the total energy (internal or enthalpy depending on the reactor type) of the gas-particle system were evaluated during soot formation processes under various pyrolysis and combustion conditions. A set of simulations[3] was performed using different combinations of PAH growth and particle dynamics models, and the relative errors in elemental and energy balances were monitored over time or reactor length. The relative error of total carbon, hydrogen and energy are calculated for CVR, CPR, PSR, and PFR using different combinations of particle dynamics and inception models. The residuals for all reactor configurations remained within acceptable limits (typically below 10^{-10}), confirming that Omnisoot accurately preserves mass and energy during the coupled evolution of gas-phase and soot particles.

3.2.2.1 Constant Volume Reactor

The pyrolysis of 30% CH_4 diluted in N_2 with the initial temperature and pressure of 2455 K and 3.47 atm, respectively, was simulated using CVR with a residence time of 40 ms. The combination of available PAH growth and particle dynamics models results in eight different cases, which were simulated to verify conservation of mass and energy. Here, we focus on the total elemental balance of carbon and hydrogen, as they are key elements involved in soot formation. Figure 3.6 shows the relative errors in total carbon, hydrogen, and energy for the different PAH growth and particle dynamics models. In all cases, the errors fall below 10^{-10}, confirming that CVR satisfies mass and energy conservation.

[3] https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/validations/mass_energy

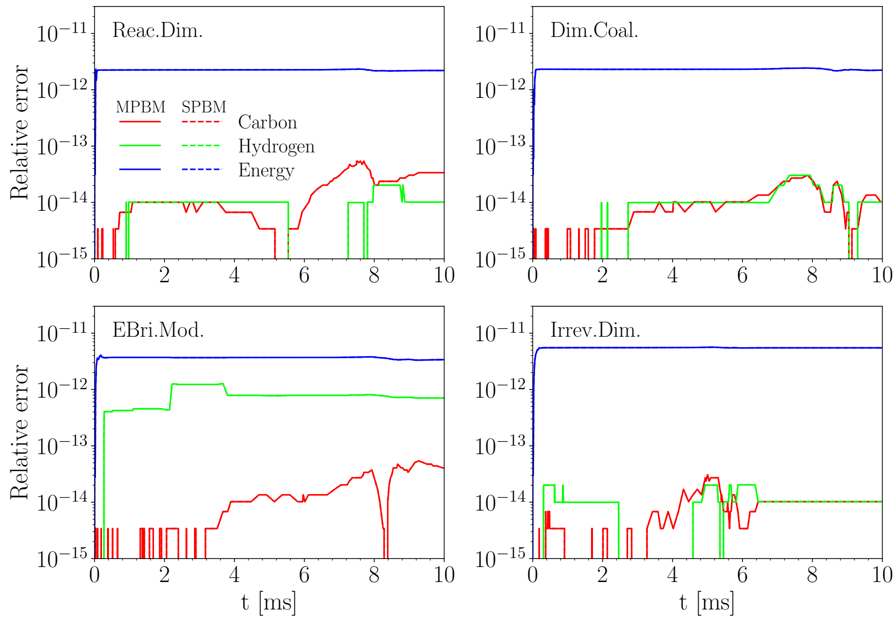

Figure 3.6: The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted against residence time during pyrolysis of 30% CH_4-N_2 at 2455 K and 3.47 atm in the constant volume reactor simulated using different PAH growth models along with MPBM (solid line) and SPBM (dashed line).

3.2.3 Imposed Pressure Reactor

The pyrolysis of 5%CH_4 in a shock-tube at post-reflected-shock temperature and pressure of T_5 =2355 K and P_5 = 4.64 atm was simulated using IPR model. The pressure was measured using an end-wall sensor by the Hanson Research Group at Stanford University. Details of the experimental setup are provided in Sec.3.3.1. The measured pressure signal was smoothed using the Savitzky–Golay filter in SciPy[169] to reduce fluctuations in the raw signal and make it suitable for the numerical solver. The residual of carbon and hydrogen mass are less than 10^{-12}, while the energy residual reaches up to 10^{-6} for energy by the end of the simulation. The relatively higher energy residual is attributed to pressure work.

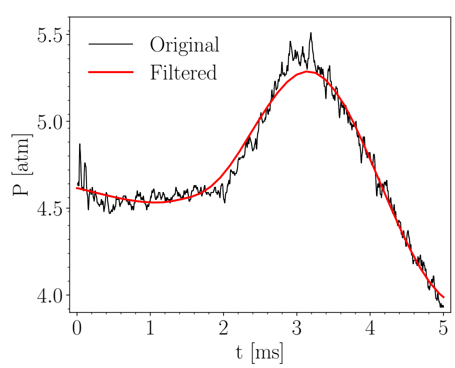

Figure 3.7: The pressure time history (black line) measured by the Stanford group for pyrolysis of 5%CH_4 at T_5 =2355 K and P_5 = 4.64 atm, and filtered (red line) with reduced fluctuations used to specify the pressure of the reactor

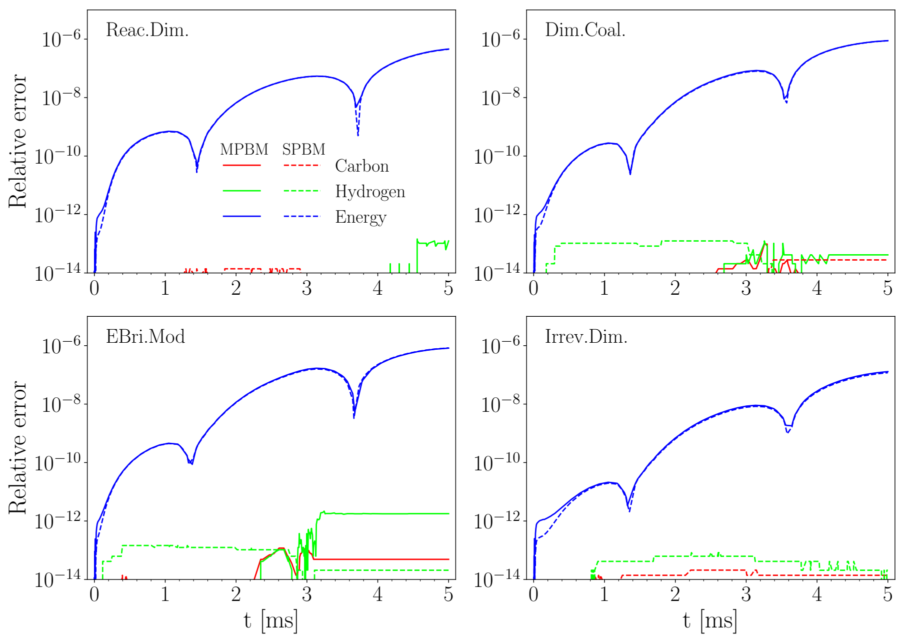

Figure 3.8: The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted against residence time during pyrolysis of 5% CH_4-Ar at post-reflected-shock temperature and pressure of T_5 =2355 K and P_5 = 4.64 atm with an imposed pressure profile simulated using different PAH growth models along with MPBM (solid line) and SPBM (dashed line)

3.2.4 Perfectly Stirred Reactor

The mass and energy balance are investigated for soot formation during ethylene-air oxidation at equivalence ratio of $\phi = 2$ in a perfectly stirred reactor. The simulation conditions were chosen based on the combustor implemented and utilized by Stouffer et al. [170]. The reactants enter the reactor with the volume of 250 ml at 300 K and atmospheric pressure. The simulation is initialized from a high temperature ($\approx$ 2000 K) to avoid trivial solution (cold reactant leaving the reactor with no chemical reactions) and to ensure the model captures a sustained combustion. The residence time of products in the reactor is 8.5 ms. Figure 3.9 shows the relative error of total elemental carbon and hydrogen mass and total enthalpy of gas and soot, which is less than 10^{-6} for all combinations of particle dynamics and PAH growth models.

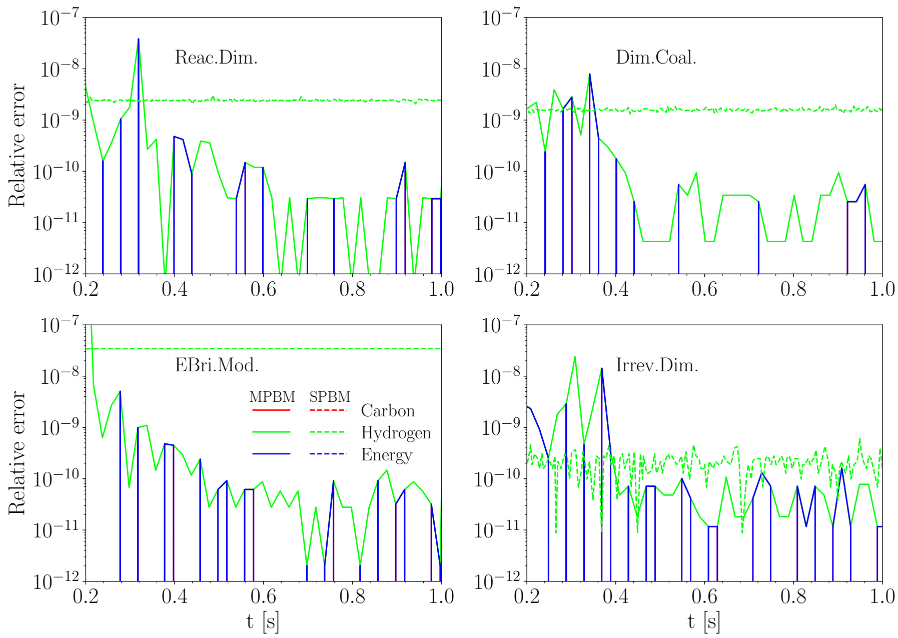

Figure 3.9: The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted in simulation time during adiabatic combustion of C_2H_4-air with $\phi = 2$ at 1 atm simulated using different combinations of PAH growth models and particle dynamics models: MPBM (solid line) and SPBM (dashed line).

3.2.5 Plug Flow Reactor

Methane pyrolysis in an adiabatic PFR was used to assess the conservation of elemental carbon, hydrogen, and energy. The inlet stream 30% CH_4 diluted in N_2 with an initial temperature of 2100 K and pressure of 1 atm. Figure 3.10 shows the residuals of total elemental carbon, hydrogen, and energy along the reactor length, up to 40 cm, for all combinations of PAH growth and particle dynamics models. The residuals remain below 10^{-11}, confirming that the PFR model in Omnisoot satisfies mass and energy conservation.

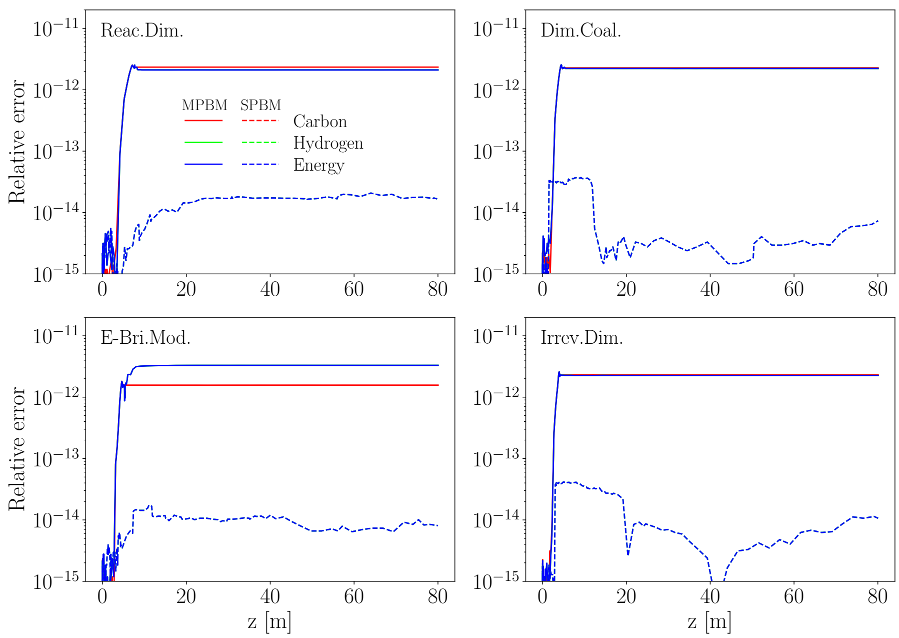

Figure 3.10: The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted against reactor length (cm) in the adiabatic flow reactor during pyrolysis of 30% CH_4-N_2 at 2100 K and 1 atm simulated using different PAH growth models and MPBM (solid line) and SPBM (dashed line)

3.3 Soot yield and morphology during methane pyrolysis in shock-tubes

In this section, we focus on simulation of soot formation in shock-tubes using the constant volume reactor of omnisoot in order to (i) assess the performance of reaction mechanisms in prediction of the species mole fraction measured by laser diagnostics, and (ii) evaluate the accuracy of inception models to predict soot yield and morphology.

Temperature ranges achieved in shock-tube experiments (T >2000 K) are relevant for plasma pyrolysis of methane (CH4) as an emerging alternative production method for co-generation of CB and hydrogen [171]. Achieving specific grades of CB for different target applications requires a good control over its properties such as its morphology (quantified by primary particle, d_p and mobility diameter, d_m), composition (elemental carbon to hydrogen ratio), specific surface area (inversely proportional to d_p, SSA = $6/(\rho_{soot}d_p)$) and internal nanostructure (composed of aligned graphitic shell and disordered core). However, this is challenging because of the complexity of soot inception and mass growth processes and its coupling with gas phase chemistry, dependence on local temperature and pressure, and short time scales of soot inception and surface growth [42]. Therefore, robust particle dynamics models coupled with detailed chemical mechanisms are essential to better understand methane-to-CB conversion process under different feedstock compositions, and temperature and pressure time-histories.

While methane combustion has been extensively studied [172] due to the need for a detailed chemical description of natural gas combustion, the formation of carbonaceous particles such as soot and CB from methane pyrolysis are not well understood yet. Significant differences exist between reaction mechanisms in prediction of CH_4 and intermediate species such as C_2H_2, and C_2H_4 in flame and reactors [172]. These uncertainties are amplified for larger hydrocarbons, especially polycyclic aromatic hydrocarbons (PAH) due to multiplicity of pathways that are not identified in available reaction mechanisms.

Shock tubes can be effective for kinetic studies and measurements of methane pyrolysis and the subsequent soot inception and surface growth due to the uniform and instant heating of the mixture behind the reflected shock to the desired temperature and pressure as well as the absence of complicating factors such as diffusion and mixing. The shock-wave propagation and reflection creates nearly isothermal and isobaric conditions ideal for kinetic studies of pyrolysis and particle formation. However, these conditions are limited to short residence times usually 1-3 ms, so they can only be used to study early stages of soot formation such as inception and surface growth, and not for processes occurring at longer residence times such as coagulation, graphitization and carbonization.

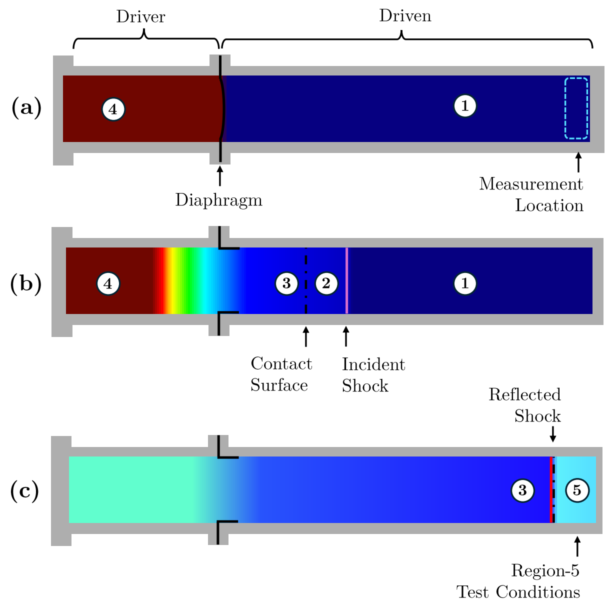

Figure 3.11: Different stages of shock wave propagation during shock-tube test from (a) before rupture of diaphragm leading to (b) propagation of incident shock towards the end of shock-tube and (c) its reflection from the end wall that increases pressure and temperature of test gas (Reprinted from Ref. [10]

Different stages of shock-wave propagation is illustrated in Fig.3.11. A shock tube consists of a driver and a driven section separated by a single-use diaphragm. The driver section contains usually an inert gas with a low molecular weight such as He, and the driven section is filled with the studied test gas. The driver section is pressurized to the desired pressure until the rupture of the diaphragm. This creates a shock wave that propagates through the driven section causing a pressure and temperature jump in the test gas.

After reaching the end wall, the shock wave reflects back towards the driver end of the shock-tube resulting in a secondary compression and heating of the mixture, and ultimately stagnating the test gas. In nearly 10 μs, this shock-heating process can bring the test gas from room-temperature to temperatures upwards of 10,000 K, and pressures more than 1000 atm.

The passage of reflected shock, referred to as region 5 with T_5 and P_5, is the reference for calculation of residence time and the starting point for simulations. The delay in appearance of soot (often quantified by a threshold for LII or extinction signals) is known as induction time, τ_{ind}. There has been a lot of research in the literature on induction time (similarly on ignition delay time) in shock tubes [173], but it is not the focus of this work. Instead, we mainly focus on species concentrations and soot characteristics during the pyrolysis of methane, at atmospheric and higher pressures which can be used for the design and optimization of CB in plasma reactors [171].

3.3.1 Experimental setup and data collection

The experiments on CH_4 pyrolysis were conducted by Hanson Research Group at Stanford University. The data has not been published yet (at the time of writing the document) and were provided through the collaboration with Monolith Materials and Stanford University.

Mole fraction time histories of methane (CH_4), acetylene (C_2H_2), and ethylene (C_2H_4) were captured using continuous wave (CW) laser absorption at 3.365 μm, 2.998 μm, and 10.532 μm, respectively [174, 175, 176]. Soot volume fraction was measured using laser light extinction at λ=633 nm and 1064 nm with absorption function, E(m) of 0.174 and 0.203 [177], respectively. Additionally, soot samples were collected onto imaging stubs mounted in the shock-tube end-wall. A schematic of the experimental setup for laser diagnostics in the shock tube is shown in Fig. 3.12. Samples were extracted from the interior surface of the shock tube endwall after each experiment to allow for imaging and analysis of the particulates. TEM images were recorded with a FEI Tecnai G2 F20 X-TWIN microscope and Gatan SC200 camera for the test case of P_5=4.5 atm and T_5=2217 K.

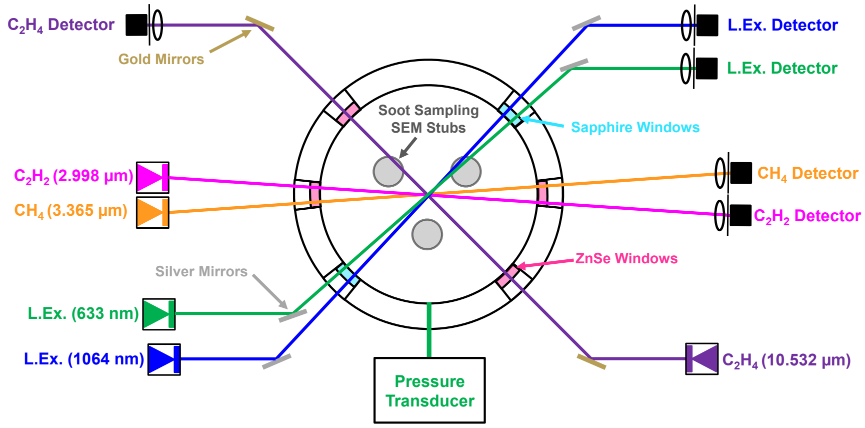

Figure 3.12: Layout of the laser diagnostics and shock-tube setup. Spatial and spectral filtering is employed to ensure high-quality absorbance measurements to infer species mole fractions and soot volume fraction. Soot samples are collected at the shock-tube endwall. All lasers are aligned in a plane 1 cm from the endwall

3.3.2 30% Methane Data-set

The first data set includes eight data points with the fuel loading of 30% CH_4-Ar at P=4±0.5 atm in the temperature range of 1800-2500 K. Table 3.1 lists the test conditions including pressure, temperature and composition of all data points.

Table 3.1: The pressure, temperature and composition of simulation data points for 30% CH_4

	Datapoints							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T [K]	1861	1917	2030	2155	2184	2313	2375	2455
P [atm]	4.12	3.74	3.56	3.93	3.62	3.58	3.75	3.47
Composition	CH_4: 0.3, Ar: 0.7							

The time history of CH_4, C_2H_4 and C_2H_2 mole fraction as well as soot yield and volume fraction were reported up to 0.5 ms. The constant volume reactor of omnisoot was used with all PAH growth and particle dynamics models. The performance of reaction mechanisms are assessed by comparing the mole fraction of species captured by laser diagnostics, CH_4, C_2H_4, and C_2H_2 with model predictions using Caltech [7], KAUST [172] and ABF [64] mechanisms. The analysis of simulation results (see Figs. B.2-B.2) showed that soot formation described by different PAH growth and particle dynamcis models have negligible impact on prediction the mole fraction of measured species in the studied temperature range. Therefore, the assessment of reaction mechanisms for prediction of major species is performed without soot to simply the analysis. Figs. 3.13-3.15 compares the mole fraction of CH_4, C_2H_4, C_2H_2 predicted by Caltech, KAUST, and ABF with data from laser diagnostics. Note that only a subset of simulations are shown for brevity, and results for the whole temperature range can be found in Figs. B.8-B.10. CH_4 mole fraction starts with a sharp drop from 0.3 at the beginning (corresponding to fast decomposition of methane) followed by a more gradual decrease. The initial decomposition rate increases with temperature. All mechanisms underpredict CH_4 mole fraction, but ABF mechanism predictions are closer to the measurements than KAUST and Caltech. In other words, KAUST and Caltech overestimate CH_4 conversion rate. The mole fraction of C_2H_4 (Fig.3.14) starts with a quick jumps after methane decomposition, reaches the maximum value and gradually decreases towards its final value. The simulations extended beyond test time of shock-tube calculations showed that C_2H_4 mole fraction predicted by KAUST and Caltech mechanisms continues to decrease to the equilibrium values, but ABF predicts an increase at longer residence times indicating reversal of C_2H_4 decomposition which is more pronounced at high temperatures. The C_2H_2 mole fraction increases in two steps, a rapid increase at the beginning followed by a gradual one. ABF mechanism predictions are in good agreement with measurements, but Caltech and KAUST underpredict the C_2H_2 mole fraction. The difference between mechanisms are more pronounced at equilibrium conditions. While ABF predicts a plateau beyond 1.5 ms, KAUST and Caltech predict a decrease towards a lower value indicating conversion of acetylene to larger hydrocarbons and PAHs. The carbon mass fraction (CMF) of measured species, CH_4, C_2H_4 and C_2H_2 combined is shown in Fig. 3.16. The CMF of different mechanisms decreases up to a certain time that shortens with temperature. Beyond that, ABF mechanism predicts an increasing trend indicating that carbon returns to major species, but KAUST and Caltech continuously transfer carbon from measured species to larger hydrocarbons. This is consistent with the CMF of soot precursors shown in Fig. 3.17, which is significantly larger for KAUST and Caltech compared with ABF by two to three orders of magnitude, so it is expected that ABF mechanism cannot provide enough precursors for soot model to predict volume fraction comparable with the measurements.

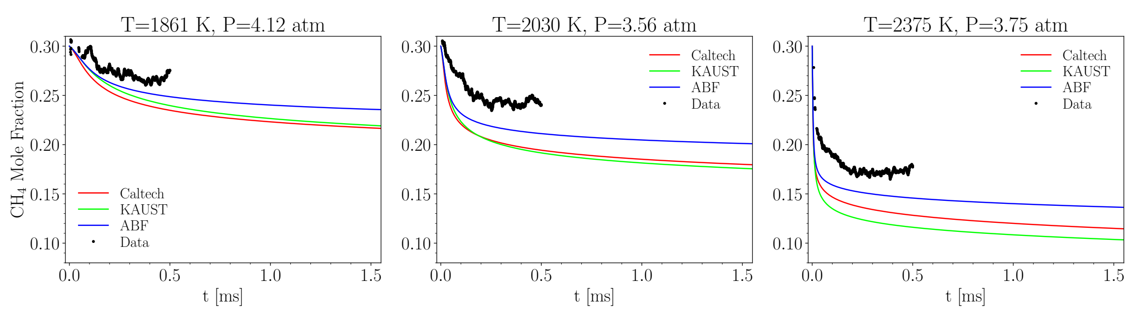

Figure 3.13: The time history of carbon mass fraction of CH_4 of 30% CH_4 pyrolysis at T=1861, 2083, and 2375 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

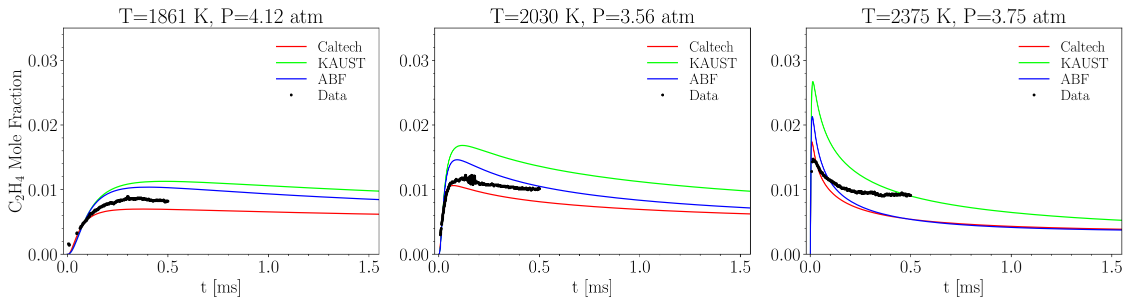

Figure 3.14: The time history of carbon mass fraction of C_2H_4 of 30% CH_4 pyrolysis at T=1861, 2083, and 2375 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

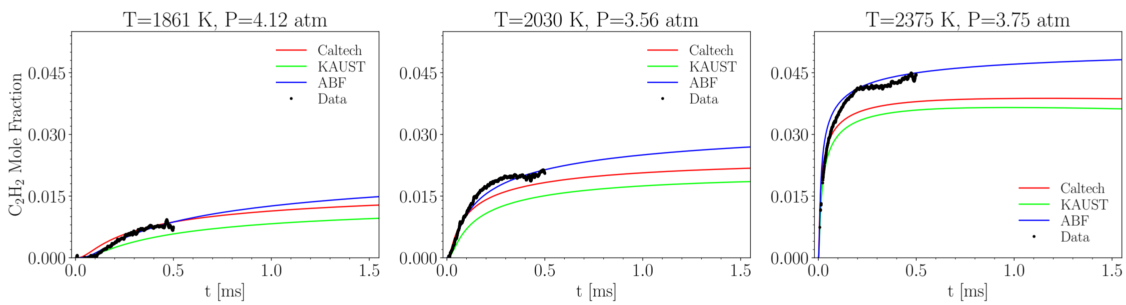

Figure 3.15: The time history of carbon mass fraction of C_2H_2 of 30% CH_4 pyrolysis at T=1861, 2083, and 2375 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

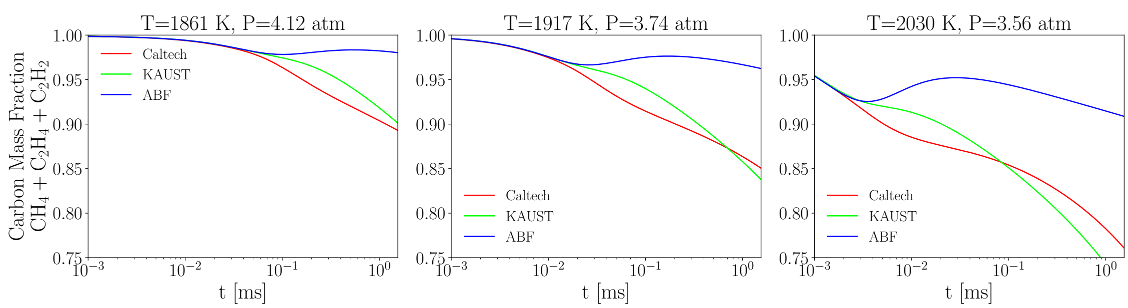

Figure 3.16: The time history of carbon mass fraction of CH_4, C_2H_4, C_2H_2 of 30% CH_4 pyrolysis at T=1861, 2083, and 2375 K using Caltech, KAUST, and ABF

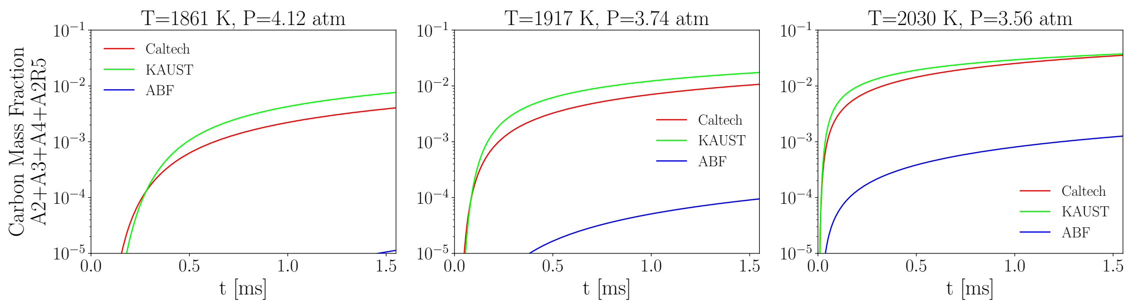

Figure 3.17: The time history of carbon mass fraction of soot precursors, A2, A3, A4 and A2R5 of 30% CH_4 pyrolysis at T=1861, 2083, and 2375 K using Caltech, KAUST, and ABF mechanisms

Comparing f_v predicted by three reaction mechanisms at T=2375 K and P=3.75 atm with the data from extinction measurements confirms the above-mentioned conclusion. As shown in Fig. 3.18, the f_v by ABF follows a similar trend as data, but the values are nearly two orders of magnitude lower indicating small PAH concentrations. Although both Caltech and KAUST predict f_v comparable with data, KAUST will be used for detailed soot simulations as its combination with E-Bridge Modified accurately captures soot volume fraction. Having said that, any conclusions drawn using KAUST can be easily generalized to Caltech as well.

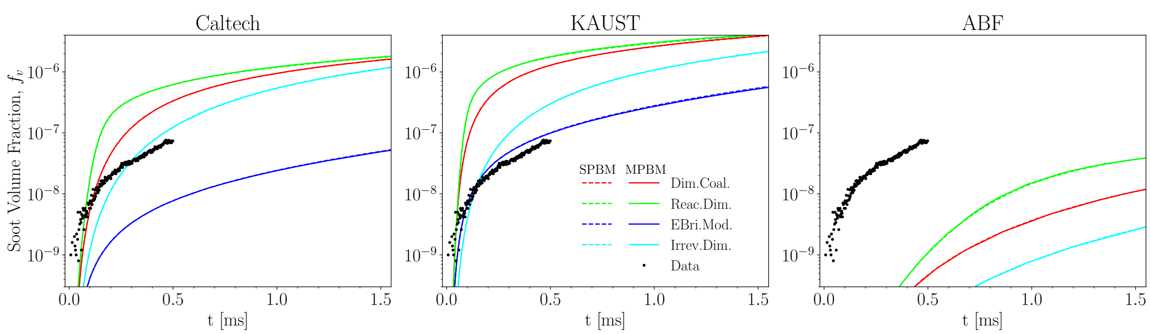

Figure 3.18: The time history of soot volume fraction of 30% CH_4 pyrolysis at T=2375 K and P=3.75 atm using Caltech, KAUST, and ABF mechanism and different PAH growth and particle dynamics models

Fig. 3.19 shows that the time variation of soot volume fraction in T=2155, 2313, and 2455 K. The measurement data was not available (or too noisy) below 2155 K due to lack of enough soot particles leading to weak extinction signals (see Fig. B.13). SPBM and MPBM results coincide indicating that taking polydispersity into account by the particle dynamics model does not affect soot mass during 1.5 ms of simulation. f_v by EBridge Modified is in good agreement with data and the lowest among inception models across the temperature range. On the other hand, Reactive Dimerization and Dimer Coalescence yields the largest f_v nearly two orders of magnitude larger than data (and EBridge Modified). There is a remarkable similarity between trend of f_v and CMF of soot precursors (Fig. 3.17) that highlights the major role of inception and PAH adsorption in soot mass growth. Temperature promotes soot formation in the studied shock tube. Fig. 3.20 depicts that f_v computed at 1.5 ms increases with temperature in a nearly linear fashion expect for EBridge Modified.

Fig.3.21 shows the time history of primary particle diameter, d_p at different temperatures. For Sectional model, d_p is the average primary particle diameter calculated from arithmetic mean of d_p of all sections. As expected, EBridge Formation and Irreversible Dimerization predict the lowest d_p due to low soot mass growth rate leading to small f_v values. Dimer Coalescence and Reactive Dimerization generate a relatively close f_v values (corresponding to similar soot mass growth), but Reactive Dimerization predicts a larger d_p indicating lower inception flux and stronger PAH adsorption rate of this model compared to Dimer Coalescence. There is a noticeable difference between sectional and monodisperse models in the d_p predicted by Reactive Dimerization, but other inception models do not exhibit such a sensitivity to particle dynamics model. Both models predict the initial rapid rise in d_p. While MPBM predicts a gradual increase to final value, d_p by SPBM decreases after the initial rise due to stronger inception rate by SPBM that generates particle with d_p=2 nm bringing down the average d_p. Fig. 3.22 depicts temperature dependence of d_p at 1.5 ms where SPBM predicts a decreasing trend as opposed to MPBM.

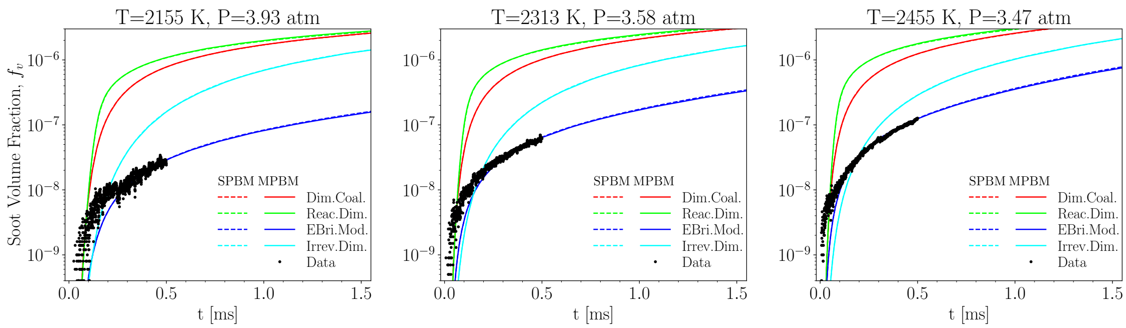

Figure 3.19: The time history soot volume fraction, f_v of 30% CH_4 pyrolysis for T=2155, 2313, and 2455 K using KAUST with different combinations of PAH growth and particle dynamics models compared with extinction measurements

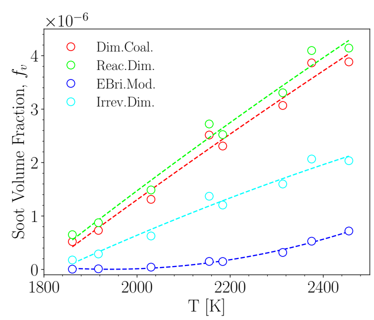

Figure 3.20: The soot volume fraction, f_v at 1.5 ms increasing with temperature during 30% CH_4 pyrolysis using KAUST and SPBM. The dashed lines are added to guide the eye

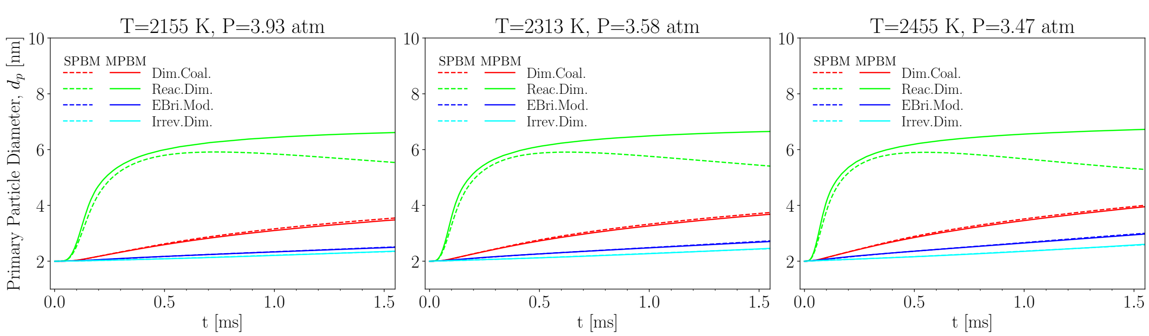

Figure 3.21: The time history of primary particle diameter, d_p of 30% CH_4 pyrolysis for T=2155, 2313, and 2455 K using KAUST with different combinations of PAH growth and particle dynamics models

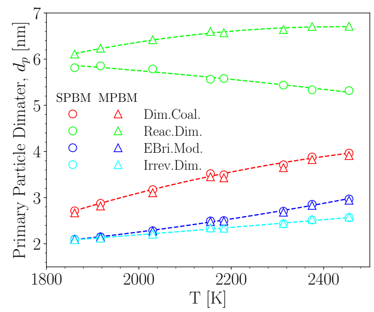

Figure 3.22: The primary particle diameter, d_p at 1.5 ms increasing with temperature during 30% CH_4 pyrolysis using KAUST with MPBM and SPBM. The dashed lines are added to guide the eye

As shown in Fig. 3.23, the inception flux of Reactive Dimerization starts with a quick rise followed by a quick drop until t≈0.25 ms which is nearly the same for both particle dynamics models, but beyond that SPBM predicts an increasing inception flux. This stems from a lower PAH adsorption rate by SPBM (with Reactive Dimerization) directing more PAHs to inception. All models predict an initial rapid rise due to PAH production by CH_4 decomposition. Dimer Coalescence reaches the highest peak and continuously decreases afterward because of consumption of PAHs. However, Irreversible Dimerization and EBridge Modified nearly stay constant towards the end of simulation time.

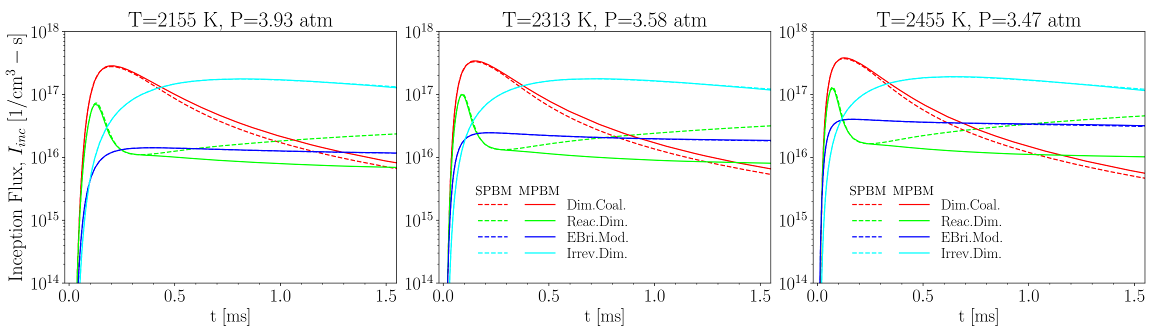

Figure 3.23: The time history of inception flux, I_{inc} of 30% CH_4 pyrolysis for T=2155, 2313, and 2455 K using KAUST with different combinations of PAH growth and particle dynamics models

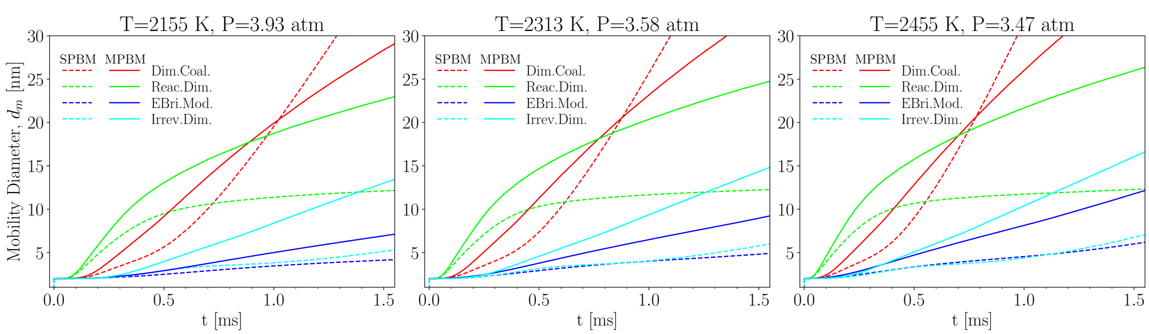

Figure 3.24: The time history of mobility diameter, d_m of 30% CH_4 pyrolysis for T=2155, 2313, and 2455 K using KAUST with different combinations of PAH growth and particle dynamics models

As shown Fig. 3.24, d_m exhibits a stronger sensitivity to the employed particle dynamics model compared to d_p because MPBM cannot capture the initial transition of particle population before the

Table 3.2: The pressure, temperature and composition of simulation data points for 10% CH_4

	Datapoints									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T [K]	1873	1927	2074	2030	2348	1776	1897	2085	2277	2406
P [atm]	1.09	1.37	1.27	1.22	1.13	4.76	4.49	4.85	4.43	3.89
Composition	CH_4: 0.1, CO_2: 0.01, Ar: 0.7									

attainment of self-preserving size distribution. As a result, morphology (quantified by characteristic diameters) predicted by MPBM diverges from the SPBM results that are closer to actual polydisperse morphology of soot. Overall, Dimer Coalescence predicts a larger mean mobility diameter because of stronger inception flux at the beginning of simulation time (see Fig. 3.23) leading to more primary particles and higher coagulation rate that generates larger agglomerates. Fig. 3.25 shows the carbon addition rate by PAH adsorption is comparable and greater than that of HACA. This difference is more pronounced for Reactive Dimerization where PAH adsorption is larger by more than one order of magnitude.

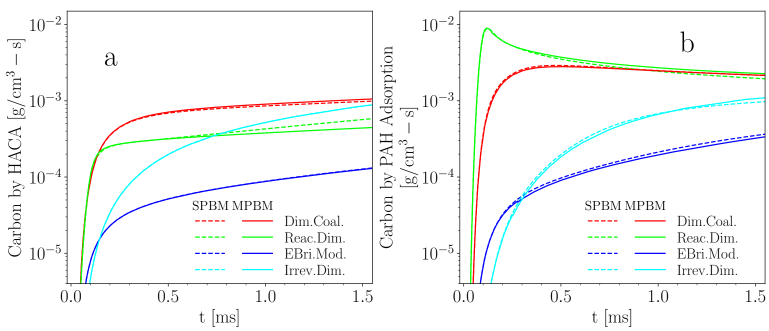

Figure 3.25: The time history of rate of carbon mass addition by (a) HACA and (b) PAH adsorption at T=2313 K using KAUST with different combinations of PAH growth and particle dynamics models

3.3.3 10% Methane Data-set

The second data set from Stanford's group includes the measurements on 10% CH_4 pyrolysis at P=1±0.5 atm and 4±0.5 atm in the temperature range of 1800-2500 K. The first step is the examination of chemistry by comparing the predicted species mole fraction using different reaction mechanisms with data from laser diagnostics. Then, we look into soot volume fraction and morphology for a single data point at which TEM measurements were conducted, analyze the collected TEM images, identify main factors contributing to the discrepancies in volume fraction and morphology, optimize the PAH growth models to minimize the prediction error, and apply optimized model to the rest of data points. We limit our discussion to the 4-atm cases as extinction measurements showed negligible soot in 1-atm cases. The results of whole pressure and temperature ranges can be found in Section B.1.2.

3.3.3.1 Reaction Mechanism Assessment

As shown in Fig. 3.26, similar to 30% dataset, KAUST yields the highest CH_4 conversion followed by Caltech and ABF mechanism. While KAUST accurately predicts CH_4 mole fraction over the studied temperature range, Caltech and ABF overpredict it indicating that less carbon is directed toward intermediates and larger hydrocarbons.

Fig. 3.27 shows that ABF predicts the initial jump and gradual decrease towards equilibrium C_2H_4 mole fraction at T=1897 K in good agreement with the data, but underpredicts it at higher temperatures. KAUST mechanism overpredicts C_2H_4 production at the beginning leading to higher peak as well as its later consumption resulting in lower mole fraction towards the end of simulation compared to the measurements. Fig. 3.28 depicts underprediction of C_2H_2 mole fraction by all mechanisms over the whole temperature range, which is more pronounced for KAUST mechanism. Fig. 3.29 shows that carbon mass fraction in measured species decreases for all mechanisms showing conversion of carbon from major species to larger hydrocarbons, but similar to 30% simulation, ABF predicts an increase in total carbon mass fraction. As shown in Fig.3.30, the carbon mass fraction of designated soot precursors predicted by ABF are lower by more than one order of magnitude compared to KAUST and Caltech.

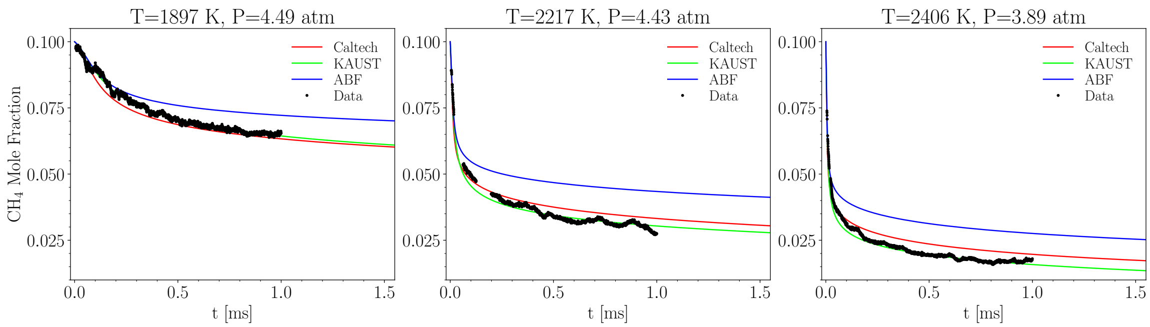

Figure 3.26: The time history of mole fraction of CH_4 of 10% CH_4 pyrolysis at T=1897, 2217, and 2406 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

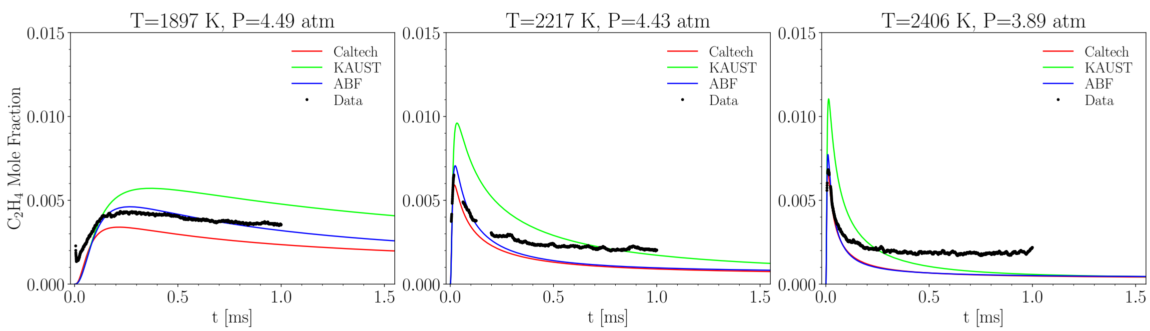

Figure 3.27: The time history of mole fraction of C_2H_4 of 10% CH_4 pyrolysis at T=1897, 2217, and 2406 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

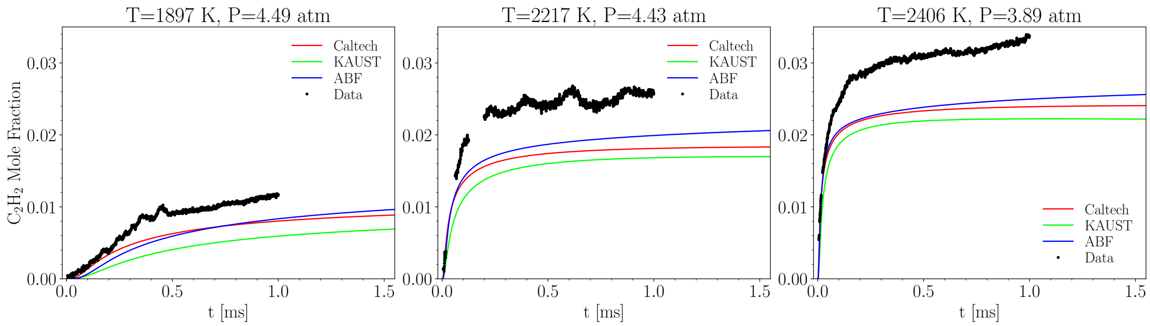

Figure 3.28: The time history of mole fraction of C_2H_2 of 10% CH_4 pyrolysis at T=1897, 2217, and 2406 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

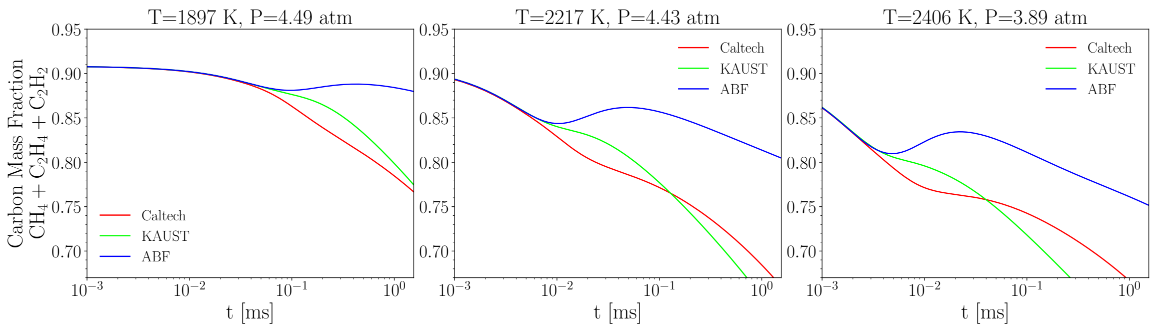

Figure 3.29: The time history of carbon mass fraction of CH_4 C_2H_4, C_2H_2 of 10% CH_4 pyrolysis at T=1897, 2217, and 2406 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

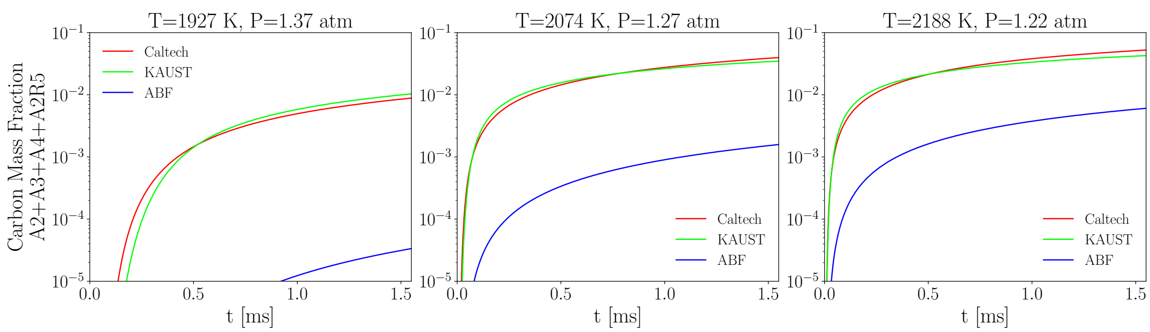

Figure 3.30: The time history of carbon mass fraction of A2, A3, A4, A2R5 of 10% CH_4 pyrolysis at T=1897, 2217, and 2406 K using Caltech, KAUST, and ABF mechanisms compared with laser diagnostics data

3.3.3.2 Characterization of soot morphology in methane pyrolysis shock-tube

The TEM images of soot made during pyrolysis of 10% CH_4 was provided by Stanford group for a single data point of T=2217 K and P=4.5 atm that can be used to characterize soot morphology from d_p and d_m. The images in addition to soot volume fraction enables assessing the performance of omnisoot in the description of soot generated during methane pyrolysis in the shock-tube. As reported by Stanford team, the expansion wave traverses the shock tube around 2 ms reducing the temperature to nearly 1000 K freezing the chemical reactions that contribute to the surface growth, but the coagulation continues until the collection of particles leading to larger agglomerates. As a result, d_p estimated from the TEM images closely represents d_p of agglomerates at the end of process, but d_m estimated from TEM images could be larger than the actual values due to post-expansion-wave growth of particles. *atems* package [11] was used for characterizing soot aggregates recorded in TEM images with different methods for evaluating the aggregate projected area, perimeter, and primary particle diameter. It creates a binary map from the raw TEM image where white and black pixels represent the agglomerates and background, respectively, and feeds the map to an agglomerate segmentation algorithm to detect individual agglomerates. Fig.3.31 shows a sample from TEM images and segmented agglomerates from the generated map.

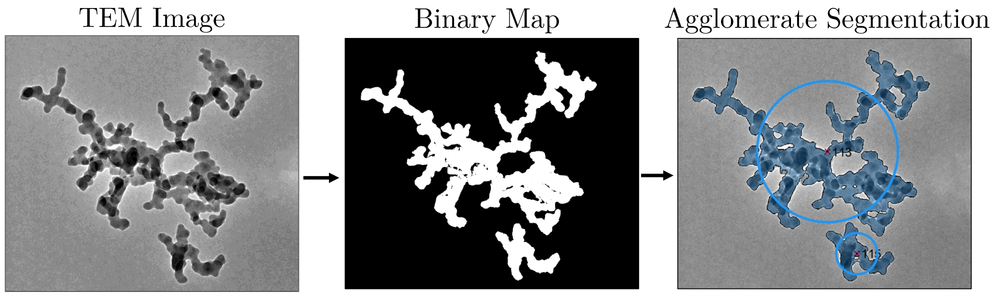

Figure 3.31: A TEM image provided by Stanford group (left pane) with the generated binary map (middle pane) and detected agglomerates using the segmentation algorithm (right pane)

We applied K-means clustering (KMC) [11] and otsu thresholding [178] to the same TEM images and compared the segmented agglomerates. A sample is shown in Fig.3.32 where KMC detected more agglomerates in the TEM image, but segments part of the background as agglomerates or divides a single agglomerate into multiple ones. On the other hand, otsu thresholding misses most of agglomerates in the sampled TEM image. Here, the K-means clustering [11] is used and 171 agglomerates were detected. d_m was calculated from the diameter of equivalent projected area , A_a of each agglomerate. The pair correlation method (PCM) [178] was applied to compute the projected primary particle area, A_p and the mean d_p assuming that primary particle are almost uniform in size within each agglomerate. The arithmetic mean of d_p of the entire samples is calculated as

$$\bar{d}_p = \frac{\sum_{Aggs} n_p d_p}{\sum_{Aggs} n_p} \tag{3.1}$$

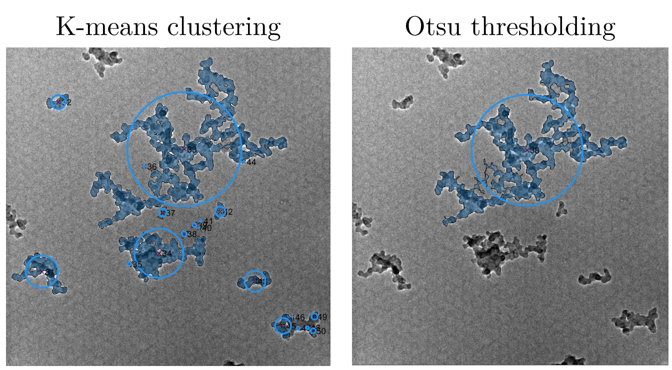

Figure 3.32: Agglomerates in single TEM image segmented by K-means clustering (left pane) and otsu thresholding (right pane)

Table 3.32 reports the arithmetic mean and median of mobility and primary particle diameter detected by atems from TEM images. The computed d_m and d_p will be compared with soot sampled

Table 3.3: The morphological characteristics of agglomerates quantified by atems using KMC and PCM

Property	Arithmetic mean	Median
d_m [nm]	97	69
d_p [nm]	22	18

from various sources. Soot morphology is mainly governed by coagulation leading to self-similar structures in which d_m scales with d_p and n_p. Olfert and Rogak [179] analyzed soot particles from flares [180], inverted burners [181], compression ignition engines [182] and other sources to support the external mixing hypothesis and related agglomerate size characterized by d_m to d_p using the following power-law:

$$d_p = d_{p,100}\left(\frac{d_m}{100nm}\right)^{D_{\mathrm{TEM}}}, \tag{3.2}$$

where $d_{p,100}$ is the average primary particle diameter for for an agglomerate with $d_m = 100$ nm, and D_{TEM} is the exponent. Both quantities were obtained by fitting Eq. (3.2) to soot sampled from different sources. Fig 3.33 demonstrates the scatter plot of d_p against d_m compared with power-law curves of Eq. (3.2) using two sets of prefactor and exponent values, i) D_{TEM}=0.32, $d_{\mathrm{p},100}$=20.6 nm, and ii) D_{TEM}=0.39, $d_{\mathrm{p},100}$=20.2 nm taken from average values reported in summary of fit parameters in Table 1 of Olfert and Rogak [179]. The power-law shows a good agreement with quantities that d_p and d_m computed by *atems* can be good representative of soot agglomerates with minimal agglomerate overlap.

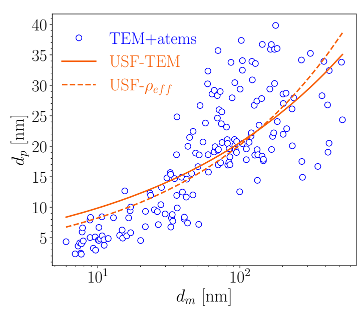

Figure 3.33: The d_m as a function of d_p from TEM image obtained by atems [11] (symbols) compared with the power law in Eq. (3.2)

The number of primary particles, N_{pri} was estimated from f_v and d_p assuming spherical primary particles with fixed size during the process. Using minimum, maximum, and mean of d_p from TEM measurements provides a range of expected N_{pri} can be obtained.

$$N_{pri} = \frac{f_v}{\pi d_p^3/6}. \tag{3.3}$$

3.3.3.3 Modelling soot yield and morphology

Fig. 3.34 compares predicted f_v and d_p with data. The horizontal scatter bars denote the uncertainty in the residence time corresponding to the TEM measurement. EBridge Modified and Irreversible Dimerization predicted soot volume fraction in close agreement with measurements, but Reactive Dimerization and Dimer Coalescence overpredict f_v by a factor of 2-3. On the other hand, d_p is significantly underestimated by all PAH growth models. Reactive Dimerization yields the largest d_p nearly 6 nm due to stronger PAH adsorption rate.

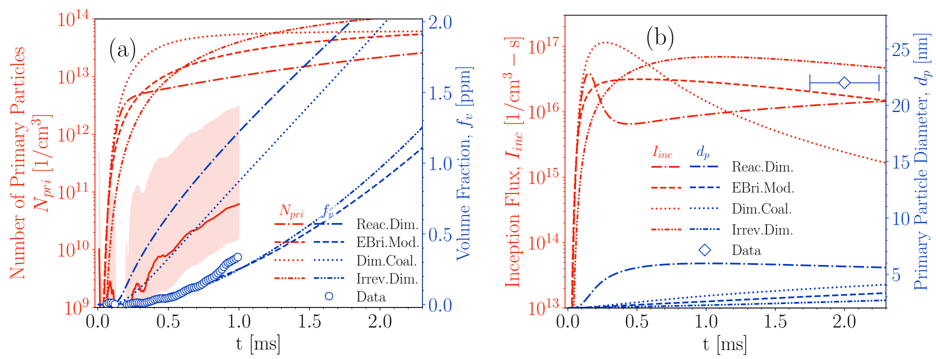

Figure 3.34: The soot volume fraction, f_v and number of primary particles, N_{pri} (left pane), and , the soot inception flux, I_{inc} and the primary particle diameter, d_p predicted by KAUST mechanism, SPBM, and different PAH growth models at T=2230 K, P=4.5 atm that corresponds to shock-tube conditions of TEM measurement

The underprediction of d_p by the PAH growth models despite producing enough or even more soot carbon mass compared with the measurement motivated a sensitivity analysis for soot yield and morphology.It should be noted that the sensitivity analysis is only focused on soot model to identify the sub-model(s) with the dominant effect on yield and morphology, and it does not include reactions in the gas phase.

First, the effect of HACA rate on soot yield and morphology is examined for the data point with T=2230 K and P=4.5 atm. To this purpose, the surface reactivity, α, in HACA formulation is modified by introducing a damping factor, ζ in Eq. (2.156) as:

$$\alpha^i = \tanh\left(\frac{12.56 - 0.00563 \cdot \zeta T}{\log_{10}\left(\frac{\rho_{soot} \cdot Av}{W_{carbon}} \frac{\pi}{6} {d_p^i}^3\right)} - 1.38 + 0.00068 \cdot \zeta T\right). \tag{3.4}$$

Note that, there are many adjustable parameters in HACA scheme such as rate constants. We picked ζ to modify α for a number of reasons: First, α was initially introduced as a tuning parameter as a function of local temperature and primary particle diameter to control surface growth rate, and it has been usually adjusted specifically for each flame [183, 134] to match the predicted volume fraction with the measurements. Second, the global empirical relation of Appel et al. [64] (used in omnisoot to quantify α) developed by fitting parameters of α to minimize the prediction error of volume fraction for various premixed flames, but shock tubes generally have larger temperature ranges compared with premixed flames, which could excessively reduce surface reactivity and HACA growth rates. Finally, larger volume fractions (in the order of few ppm) were underpredicted by the global α relation (Fig. 9 of [64]). So, the low values of α in the shock tube at T=2230 K corresponding to process conditions of TEM measurements can contribute to underprediction of surface growth rates and d_p. We introduce ζ to reduce the damping effect of temperature on surface reactivity that promote HACA growth rates resulting in larger primary particle diameters, and possibly lessen the discrepancy between predicted and measured d_p. Fig. 3.35 demonstrates the variation of α with temperature for ζ=0.6, 0.8, 1 and primary particle diameter 2 and 6 nm. α decreases with temperature, and it is inversely proportional to ζ.

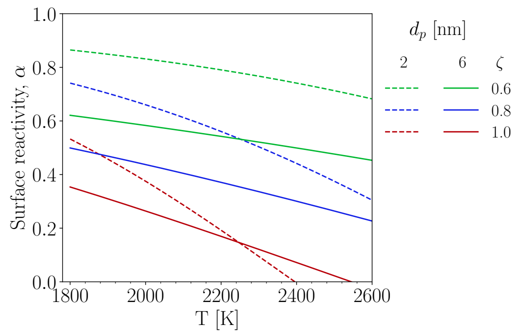

Figure 3.35: The variation of surface reactivity, α, as a function of temperature for different values of ζ and primary particle size of 2 nm (dashed lines) and 6 nm (solid lines)

A series of simulations was performed to study the damping effect of temperature on soot mass and morphology by varying ζ from 1 to 0.8 and 0.6. Note that, ζ of 1 represent the original α formulation. The reaction mechanism and particle dynamic model were set to KAUST and SPBM, respectively. Fig. 3.36 depicts the volume fraction variation for each PAH growth model using different ζ values. As expected, lowering ζ increases the soot volume fraction with a maximum of 29% at t=2 ms for Dimer Coalescence. However, as shown in Fig. 3.37 ζ has negligible effect on d_p with all PAH growth models. As a result, HACA growth rate cannot account for the observed gap between measure and predicted primary particle within 2 ms of the simulation of shock tube.

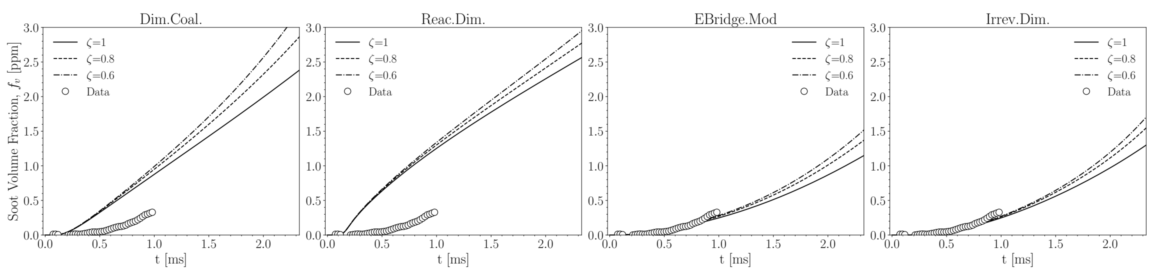

Figure 3.36: The effect of reducing ζ leading to larger HACA rates on soot volume fraction, f_v using KAUST, SPBM and different PAH growth model

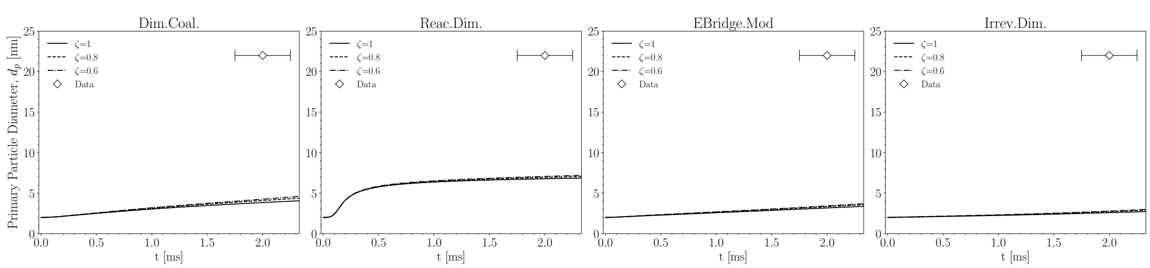

Figure 3.37: The effect of reducing ζ leading to larger HACA rates on primary particle diameter, d_p using KAUST and MPBM and different PAH growth model

There is evidence supporting the coalescence (or sintering) of soot particles at initial stages of formation. The TEM images collected thermophoretically [184] near the inception zone of a burner-stabilized laminar premixed ethylene–oxygen–argon flame ($\phi = 2.5$) showed particles with apparent sizes larger than those measured by Scanning Mobility Particle Sizer (SMPS) [185] suggesting that the particles flatten on the TEM grid upon impact. In other words, incipient and nascent soot particles may have a liquid-like state that quickly transforms to a fully solid particles due to soot aging and maturity [51]. At intial stages, these particle can coalesce (merge) upon collision forming larger primary particles. There are a few experimental studies on soot coalescence. Ono et al. [24] reheated size-selected particles with a differential mobility analyzer formed during pyrolysis of ethylene and benzene in a secondary reactor and observed the morphological changes by comparing the TEM images. Matsukawa et al. [186] proposed a relation based on solid-state sintering mechanism of metalic nano-particles to describe characteristic sintering time of soot as:

$$\tau_c^i = A(d_p^i)^4 \cdot T \cdot exp(\frac{E_a}{RT}), \tag{3.5}$$

where $A = 3.0110^{22}[\mathrm{s/m^4 - K}]$ and $E_a = 1.4010^5[\mathrm{J/mol}]$ are the prefactor, and activation energy, respectively, that are determined by curve-fitting to the change in primary particle diameter of soot after reheating in a flow reactor. Coalescence does not affect the agglomerate mass, but rather decreases the surface area of an agglomerate by merging primary particles. Coalescence is described in omnisoot by adding another partial source term to Eq. 2.39 as:

$$\left(S_{N_{pri}}\right)_{coal} = -\frac{3}{\tau_c^i}\left(n_p^i - \left(n_p^i\right)^{2/3}\right) N_{agg}^i \tag{3.6}$$

Fig. 3.38 depicts the decrease in f_v by considering coalescence because of reduction in the surface area of agglomerates hence the number of active sites for HACA growth. As shown in Fig. 3.39, coalescence increases d_p with a maximum of 64% for Dimer Coalescence, but it cannot account for the significant underprediction of primary particle diameter by the model.

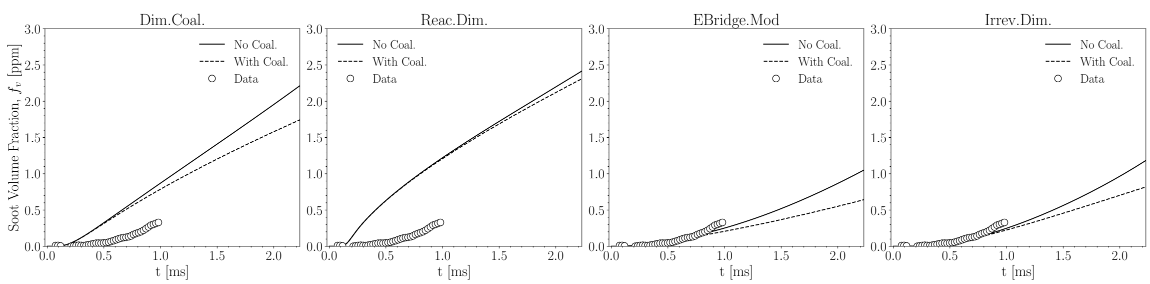

Figure 3.38: The effect of coalescence onsoot volume fraction, f_v using KAUST, SPBM and different PAH growth model

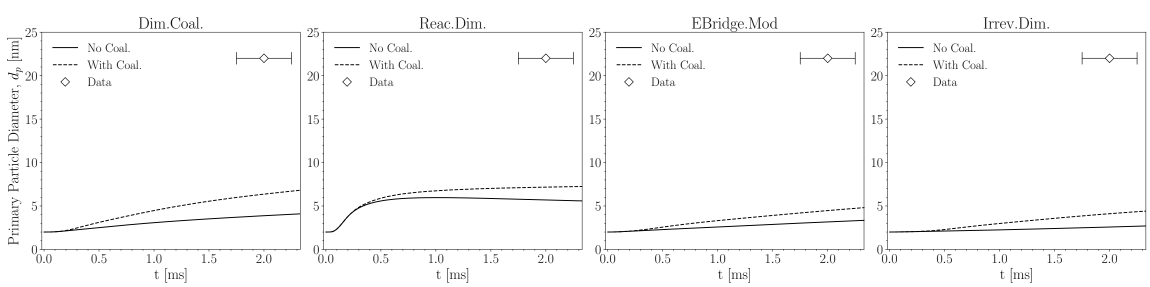

Figure 3.39: The effect of coalescence on primary particle diameter, d_p using KAUST and SPBM and different PAH growth model

The next step is examining the effect on inception and PAH adsorption on predicted soot volume fraction and primary particle diameter. This is achieved by introducing two prefactors, δ and η to adjust inception flux and PAH adsorption rate, respectively. Nine different simulation cases are designed by assigning 1, 0.01, 0.001 to δ and 1, 10, 100 to η where the case with $\delta, \eta = 1$ corresponds to the default PAH growth parameters. Fig. 3.40 and 3.41 shows modifying δ and η

values in Reactive Dimerization model can significantly change the f_v and d_p. Modified primary particle diameter values increased large enough to cover the gap between TEM measurements and simulations. By introducing the same prefactors in the rest of PAH growth model, we noticed that d_p exhibits a similar sensitivity with respect to δ and η values. Therefore, the inception flux and increasing the PAH adsorption rate are the major factors that determine primary particle size and soot yield in shock-tube simulations.

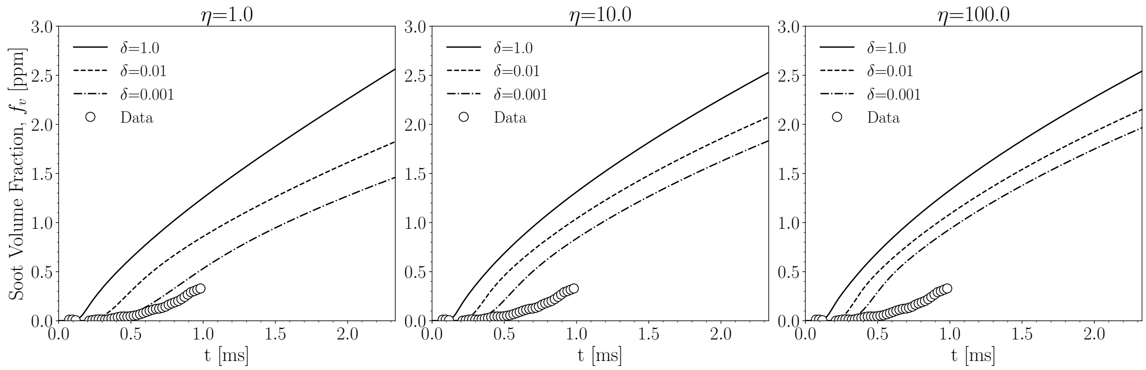

Figure 3.40: The effect of inception flux and PAH adsorption rate on soot volume fraction, f_v, using KAUST, Reactive Dimerization and SPBM

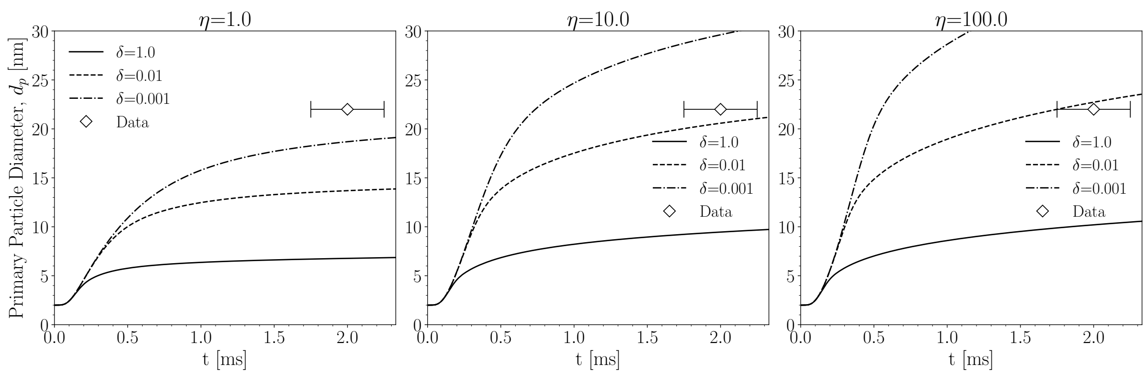

Figure 3.41: The effect of inception flux and PAH adsorption rate on primary particle diameter, d_p using KAUST, Reactive Dimerization and SPBM

Now that the importance of inception and PAH adsorption is demonstrated, a systematic analysis will be conducted for each PAH growth model by altering δ and η to determine the rate constants that minimize the prediction error of both f_v and d_p. The weighted average of mean square error of f_v and d_p was used to calculate the total error. We employ a two-step grid search for all PAH growth models. In the first step, both δ and η are divided and multiplied by a series factors ranging 1-10^4, respectively, leading to 81 simulations in total. The simulation case with the lowest error is used as the starting point for the second grid search by perturbing δ and η between -50% and +50%, leading to 45 simulations. It is important to note that the purpose of optimization is not necessarily finding the global minimum of cost function (the combined error of d_p and f_v), but rather to show that it is possible to predict both f_v and d_p in good agreement with data by adjusting the inception flux and PAH adsorption rate ensuring that the rate constants stay within their physical limit. The agreement between data and predictions using optimized PAH growth models shown in Fig. 3.42 confirms the above-mentioned conclusion. Interestingly, the predicted N_{pir} by all PAH growth models reach a plateau of 10^{11} $[1/cm^3]$ and fall in the range inferred from extinction and TEM measurements. The inception flux, I_{inc} of PAH growth models shows similar behavior spanning over the range of $\approx 3\times10^{12}\,[1/cm^3-s]$ to $\approx 3\times10^{14}\,[1/cm^3-s]$. But, they also have noticeable differences. While Dimer Coalescence starts with a quick rise in inception flux followed by a decrease toward the end of test time, Reactive Dimerization predicts a more uniform inception over the simulated time.

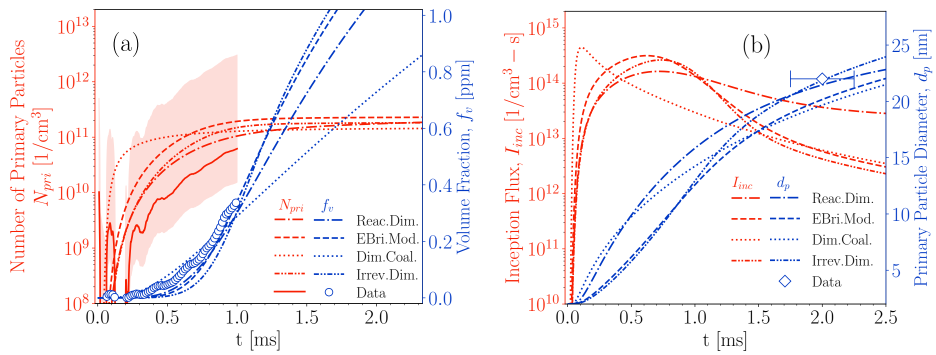

Figure 3.42: The soot volume fraction, f_v and number of primary particles, N_{pri} (left pane), the soot inception flux, I_{inc} and the primary particle diameter, d_p predicted by KAUST mechanism, SPBM, and different PAH growth models with optimized rate at T=2230 K, P=4.5 atm that corresponds to shock-tube conditions of TEM measurement

The PAH growth models with optimized rates were applied to the whole temperature and pressure ranges (see Fig. B.23 and B.24). As shown in Fig. 3.43, the measured f_v demonstrates a relatively strong sensitivity to temperature best captured by EBridge Formation. This model start with a temperature-dependent dehydrogenation step, but other model are initialed with PAH collision that weakly depends on temperature. On the other hand, Dimer Coalescence is least sensitive to temperature. While it slightly underestimates f_v at T=2085 K, it predicts soot volume fraction four time larger than data at T=2406 K. The evolution of primary particle diameter, d_p seems to not be considerably affected with temperature in 2000 K<T<2400 K. The d_p using optimized PAH growth models is calculated at t=2 ms and plotted for the studied T_5 range in Fig. 3.45 that shows a bell-like temperature dependency with peak around 2200 K for all PAH growth models.

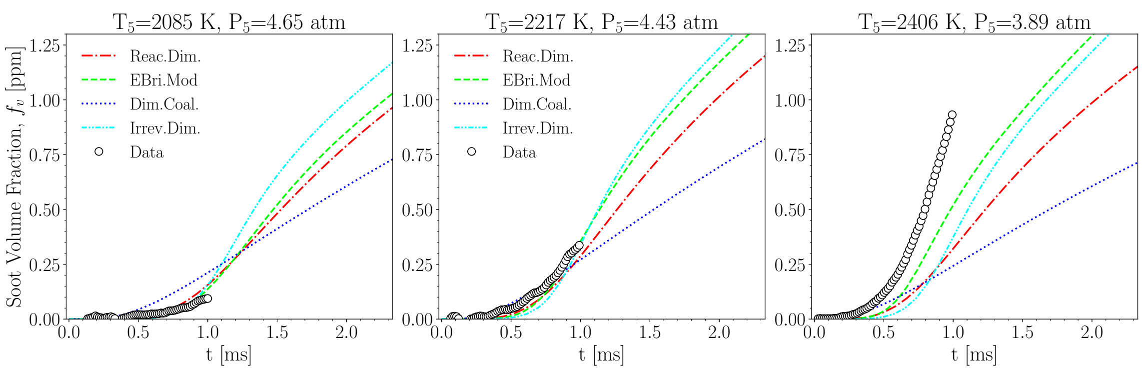

Figure 3.43: The soot volume fraction, f_v over 2000 K<T<2400 K, using KAUST, SPBM, and optimized inception models

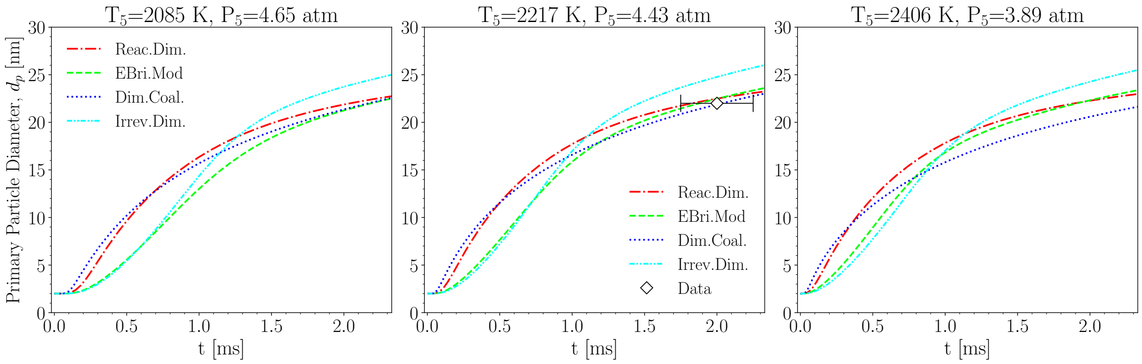

Figure 3.44: The primary particle diameter, d_p over 2000 K<T<2400 K, using KAUST, SPBM, and optimized inception models

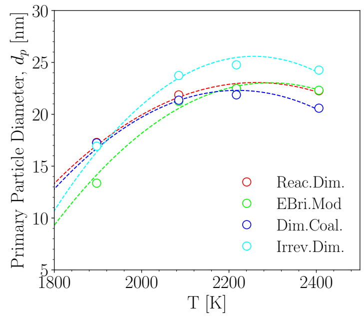

Figure 3.45: The primary particle diameter, d_p at t=2 ms over the studied temperature range using KAUST, SPBM, and optimized inception models. The dashed line is added to guide the eye.

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Appendix A

Derivation of Transport Equations

This chapter explains the derivation of transport equations from first principles for mass, species, energy and soot variables for the reactors employed in omnisoot. The conventional reactor transport equation have been derived for a reacting gas without solid particle [187], so we provide a step by step derivation of equations that describes transport/evolution of mass, species and energy of gas as well as those of soot variables.

A.1 Constant Volume Reactor

A.1.1 Continuity

The rate change of mass of gas mixture is equal to the production rate of soot.

$$\frac{d}{dt}(m) = V(1-\varphi)\sum_i \dot{s}_i W_i,$$

where V is the reactor volume that stays constant during the process, and gas occupies a fraction of volume reactor without soot, so m = ρV$(1-\varphi)$.

$$\frac{d}{dt}(\rho V(1-\varphi)) = V(1-\varphi)\sum_i \dot{s}_i W_i$$

$$\Downarrow$$

$$\frac{d}{dt}(\rho(1-\varphi)) = (1-\varphi)\sum_i \dot{s}_i W_i \tag{A.1}$$

A.1.2 Species

The rate change of mass of each species can be described as:

$$\frac{d}{dt}(m_k) = V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

The mass of each species is defined as the total gas mass multiplied by the mass fraction as

$$\frac{dY_k}{dt}\rho\, V(1-\varphi) + Y_k \frac{d}{dt}(\rho V(1-\varphi)) = V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

where the colored term can be substituted from Equation (A.1).

$$\frac{dY_k}{dt}\rho\, V(1-\varphi) + Y_k V(1-\varphi)\sum_i \dot{s}_i W_i = V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow \times \frac{1}{\rho V(1-\varphi)}$$

$$\frac{dY_k}{dt} = \frac{1}{\rho}\left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i\right] \tag{A.2}$$

A.1.3 Energy

In the constant volume reactor, the internal energy is favorable since the external boundaries of the control volume are not changed,and the work from soot volume is neglected, so formulating the energy balance based on the internal energy is preferred. The energy passes through the boundaries of the control volume via soot generation and external heat source.

$$\frac{dE}{dt} = -\frac{dE_{soot}}{dt} + \dot{Q}$$

$$\Downarrow E = m \sum Y_k e_k$$

$$\frac{dm}{dt} \sum_k Y_k e_k + m \sum_k \frac{dY_k}{dt} e_k + m \sum_k Y_k \frac{de_k}{dt} = -\frac{dE_{soot}}{dt} + \dot{Q}$$

$$\Downarrow \text{from continuity}$$

$$V(1-\varphi) \sum_i \dot{s}_i W_i \sum_k Y_k e_k + m \sum_k \frac{dY_k}{dt} e_k + m \sum_k Y_k \frac{de_k}{dt} = -\frac{dE_{soot}}{dt} + \dot{Q}$$

$$\Downarrow \text{from species}$$

$$V(1-\varphi) \sum_i \dot{s}_i W_i \sum_k Y_k e_k + m \sum_k \frac{1}{\rho}\left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i\right] e_k + m \sum_k Y_k \frac{de_k}{dt} = -\frac{dE_{soot}}{dt} + \dot{Q}$$

$$\Downarrow Y_k \frac{de_k}{dt} = Y_k c_{c,k} \frac{dT}{dt}$$

$$\Downarrow \sum_k^{max} Y_k c_{c,k} = c_p$$

$$V(1-\varphi) \sum_i \dot{s}_i W_i \sum_k Y_k e_k + m \sum_k \frac{1}{\rho}\left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i\right] e_k + m c_v \frac{dT}{dt} = -\frac{dE_{soot}}{dt} + \dot{Q}$$

$$\Downarrow \frac{dE_{soot}}{dt} = \left(\rho_{soot} c_{v,soot} \varphi \frac{dT}{dt} - (1-\varphi) \sum_i \dot{s}_i W_i \; e_{soot}\right) V$$

$$V(1-\varphi) \sum_i \dot{s}_i W_i \sum_k Y_k e_k + m \sum_k \frac{1}{\rho}\left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i\right] e_k + m c_v \frac{dT}{dt} =$$
$$- \left(\rho_{soot} c_{v,soot} \varphi \frac{dT}{dt} - (1-\varphi) \sum_i \dot{s}_i W_i \; e_{soot}\right) V + \dot{Q}$$

$$\Downarrow m = \rho V(1-\varphi)$$

$$V\left(1-\varphi\right)\sum_i \dot{s}_i W_i \sum_k Y_k e_k + V(1-\varphi)\sum_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k e_k - V(1-\varphi)\sum_k Y_k e_k \sum_i \dot{s}_i W_i$$
$$+\rho V(1-\varphi) c_v \frac{dT}{dt} = -\left(\rho_{soot} c_{v,soot} \varphi \frac{dT}{dt} - (1-\varphi)\sum_i \dot{s}_i W_i \; e_{soot}\right) V + \dot{Q}$$

$$\Downarrow \text{Eliminating } V\left(1-\varphi\right)\sum_i \dot{s}_i W_i \sum_k Y_k e_k$$

$$\Downarrow \times 1/V$$

$$(1-\varphi)\sum_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k e_k + \rho(1-\varphi) c_v \frac{dT}{dt} = -\rho_{soot} c_{v,soot} \varphi \frac{dT}{dt} +$$
$$(1-\varphi)\sum_i \dot{s}_i W_i \; e_{soot} + \dot{Q}$$

$$\Downarrow$$

$$\left(\rho\left(1-\varphi\right) c_v + \rho_{soot} c_{v,soot} \varphi\right) \frac{dT}{dt} = -\left(1-\varphi\right)\sum_k e_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k + \left(1-\varphi\right)\sum_i \dot{s}_i W_i \; e_{soot} + \frac{\dot{Q}}{V}$$

$$\Downarrow$$

$$\frac{dT}{dt} = \frac{1}{\rho\left(1-\varphi\right) c_v + \rho_{soot} c_{v,soot} \varphi} \left[-\left(1-\varphi\right)\sum_k e_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k \right.$$
$$\left. + \left(1-\varphi\right)\sum_i \dot{s}_i W_i \; e_{soot} + \frac{\dot{Q}}{V}\right] \qquad \text{(A.3)}$$

A.1.4 Soot Variables

The derivation of equations for soot variables can be accomplished similar to species evolution.

$$\frac{d}{dt}\left(m\psi\right) = \; \rho \; V(1-\varphi) S_\psi$$

$$\Downarrow$$

$$\frac{d}{dt}\left(\rho V(1-\varphi\psi)\right) = \; \rho \; V(1-\varphi) S_\psi$$

$$\Downarrow \times 1/V$$

$$\rho\left(1-\varphi\right)\frac{d\psi}{dt} + \psi \frac{d}{dt}\left(\rho(1-\varphi)\right) = \; \rho(1-\varphi) S_\psi$$

$$\Downarrow \text{from continuity}$$

$$\rho\,(1-\varphi)\,\frac{d\psi}{dt}+\psi V\,(1-\varphi)\sum_i \dot{s}_i W_i=\rho(1-\varphi)S_\psi$$

$$\Downarrow \times \frac{1}{\rho(1-\varphi)}$$

$$\frac{d\psi}{dt}= \; S_\psi-\frac{1}{\rho}\left[\psi\sum_i \dot{s}_i W_i\right] \tag{A.4}$$

A.2 Imposed Pressure Reactor

A.2.1 Energy

A.3 Plug Flow Reactor

A.3.1 Continuity

The gas flow that passes through the cross-section of a differential element along the flow reactor is described as:

$$\rho u A\,(1-\varphi)\,|_{z+dz}-\rho u A\,(1-\varphi)\,|_z=Adz(1-\varphi)\sum_i \dot{s}_i W_i$$

$$\Downarrow$$

$$\rho\; u(1-\varphi)A=Adz(1-\varphi)\sum_i \dot{s}_i W_i$$

$$\Downarrow$$

$$\frac{d}{dz}\,(\rho u(1-\varphi))\;\;=(1-\varphi)\sum_i \dot{s}_i W_i \tag{A.5}$$

A.3.2 Momentum

$$\frac{d}{dz}(\dot{m}u)=-\frac{d}{dz}(P(1-\varphi))-\tau_w\frac{P_c}{A}$$

$$\Downarrow$$

$$u\frac{d}{dz}(\dot{m})+\dot{m}\frac{d}{dz}(u)=-\frac{d}{dz}(P(1-\varphi))-\tau_w\frac{1}{R_H}$$

$$\Downarrow$$

$$u(1-\varphi)\sum_i \dot{s}_i W_i+\rho u(1-\varphi)\frac{d}{dz}(u)=-\frac{d}{dz}(P(1-\varphi))-\frac{\tau_w}{R_H} \tag{A.6}$$

A.3.3 Species

$$\frac{d}{dz}(m_k) = (1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow$$

$$\frac{d}{dz}(\rho u(1-\varphi)Y_k) = (1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow$$

$$\rho u (1-\varphi)\frac{dY_k}{dz} + Y_k \frac{d}{dz}(\rho u (1-\varphi)) = (1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$\Downarrow$ from Equation (A.5)

$$\rho u(1-\varphi)\frac{dY_k}{dz} + Y_k(1-\varphi)\sum_i \dot{\omega}_i W_i = (1-\varphi)(\dot{\omega}_k + \dot{s}_k)W_k$$

$$\Downarrow \times 1/(\rho(1-\varphi))$$

$$\frac{dY_k}{dz} = \frac{1}{\rho u(1-\varphi)}\left[(1-\varphi)(\dot{\omega}_k + \dot{s}_k)W_k - Y_k(1-\varphi)\sum_i \dot{s}_i W_i\right] \tag{A.7}$$

A.3.4 Energy

$$\frac{d}{dz}(\rho u (1-\varphi) h) + \frac{d}{dz}(\rho u \varphi h_{soot}) = \dot{q}'$$

$$\Downarrow$$

$$\frac{dh}{dz}(\rho u (1-\varphi)) + h\frac{d}{dz}(\rho u (1-\varphi)) + \frac{d}{dz}(\rho u \varphi) h_{soot} + \frac{dh_{soot}}{dz}\rho\, u\varphi = \dot{q}'$$

$$\Downarrow h = \sum_k Y_k h_k$$

$$\rho u (1-\varphi)\frac{d}{dz}\left(\sum_k Y_k h_k\right) + h\frac{d}{dz}(\rho u (1-\varphi)) + \frac{d}{dz}(\rho u \varphi) h_{soot} + \frac{dh_{soot}}{dz}\rho u\varphi = \dot{q}'$$

$$\Downarrow$$

$$\rho u (1-\varphi)\sum_k \frac{dY_k}{dz} h_k + \rho u (1-\varphi)\sum_k \frac{dh_k}{dz} Y_k + h\frac{d}{dz}(\rho u (1-\varphi)) + \frac{d}{dz}(\rho u \varphi) h_{soot} + \frac{dh_{soot}}{dz}\rho\, u\varphi = \dot{q}'$$

$$\Downarrow \text{from Equation (A.5)\&(A.7)}$$

$$\rho u\left(1-\varphi\right)\sum_k \frac{1}{\rho u}\left[(\dot{\omega}_k+\dot{s}_k)W_k - Y_k\sum_i \dot{s}_i W_i\right]h_k + \rho u\left(1-\varphi\right)\sum_k \frac{dh_k}{dz}Y_k + h\frac{d}{dz}\left(\left(1-\varphi\right)\sum_i \dot{s}_i W_i\right) + \frac{d}{dz}\left(\rho u\varphi\right)h_{soot} + \frac{dh_{soot}}{dz}\rho u\varphi = \dot{q}'$$

$$\Downarrow \frac{dh_k}{dz} = c_{p,k}\frac{dT}{dz}$$

$$\Downarrow \sum_k Y_k c_{p,k} = c_p$$

$$\Downarrow \frac{dh_{soot}}{dz} = c_{p,soot}\frac{dT}{dz}$$

$$\left(1-\varphi\right)\sum_k\left[(\dot{\omega}_k+\dot{s}_k)W_k - Y_k\sum_i \dot{s}_i W_i\right]h_k + \rho u\left(1-\varphi\right)c_p\frac{dT}{dz} + h(1-\varphi)\sum_i \dot{s}_i W_i + \frac{d}{dz}\left(\rho u\varphi\right)h_{soot} + c_{p,soot}\frac{dT}{dz}\rho u\varphi = \dot{q}'$$

$$\Downarrow$$

$$\left(1-\varphi\right)\sum_k\left[(\dot{\omega}_k+\dot{s}_k)W_k\right]h_k + \rho u\left(1-\varphi\right)c_p\frac{dT}{dz} - \left(1-\varphi\right)\sum_i \dot{s}_i W_i h_{soot} + c_{p,soot}\frac{dT}{dz}\rho u\varphi = \dot{q}'$$

$$\Downarrow$$

$$\left[\rho u\left(1-\varphi\right)c_p + \rho u\varphi c_{p,soot}\right]\frac{dT}{dz} + \left(1-\varphi\right)\sum_k\left[(\dot{\omega}_k+\dot{s}_k)W_k\right]h_k - \left(1-\varphi\right)\sum_i \dot{s}_i W_i h_{soot} = \dot{q}'$$

$$\Downarrow$$

$$\frac{dT}{dz} = \frac{1}{\rho u\left(1-\varphi\right)c_p + \rho u\varphi c_{p,soot}}\left[-\left(1-\varphi\right)\sum_k\left(\dot{\omega}_k+\dot{s}_k\right)W_k h_k + \left(1-\varphi\right)\sum_i \dot{s}_i W_i h_{soot} + \dot{q}'\right]$$

A.4 Perfectly Stirred Reactor

A.4.1 Continuity

The mass conservation for the partially stirred reactor can be written as:

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out} + V\left(1-\varphi\right)\sum_i \dot{s}_i W_i \tag{A.8}$$

A.4.2 Species

$$\frac{d}{dt}(m_k) = \dot{m}_{in} Y_k^* - \dot{m}_{out} Y_k + V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow m_k = m Y_k$$

$$Y_m \frac{dm}{dt} + m\frac{dY_k}{dt} = \dot{m}_{in} Y_k^* - \dot{m}_{out} Y_k + V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow \text{ from Equation A.8}$$

$$\left(\dot{m}_{in} - \dot{m}_{out} + V(1-\varphi)\sum_i \dot{s}_i W_i\right) Y_k + \frac{dY_k}{dt} m = \dot{m}_{in} Y_k^* - \dot{m}_{out} Y_k + V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow$$

$$V(1-\varphi) Y_k \sum_i \dot{s}_i W_i + \frac{dY_k}{dt} m = \dot{m}_{in}(Y_k^* - Y_k) + V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k$$

$$\Downarrow$$

$$\frac{dY_k}{dt}\rho\, V(1-\varphi) = \dot{m}_{in}(Y_k^* - Y_k) + V(1-\varphi)(\dot{\omega}_k + \dot{s}_k) W_k - V(1-\varphi) Y_k \sum_i \dot{s}_i W_i$$

$$\Downarrow$$

$$\frac{dY_k}{dt} = \frac{\dot{m}_{in}}{\rho V(1-\varphi)}(Y_k^* - Y_k) + \frac{1}{\rho}(\dot{\omega}_k + \dot{s}_k) W_k - \frac{1}{\rho} Y_k \sum_i \dot{s}_i W_i \tag{A.9}$$

A.4.3 Energy

$$\frac{dH}{dt} = \dot{m}_{in} h^* - \dot{m}_{out} h - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow H = mh$$

$$\frac{dm}{dt} h + m\frac{dh}{dt} = \dot{m}_{in} h^* - \dot{m}_{out} h - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow \text{ from Equation A.8}$$

$$\left(\dot{m}_{in} - \dot{m}_{out} + V(1-\varphi)\sum_i \dot{s}_i W_i\right) h + m\frac{dh}{dt} = \dot{m}_{in} h^* - \dot{m}_{out} h - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow$$

$$V(1-\varphi)\,h\sum_i \dot{s}_i W_i + m\frac{dh}{dt} = \dot{m}_{in}\left(h^* - h\right) - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow$$

$$m\frac{dh}{dt} = \dot{m}_{in}\left(h^* - h\right) - V(1-\varphi)\,h\sum_i \dot{s}_i W_i - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow h = \sum Y_k h_k$$

$$m\sum\frac{dY_k}{dt}h_k + m\sum\frac{dh_k}{dt}Y_k = \dot{m}_{in}\left(h^* - h\right) - V(1-\varphi)\,h\sum_i \dot{s}_i W_i - \frac{dH_{soot}}{dt} + \dot{Q}$$

$\Downarrow$ from Equation A.9

$$m\sum\left(\frac{\dot{m}_{in}}{\rho V(1-\varphi)}\left(Y_k^* - Y_k\right) + \frac{1}{\rho}\left(\dot{\omega}_k + \dot{s}_k\right)W_k - \frac{1}{\rho}Y_k\sum_i \dot{s}_i W_i\right)h_k + m\sum c_{p,k}\frac{dT}{dt}Y_k = \dot{m}_{in}\left(h^* - h\right) - V(1-\varphi)\,h\sum_i \dot{s}_i W_i - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow m = \rho\ V(1-\varphi)$$

$$\dot{m}_{in}\sum\left(Y_k^* - Y_k\right)h_k + V(1-\varphi)\sum\left(\dot{\omega}_k + \dot{s}_k\right)W_k h_k + mc_p\frac{dT}{dt} = \dot{m}_{in}\left(h^* - h\right) - \frac{dH_{soot}}{dt} + \dot{Q}$$

$$\Downarrow$$

$$\dot{m}_{in}\sum\left(Y_k^* - Y_k\right)h_k + V(1-\varphi)\sum\left(\dot{\omega}_k + \dot{s}_k\right)W_k h_k + mc_p\frac{dT}{dt} = \dot{m}_{in}\left(h^* - h\right) - \left(\rho_{soot}c_{p,soot}\varphi V\frac{dT}{dt} - (1-\varphi)\,V\sum_i \dot{s}_i W_i\ h_{soot}\right) + \dot{Q}$$

$$\Downarrow$$

$$\dot{m}_{in}\sum\left(Y_k^* - Y_k\right)h_k + V(1-\varphi)\sum\left(\dot{\omega}_k + \dot{s}_k\right)W_k h_k + mc_p\frac{dT}{dt} = \dot{m}_{in}\left(h^* - h\right) - \left(\rho_{soot}c_{p,soot}\varphi V\frac{dT}{dt} - (1-\varphi)\,V\sum_i \dot{s}_i W_i\ h_{soot}\right) + \dot{Q}$$

$$\Downarrow m = \rho V(1-\varphi)$$

$$\Downarrow \times 1/V$$

$$\dot{m}_{in} \sum \left(Y_k^* - Y_k\right) h_k + V\left(1-\varphi\right) \sum \left(\dot{\omega}_k + \dot{s}_k\right) W_k h_k + \rho\ V\left(1-\varphi\right) c_p \frac{dT}{dt} = \\ \dot{m}_{in}\left(h^* - h\right) - \left(\rho_{soot} c_{p,soot} \varphi V \frac{dT}{dt} - \left(1-\varphi\right) V \sum_i \dot{s}_i W_i\ h_{soot}\right) + \dot{Q}$$

$$\Downarrow$$

$$\left(\rho\left(1-\varphi\right) c_p + \rho_{soot} c_{p,soot} \varphi\right) \frac{dT}{dt} = \frac{\dot{m}_{in}}{V}\left(h^* - h\right) - \frac{\dot{m}_{in}}{V} \sum \left(Y_k^* - Y_k\right) h_k \\ - \left(1-\varphi\right) \sum \left(\dot{\omega}_k + \dot{s}_k\right) W_k h_k + \left(1-\varphi\right) \sum_i \dot{s}_i W_i\ h_{soot} + \frac{\dot{Q}}{V}$$

$$\Downarrow$$

$$\frac{dT}{dt} = \frac{1}{\rho\left(1-\varphi\right) c_p + \rho_{soot} c_{p,soot} \varphi} \left[\frac{\dot{m}_{in}}{V}\left(h^* - h\right) - \frac{\dot{m}_{in}}{V} \sum \left(Y_k^* - Y_k\right) h_k - \left(1-\varphi\right) \sum \left(\dot{\omega}_k + \dot{s}_k\right) W_k h_k \right. \\ \left. + \left(1-\varphi\right) \sum_i \dot{s}_i W_i\ h_{soot} + \frac{\dot{Q}}{V}\right] \tag{A.10}$$

Appendix B

Additional Results

B.1 Methane pyrolysis data from Hanson Research Group

B.1.1 30% CH_4 dataset

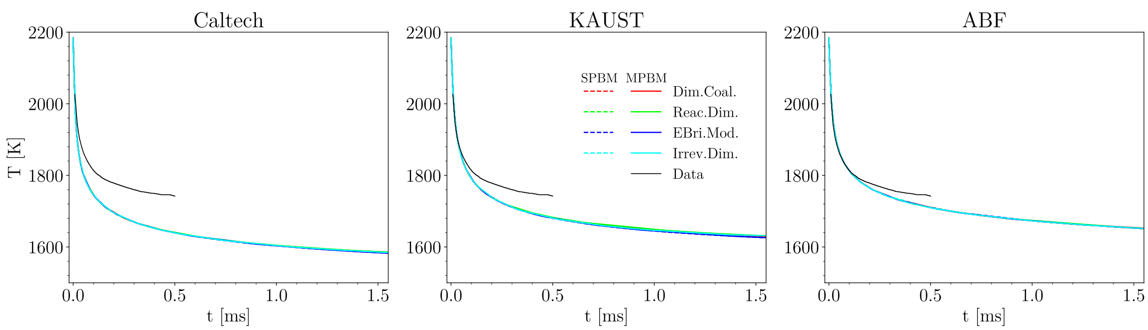

Figure B.1: The time history of temperature of 30% CH_4 pyrolysis at T=2184 K and P=3.62 atm using Caltech, KAUST, and ABF mechanisms and different PAH growth and particle dynamics models compared with the simulation results reported by Hanson Research Group

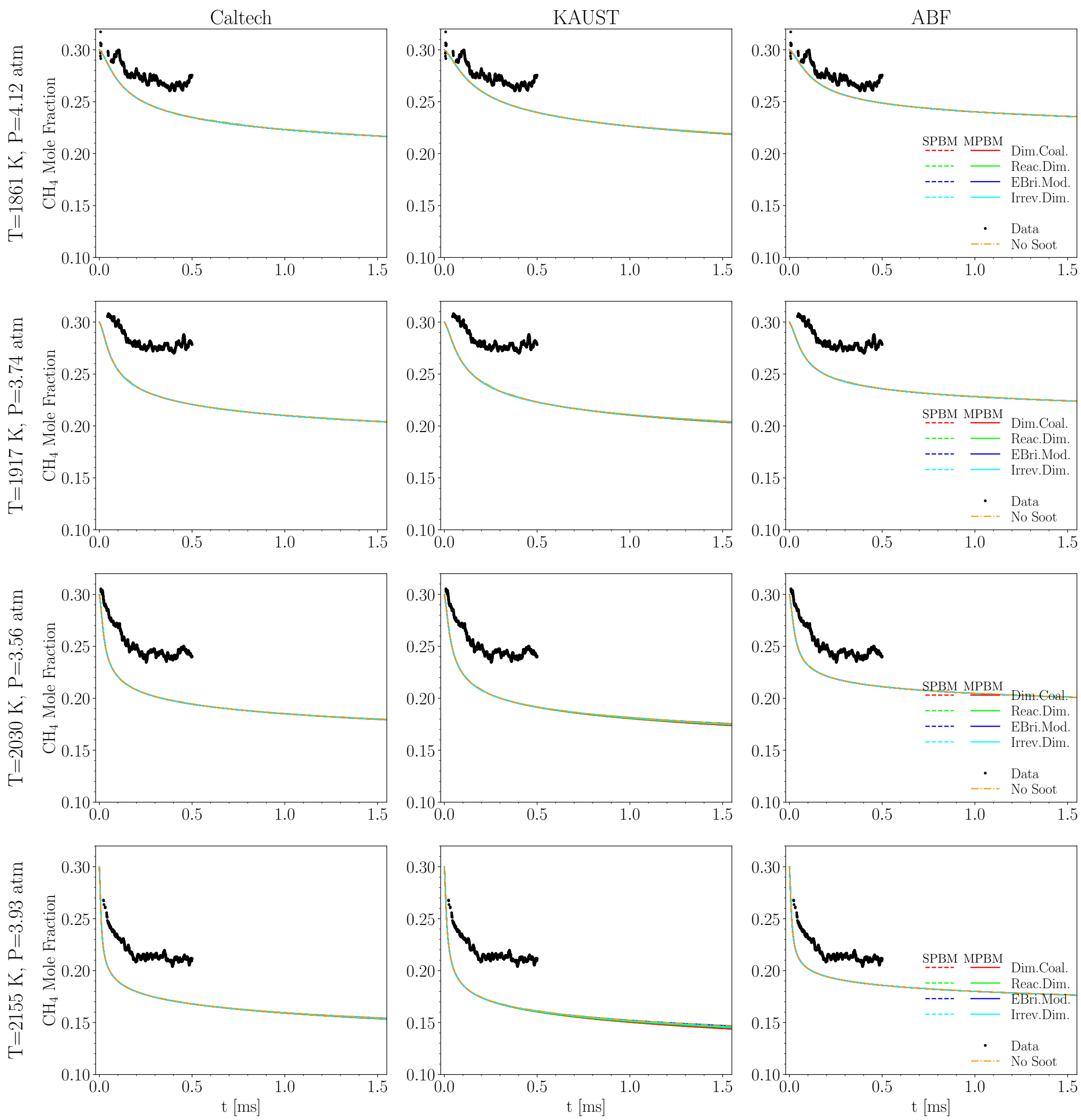

Figure B.2: The time history of CH_4 mole fraction of 30% CH_4 pyrolysis for 1861 K< T <2155 K using Caltech, KAUST, and ABF mechanisms without soot (only chemistry) and with soot modeling described by different PAH growth and particle dynamics models compared with laser diagnostic data

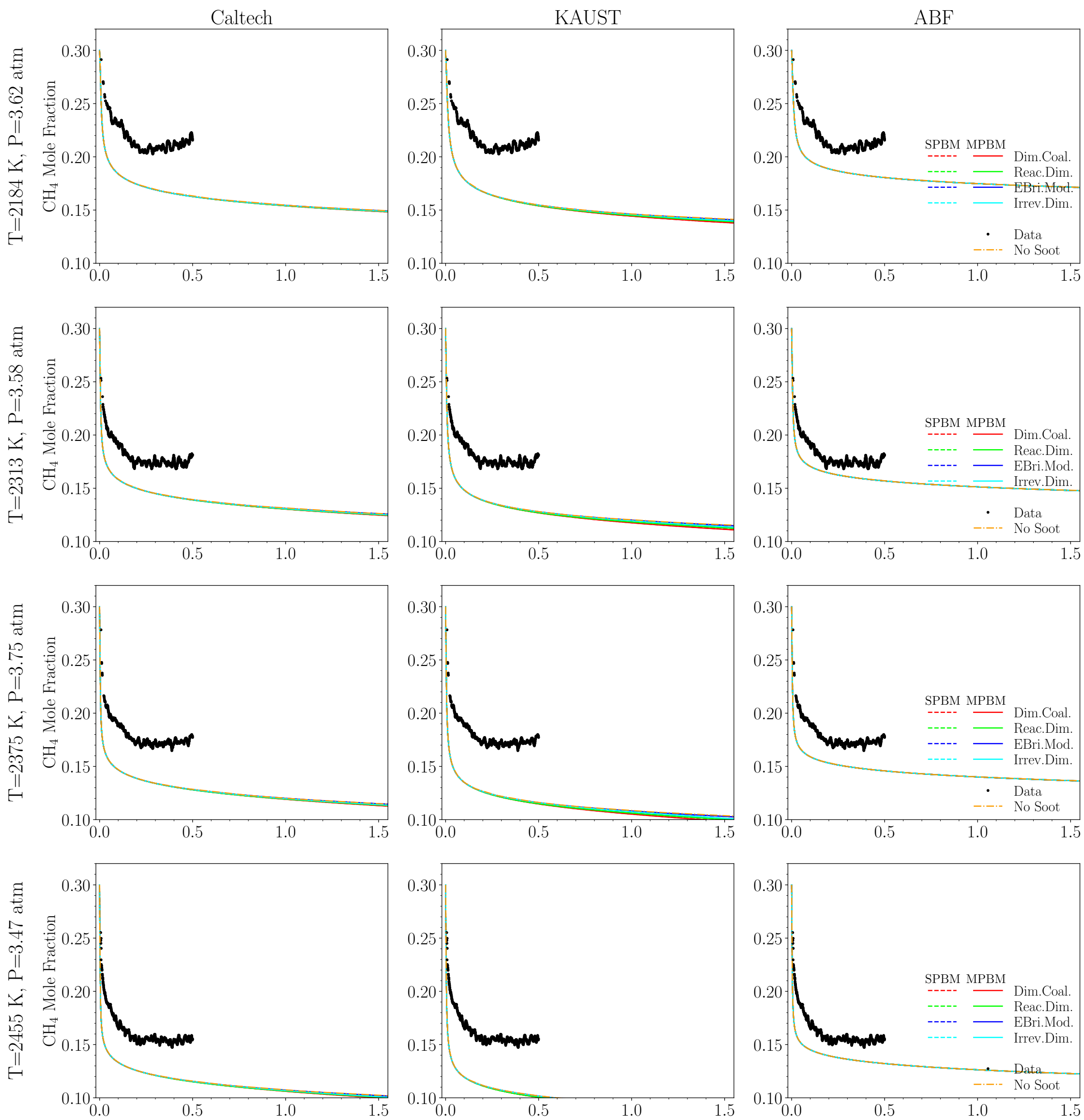

Figure B.3: The time history of CH_4 mole fraction of 30% CH_4 pyrolysis for 2184 K< T <2455 K using Caltech, KAUST, and ABF mechanism without soot (only chemistry) and with soot modeling described by different PAH growth and particle dynamics models compared with laser diagnostic data

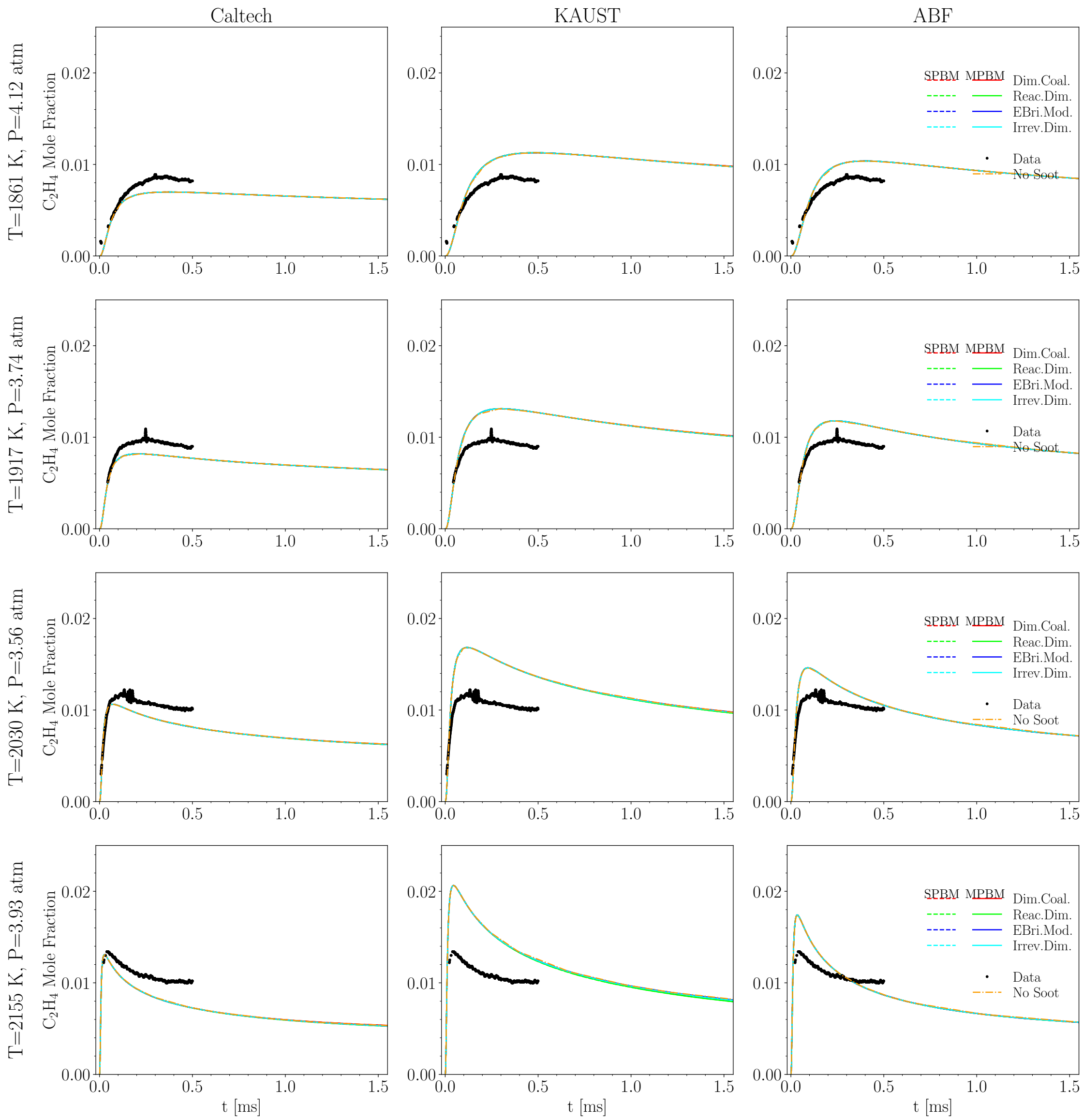

Figure B.4: The time history of C_2H_4 mole fraction of 30% CH_4 pyrolysis for 1861 K< T <2155 K using Caltech, KAUST, and ABF mechanisms without soot (only chemistry) and with soot modeling described by different PAH growth and particle dynamics models compared with laser diagnostic data

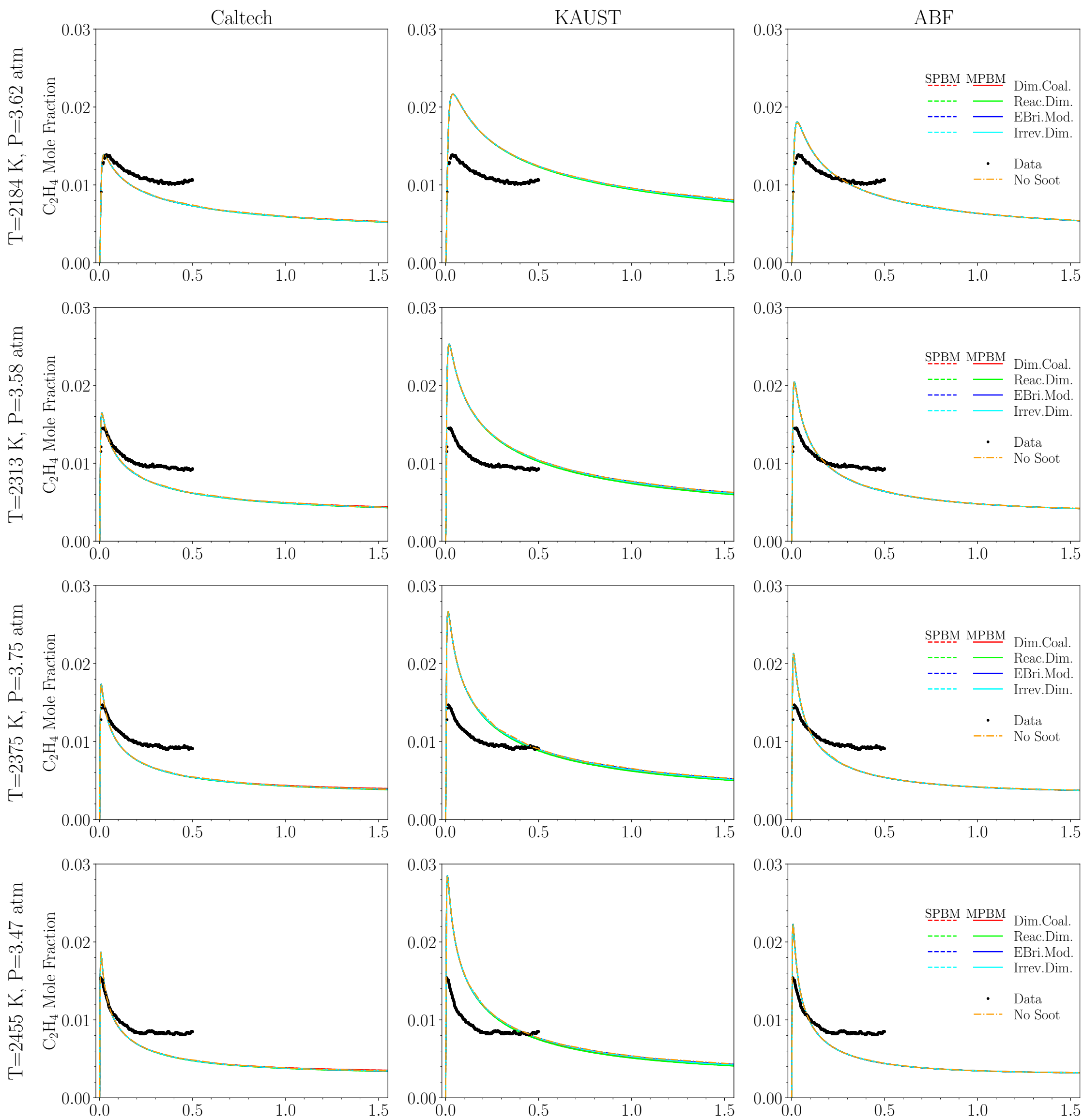

Figure B.5: The time history of C_2H_4 mole fraction of 30% CH_4 pyrolysis for 2184 K< T <2455 K using Caltech, KAUST, and ABF mechanism without soot (only chemistry) and with soot modeling described by different PAH growth and particle dynamics models compared with laser diagnostic data

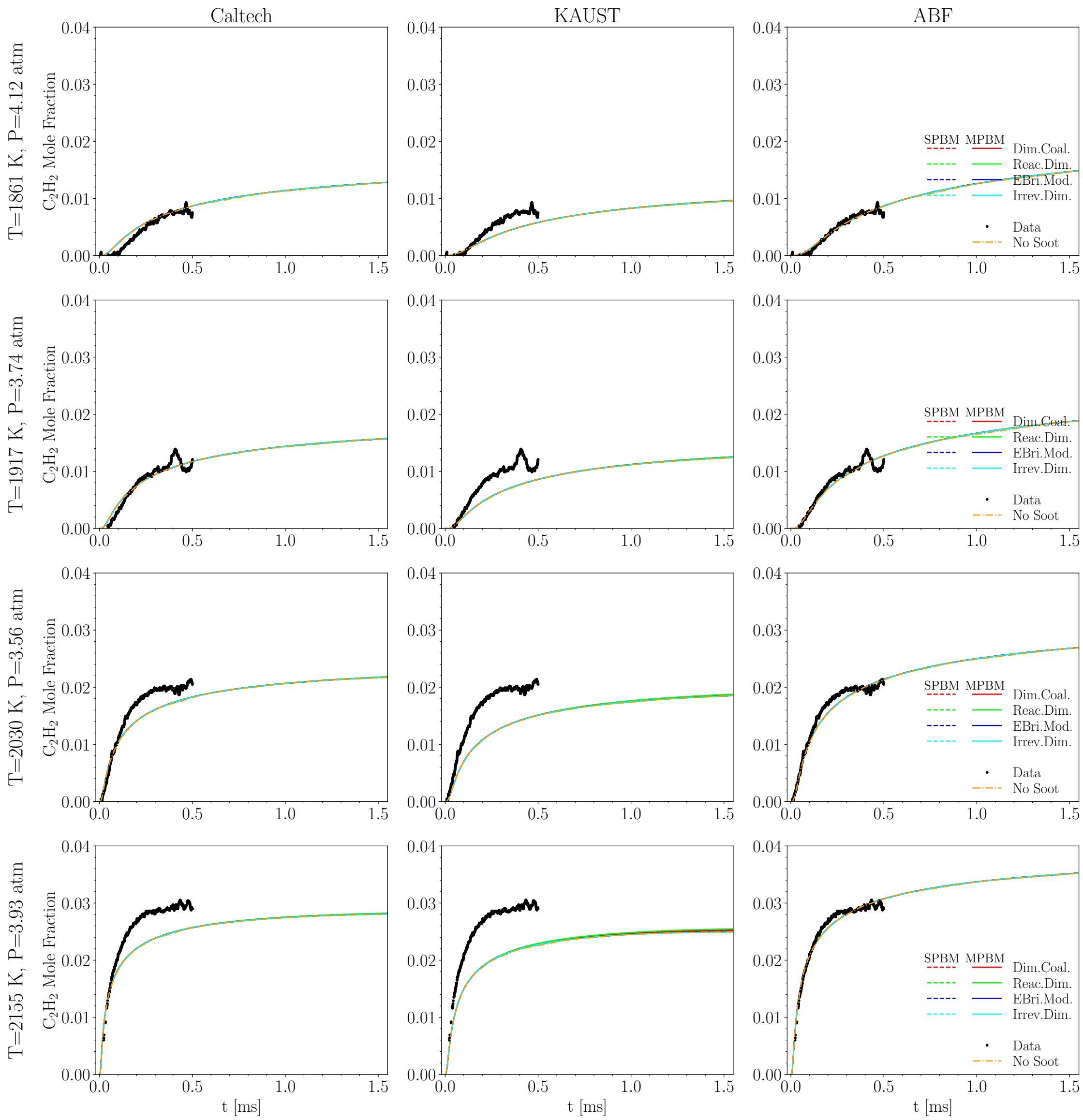

Figure B.6: The time history of C_2H_2 mole fraction of 30% CH_4 pyrolysis for 1861 K< T <2155 K using Caltech, KAUST, and ABF mechanisms without soot (only chemistry) and with soot modeling described by different PAH growth and particle dynamics models compared with laser diagnostic data

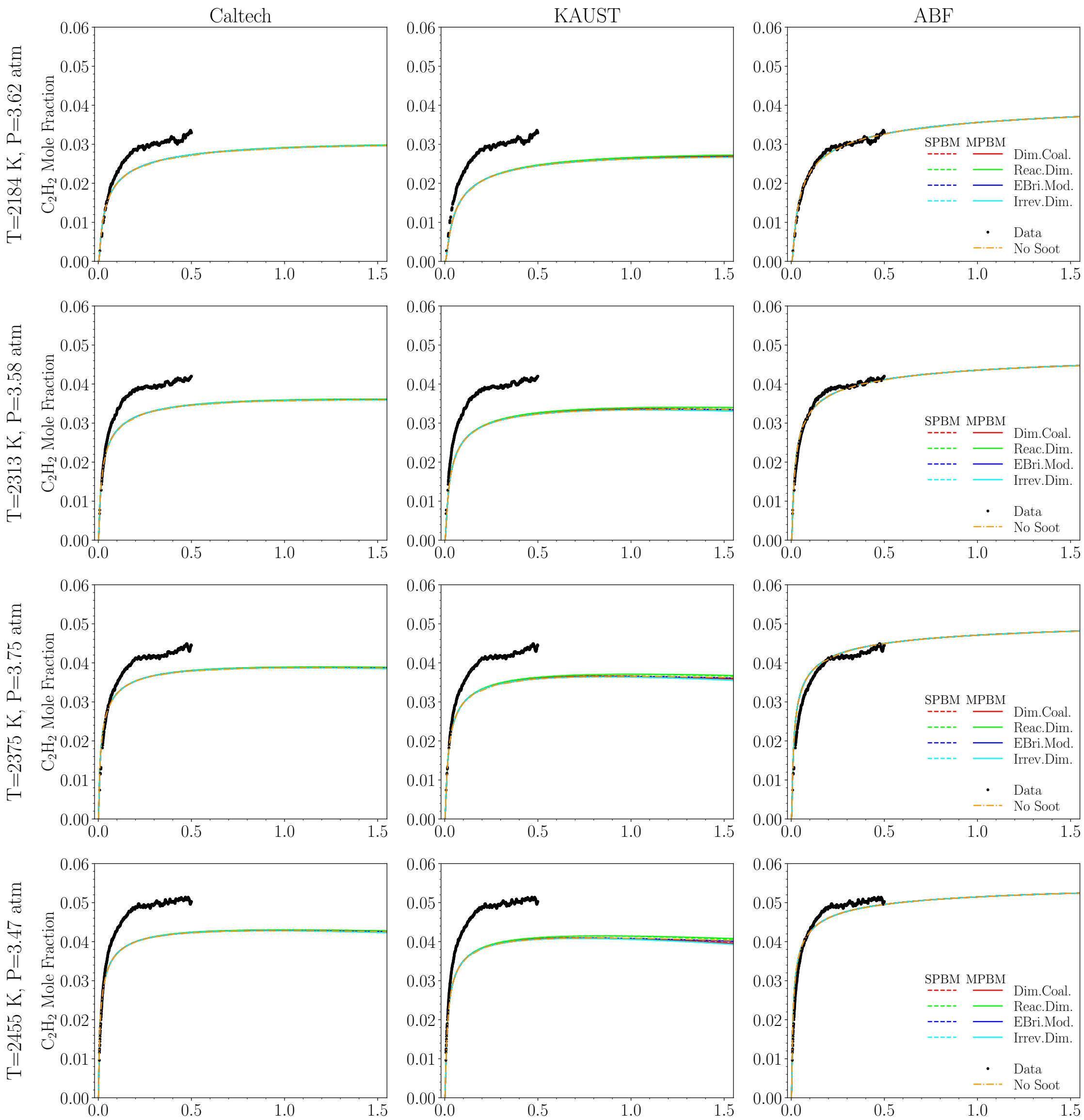

Figure B.7: The time history of C_2H_2 mole fraction of 30% CH_4 pyrolysis for 2184 K< T <2455 K using Caltech, KAUST, and ABF mechanism without soot (only chemistry) and with soot modeling described by different PAH growth and particle dynamics models compared with laser diagnostic data

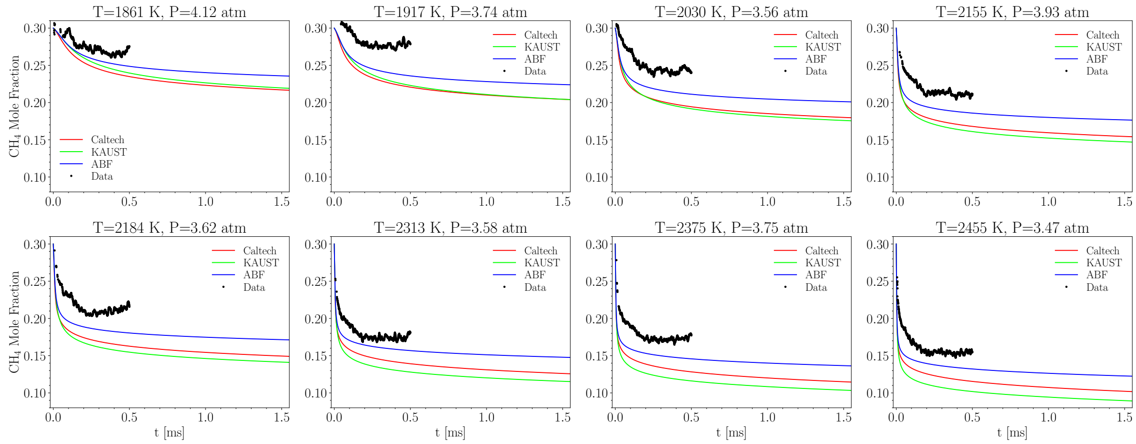

Figure B.8: The time history of CH_4 mole fraction of 30% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot compared with laser diagnostic data

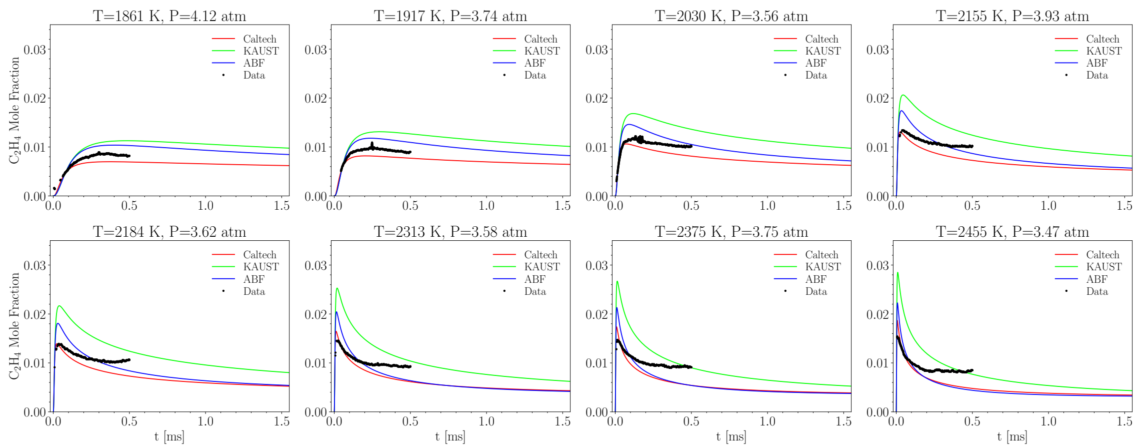

Figure B.9: The time history of C_2H_4 mole fraction of 30% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot compared with laser diagnostic data

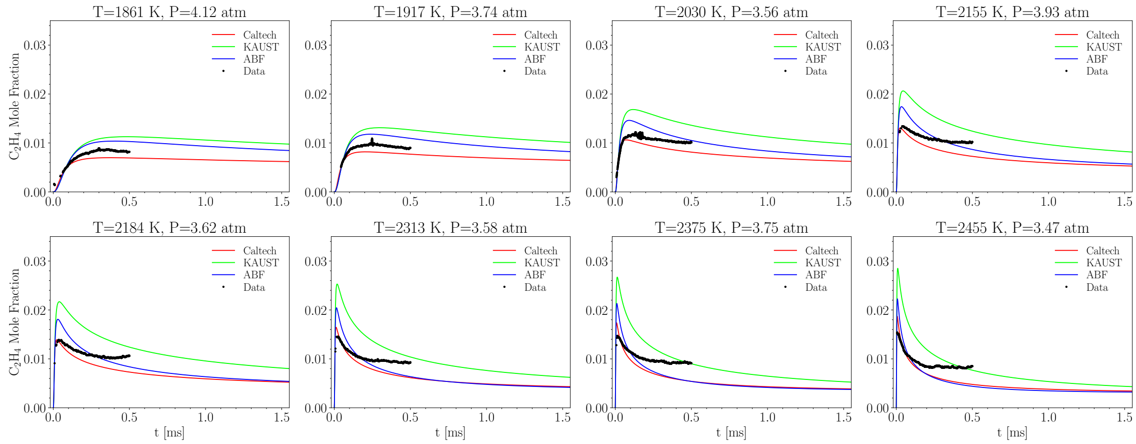

Figure B.10: The time history of C_2H_2 mole fraction of 30% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot compared with laser diagnostic data

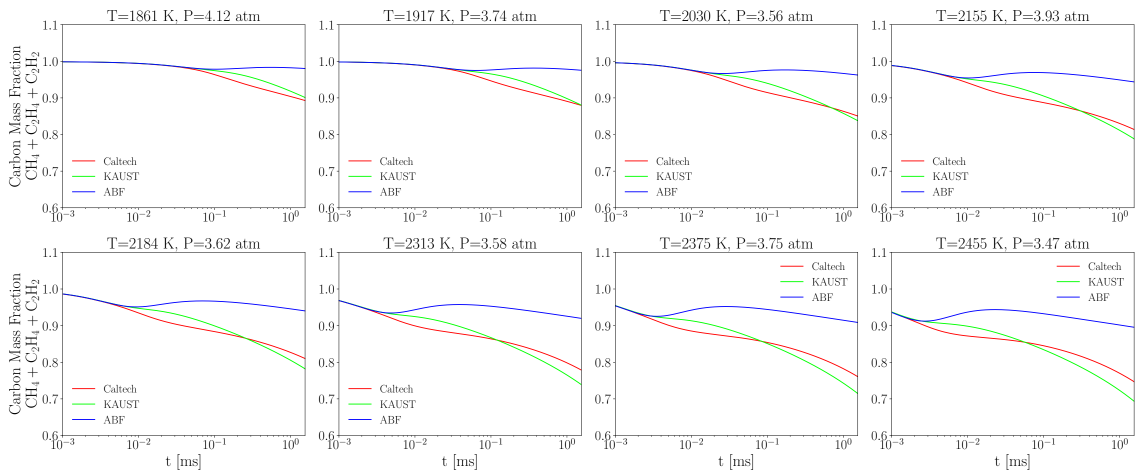

Figure B.11: The time history of carbon mass fraction of CH_4, C_2H_4, C_2H_2 of 30% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot

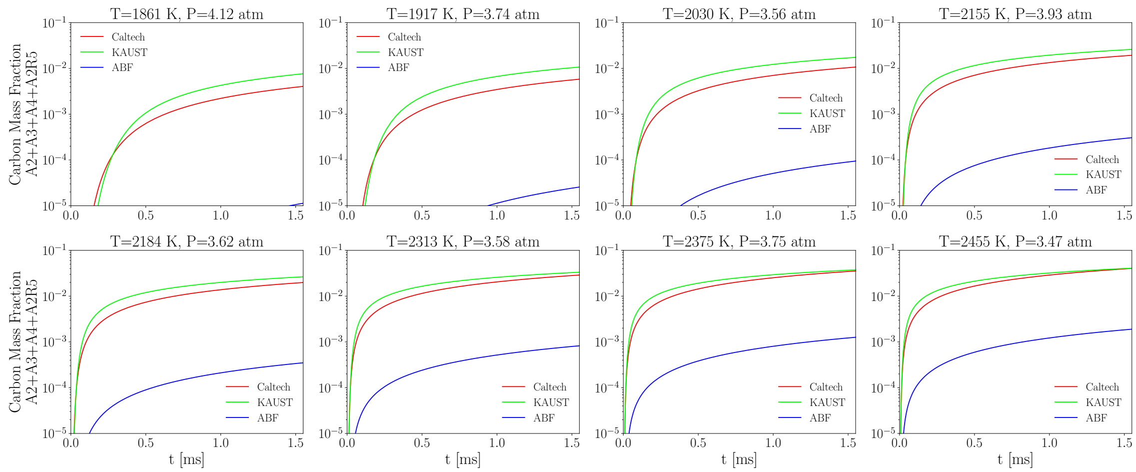

Figure B.12: The time history of carbon mass fraction of soot precursors, A2, A3, A4, and A2R5 of 30% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot

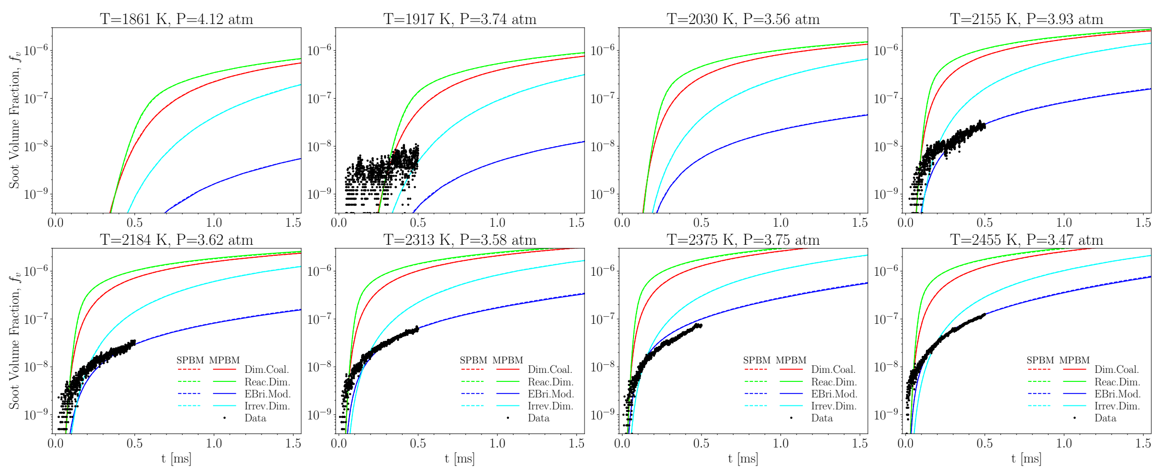

Figure B.13: The time history of soot volume fraction, f_v of 30% CH_4 pyrolysis for 1861 K< T <2455 K using KAUST with different combinations of PAH growth and particle dynamics models compared with extinction measurements

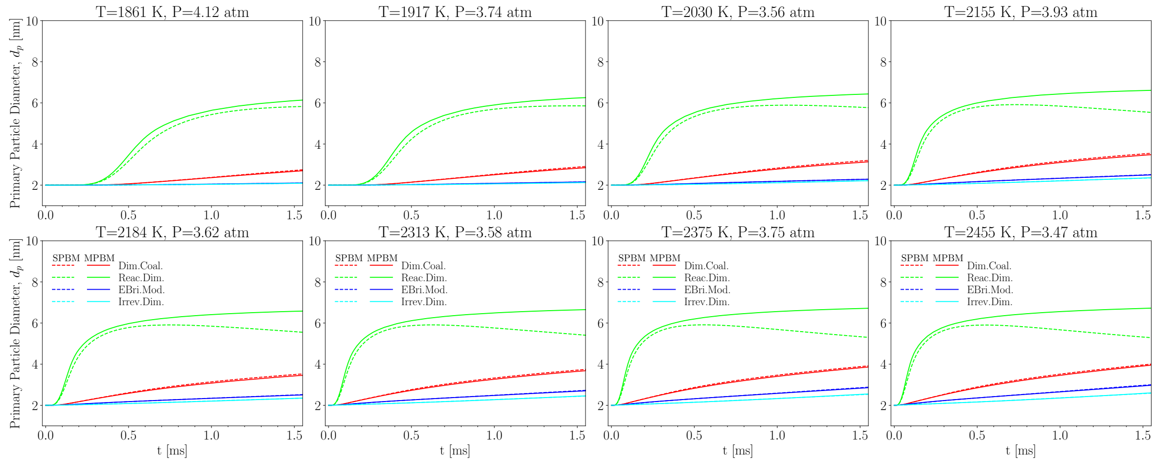

Figure B.14: The time history of primary particle diameter, d_p of 30% CH_4 pyrolysis for 1861 K< T <2455 K using KAUST with different combinations of PAH growth and particle dynamics models compared with extinction measurements

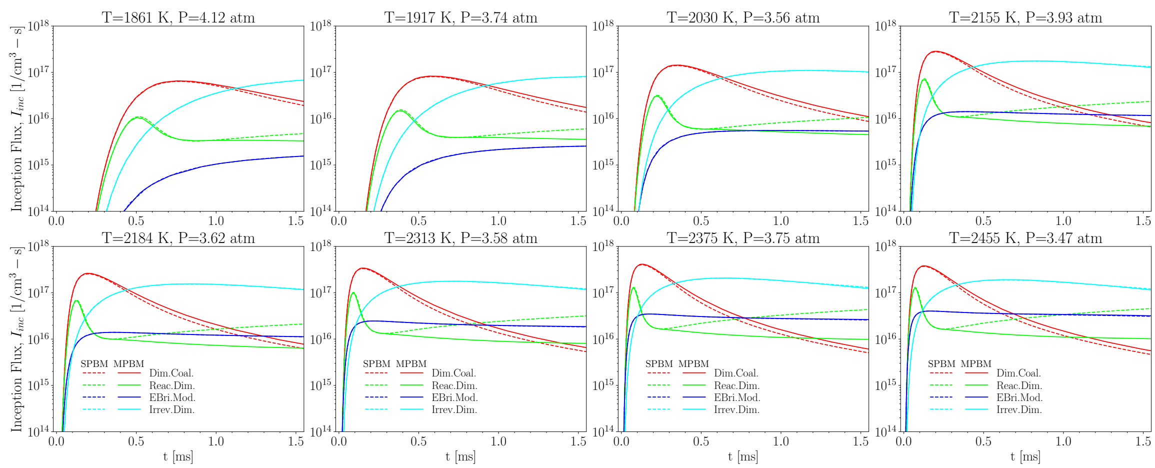

Figure B.15: The time history of inception flux, I_{inc} of 30% CH_4 pyrolysis for 1861 K< T <2455 K using KAUST with different combinations of PAH growth and particle dynamics models compared with extinction measurements

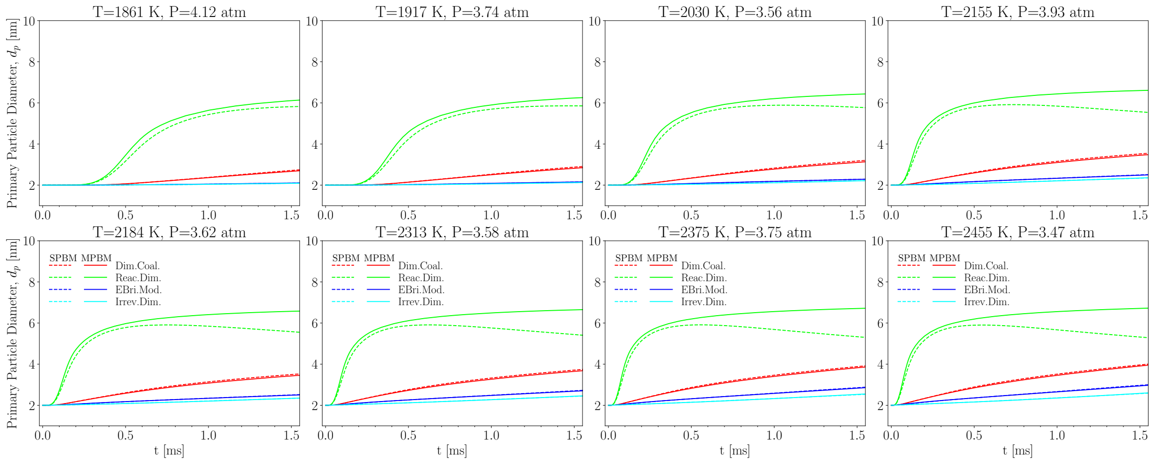

Figure B.16: The time history of mobility diameter, d_m of 30% CH_4 pyrolysis for 1861 K< T <2455 K using KAUST with different combinations of PAH growth and particle dynamics models compared with extinction measurements

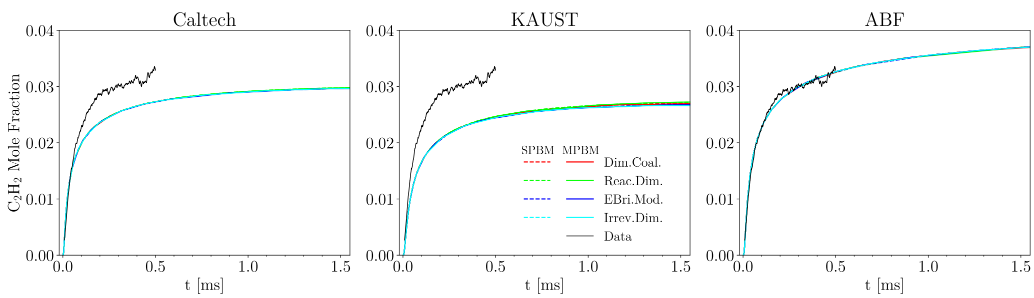

Figure B.17: The time history of C_2H_2 mole fraction of 30% CH_4 pyrolysis at T=2184 K and P=3.62 atm using Caltech, KAUST, and ABF mechanism and different PAH growth and particle dynamics models compared with laser diagnostic data

B.1.2 10% CH_4 dataset

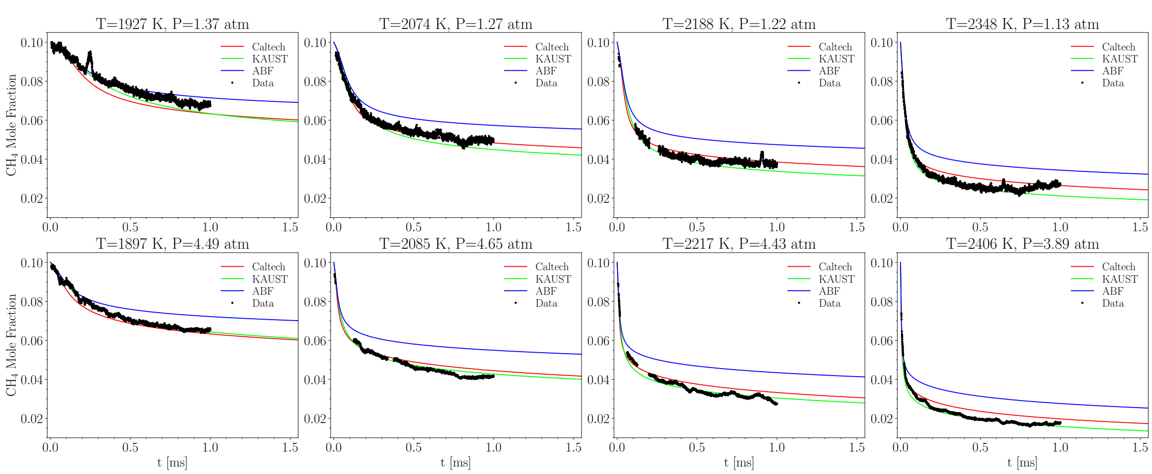

Figure B.18: The time history of CH_4 mole fraction of 10% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot compared with laser diagnostic data

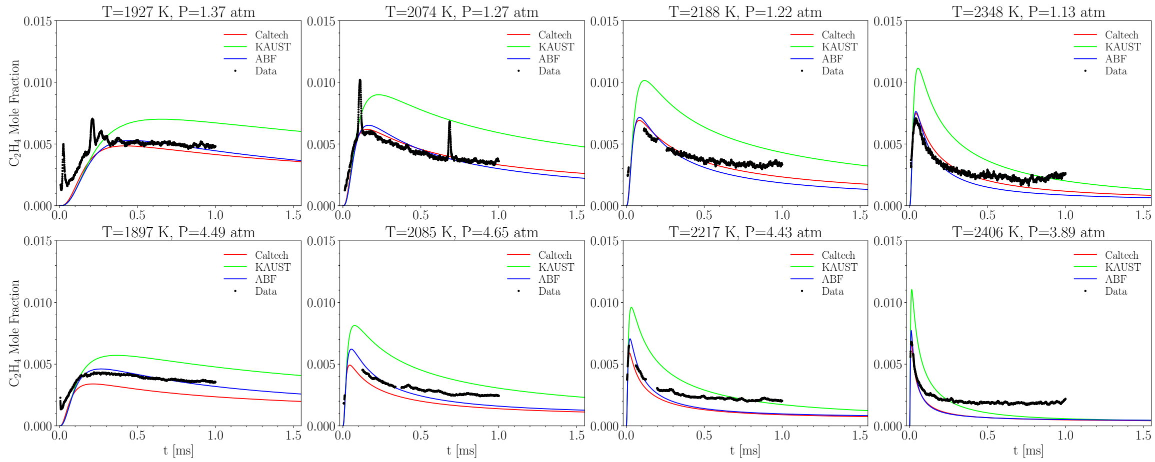

Figure B.19: The time history of C_2H_4 mole fraction of 10% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot compared with laser diagnostic data

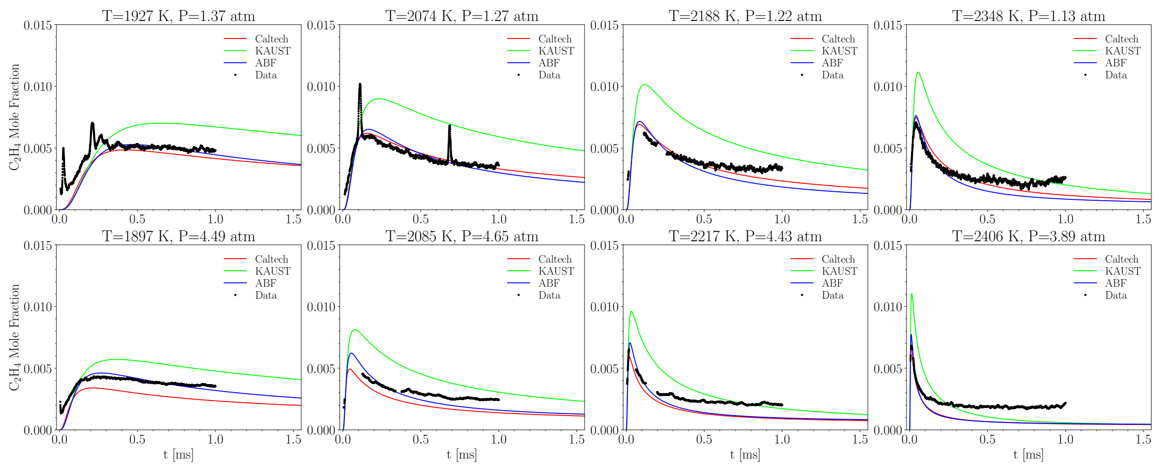

Figure B.20: The time history of C_2H_2 mole fraction of 10% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot compared with laser diagnostic data

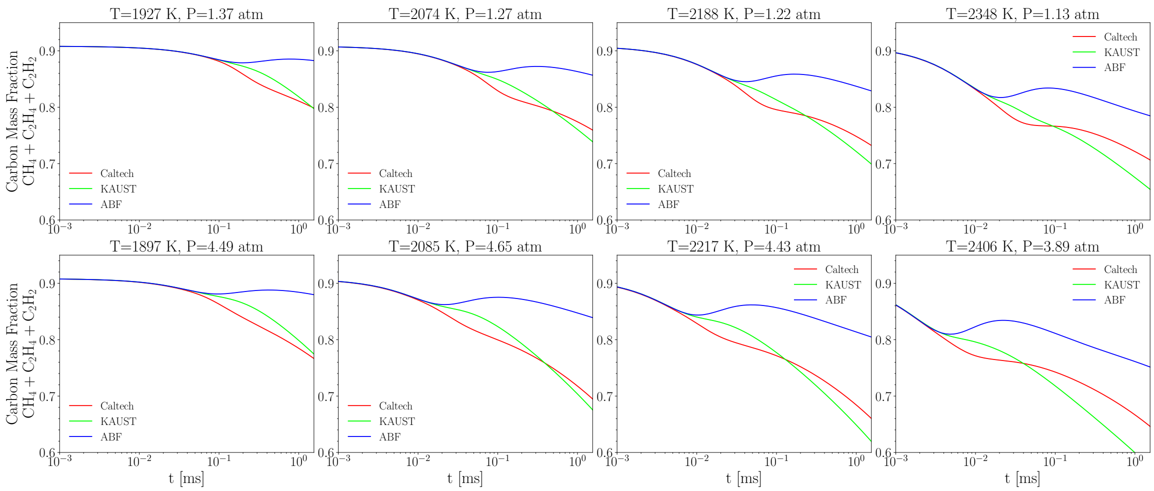

Figure B.21: The time history of carbon mass fraction of CH_4, C_2H_4, C_2H_2 of 10% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot

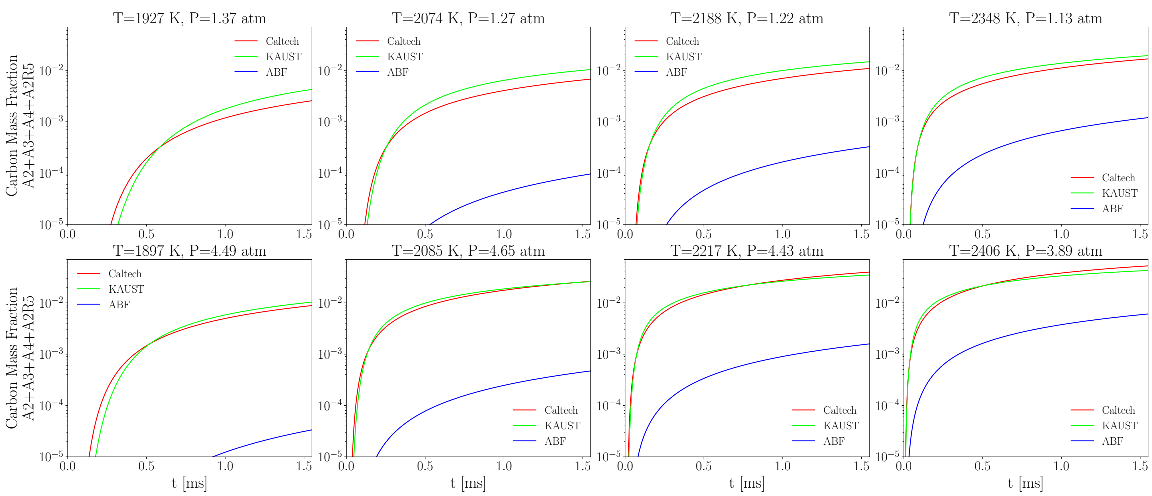

Figure B.22: The time history of carbon mass fraction of soot precursors, A2, A3, A4, and A2R5 of 10% CH_4 pyrolysis for 1861 K< T <2455 K using Caltech, KAUST, and ABF mechanisms without soot

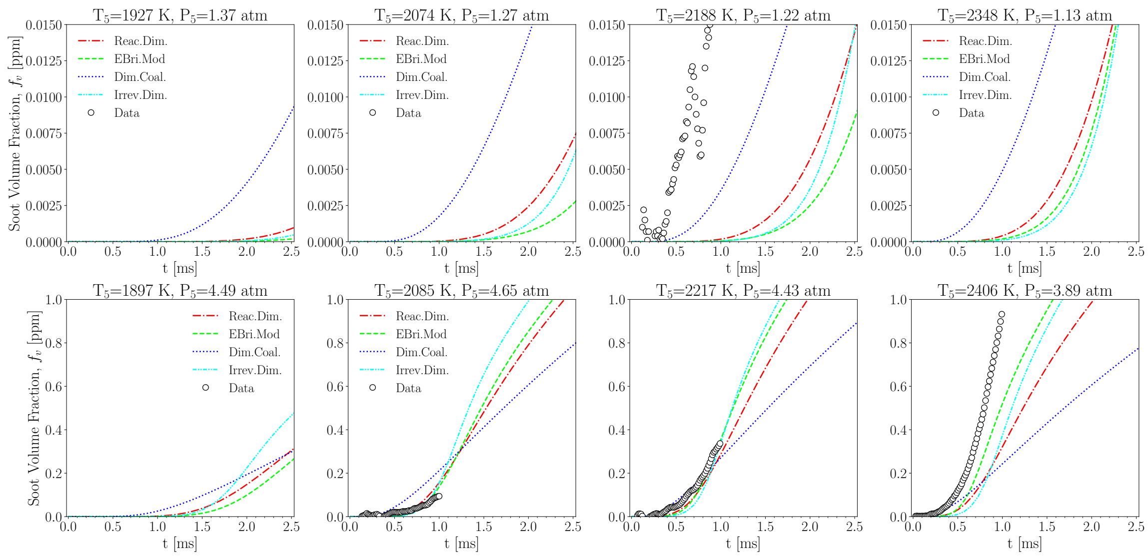

Figure B.23: The soot volume fraction, f_v over 1800 K<T<2400 K and P near 1 and 4 atm for 10% CH_4 pyrolysis using KAUST, SPBM and optimized PAH growth models that minimizes f_v and d_p at T=2217 and P=4.43 atm

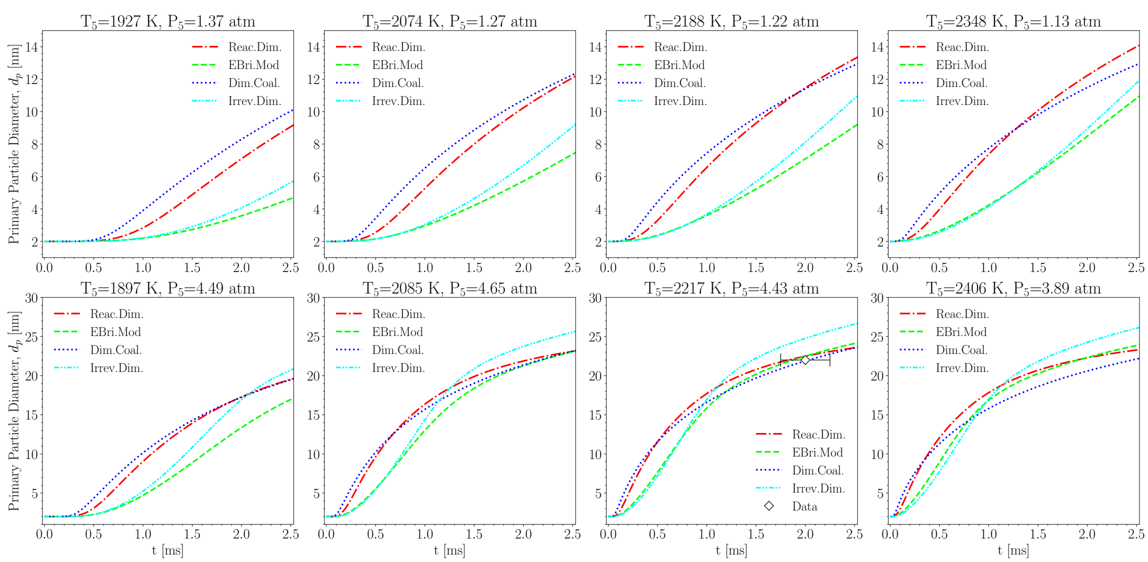

Figure B.24: The primary particle diameter, d_p over 1800 K<T<2400 K and P near 1 and 4 atm for 10% CH_4 pyrolysis using KAUST, SPBM and optimized PAH growth models that minimizes f_v and d_p at T=2217 and P=4.43 atm

B.2 Methane pyrolysis data of Agafonov et al. [1]

The pyrolysis of 5% and 10% CH_4 diluted with Ar is investigated for the post-shock temperature range of 1800-3000 K and the pressure range of 4.7-7.1 bar. We assume the pressure linearly increase by temperature across the simulation cases. The obtained soot yields by were compared with the soot yield measured by Agafonov et al. [1] using a dual-beam absorption–emission technique. Agafonov et al. [1] reported yield×E(m) at λ=632 nm, and yield data was retrieved using E(m)=0.37 suggested therein.

Figure B.25 depicts soot yield (SY) at t=1.5 ms for 5% and 10% CH_4 over the temperature range of 1800-3000 K using MPBM and SPBM and different inception model and Caltech [7] mechanism. Note that, the simulations were also performed using ABF [64] and KAUST [172] mechanisms, and the results were included in Appendix****. The soot yield predictions exhibit a close agreement with measurements [1] considering the uncertainties in residence time from experiments. The model also successfully captures the bell-shape dependence of soot yield on temperature that has been observed in a variety of hydrocarbons in shock tube [188, 189]. The particle dynamics model has minimal effect on the predicted soot yield. In the vicinity of peak soot yield, SPBM results in slightly lower yield than MPBM, but they are indistinguishable in the rest of the temperature range.

In %5 CH_4, the SY peak temperature obtained from the model is slightly shifted towards higher temperatures (2300 K< T_{peak} <2400 K) compared to the measurements T_{peak} =2200 K. There are noticeable differences in the behavior of PAH growth model depending on the shock-tube initial temperature. When T<2100 K, Reactive Dimerization and EBridge Formation have the highest and lowest soot yield, respectively with Dimer Coalescence predicting soot yields that always falls between Reactive and Irreversible Dimerization. However, The soot yield of EBridge Formation rapidly rises with temperature and exceeds that of Reactive Dimerization and stays higher for the rest of temperature range.

The SY noticeably increases for higher initial CH_4 mole fraction of %10 because more PAH and C_2H_2 are formed leading to stronger carbon conversion rate to soot via inception and surface growth. The peak SY obtained from the model occurs in a higher temperature range 2600-2700 K compared to %5 CH_4. Soot yield trends can be better understood by examining carbon mass fraction of species directly contributing to soot mass. Figure B.26 depicts the bell-shape distribution of the carbon mass fraction of soot precursors (A2, A2R5, A3, and A4) over the studied temperature range. In low (T=1800 K) and high (T=2300 K) end of distribution, a low amount of precursors are formed resulting in low inception rates, particle number concentration and surface growth sites that reduces the soot yield. Additionally, %10CH_4 has a wider spread with peaks at higher temperature compared to %10CH_4 which explains the shift in peak yield temperature in Figure B.25. The effect of particle dynamics model is only noticeable in Reactive Dimerization where SPBM results in higher precursor CMF (lower consumption) due to its lower PAH adsorption rates.

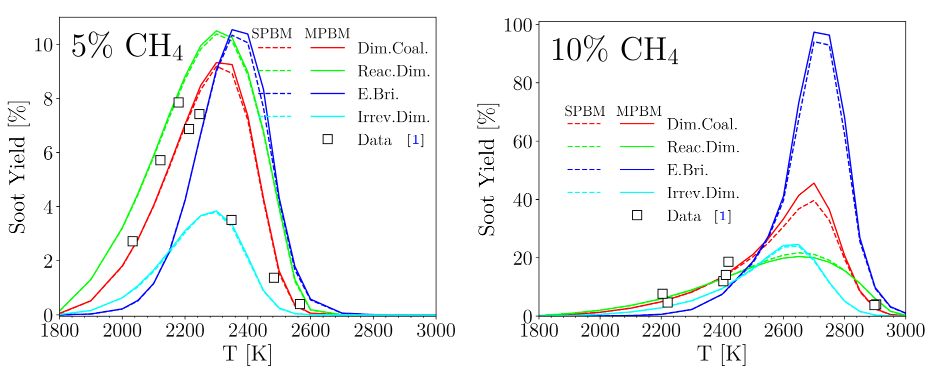

Figure B.25: The temperature dependence of soot yield during pyrolysis of 5% CH_4-Ar (left pane) and 10% CH_4-Ar (right pane) at P = 4.5–6.7 bar obtained using Caltech mechanism and different inception models compared with measurements at 1.5ms [1] where the absorption funtion of E(m)=0.37 is used to estimate soot yield.

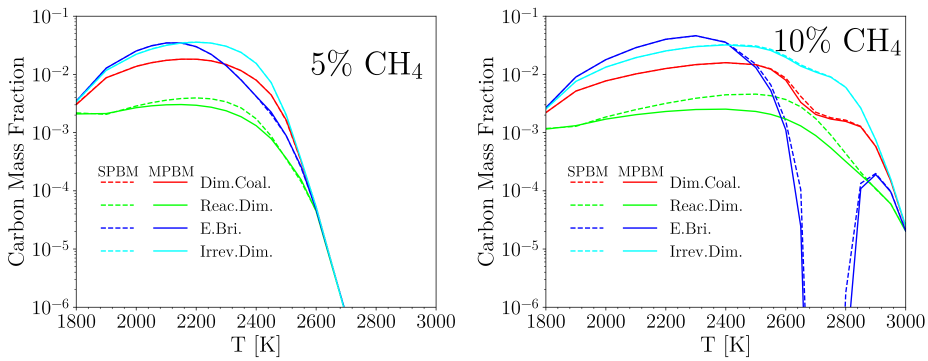

Figure B.26: The temperature dependence of carbon mass fraction of soot precursors (A2, A2R5, A3, and A4) combined during pyrolysis of 5% CH_4-Ar (left pane) and 10% CH_4-Ar (right pane) at P = 4.5–6.7 bar obtained using Caltech mechanism and different inception models at t=1.5 ms

EBridge Formation exhibits a similar behavior in %10 CH_4 simulations where it starts with the lowest SY at T<2500 K and then quickly increases reaching its peak at 100% which is significantly larger that other PAH growth models. The remarkable drop in carbon mass fraction of precursors with EBridge Formation corresponds to %100 yield meaning that all gaseous carbon including the precursors are directed towards soot particles.The higher precursor CMF with Irreversible Dimerization near the peak yield temperature region (2200-2400 K for %5 CH_4 and 2600-2800 K for %10 CH_4) indicates less consumption of precursors via inception and PAH adsorption.

Figure B.27 shows the CMF of C_2H_2 that has an overall increasing trend in the temperature range, but it reaches a plateau for %5 CH_4. There is also a remarkable drop in C_2H_2 CMF in 2600-2800 K due to strong mass growth rate of soot particles that drains C_2H_2 from the gas mixture leading to high soot yield ≊ 100%.

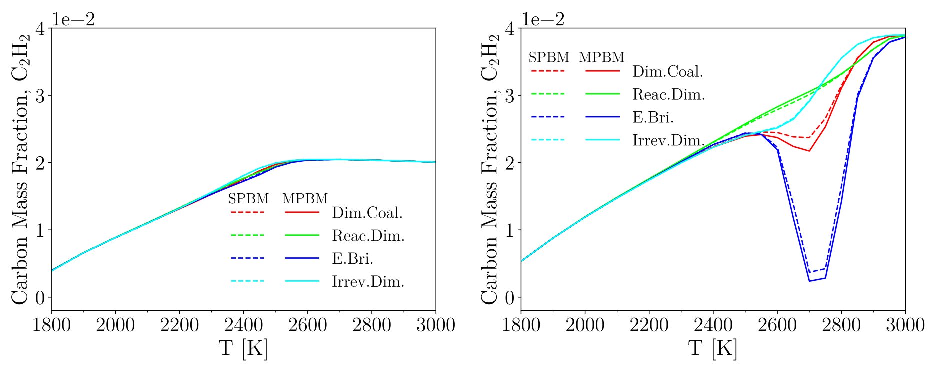

Figure B.27: The temperature dependence of carbon mass fraction of C_2H_2 during pyrolysis of 5% CH_4-Ar (left pane) and 10% CH_4-Ar (right pane) at P = 4.5–6.7 bar obtained using Caltech mechanism and different inception models at t=1.5 ms

Although soot yield, and precursor and acetylene CMF are not sensitive to particle dynamics model, there is a significant difference in agglomerate morphology between SPBM and MPBM predictions. Fig. B.28 shows the average number of primary particles per agglomerate, which is larger for SPBM by the maximum factor of 5 for EBridge Formation and Dimer Coalescence in 10% CH_4. SPBM predicts larger agglomerates due to accounting for polydispersity of particles that results in higher overall collision rate and faster growth by coagulation. n_p follows a bell-shape trend similar to soot yield (Fig. B.25). The n_p values and the difference between the particle dynamics models reaches their maximum in 2200-2400 K for %5 CH_4 and 2600-2800 K for %10 CH_4 because of a stronger inception flux leading to larger number concentration of particles and higher coagulation rate. n_p is larger for Dimer Coalescence and EBridge Formation reaching the maximum of nearly

100 and 1000 in %5 and %10 CH_4, respectively using SPBM. The effect of particle dynamics is minimum for Reactive and Irreversible Dimerization at 5% CH_4 over the whole temperature range.

Fig.B.29 shows the standard geometric deviation of mobility diameter, σ_g obtained by SPBM that reaches the maximum of 5 and 10 for %5 and %10 CH_4, respectively indicating a significant degree of polydispersity in the generated particles at t=1.5ms.

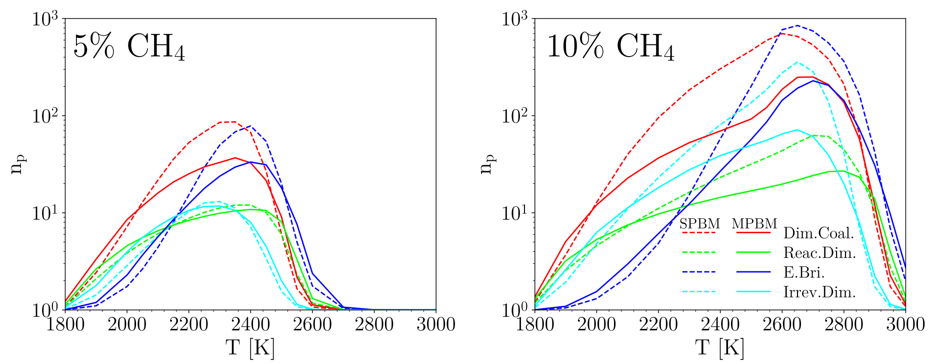

Figure B.28: The temperature dependence of average number of primary particle per agglomerate, n_p during pyrolysis of 5% CH_4-Ar (left pane) and 10% CH_4-Ar (right pane) at P = 4.5–6.7 bar obtained using Caltech mechanism and different inception models at t=1.5 ms

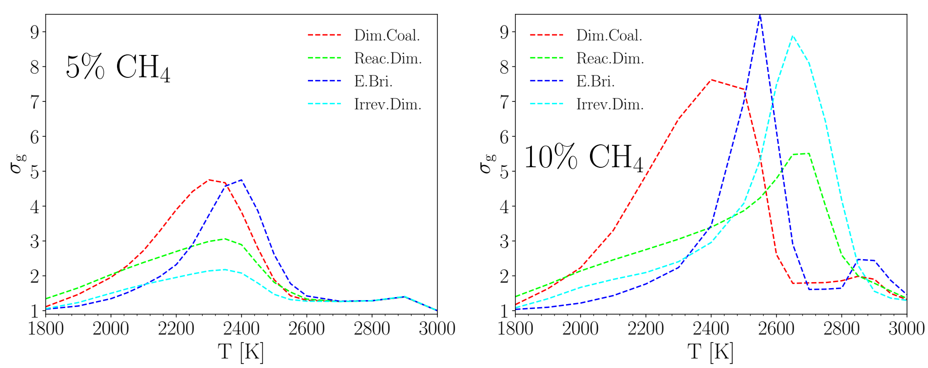

Figure B.29: The time variation of standard geometric deviation of mobility diameter, σ_g during pyrolysis of 5% CH_4-Ar (left pane) and 10% CH_4-Ar (right pane) at P = 4.5–6.7 bar obtained using Caltech mechanism and different inception models at t=1.5 ms

The σ_g values from the SMPS measurements of soot particles at t≈=45ms in a benchmark burner-stabilized premixed are close to 1.1, which is significantly lower than values observed here. So, the evolution of σ_g in the studied shock tube is examined in an extended time frame up to 4 msf for 10% CH_4 at T=2500 and 2700 K. As shown in Fig.B.30, for all PAH growth models σ_g rises in the beginning due to simultaneous inception and coagulation that increases polydispersity and rapidly drops and approaches 1.5 before t=3ms when coagulation becomes dominant and inception weaken due to consumption of precursors by PAH adsorption and HACA.

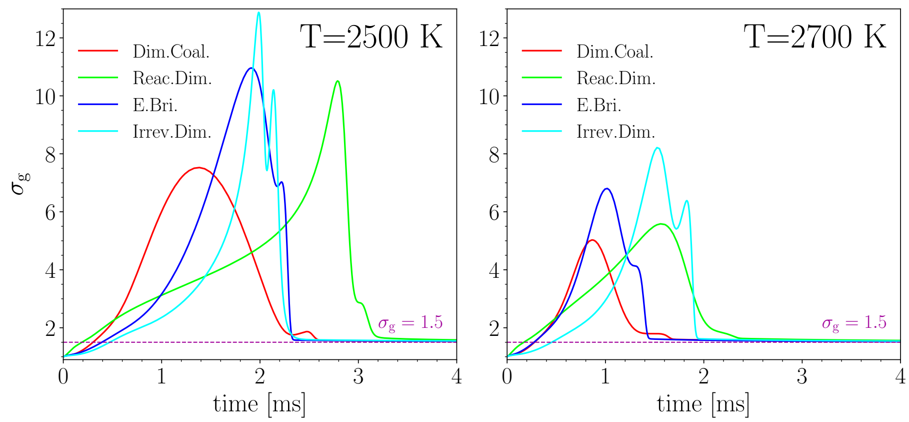

Figure B.30: The temperature dependence of standard geometric deviation of mobility diameter, σ_g during pyrolysis of 10% CH_4-Ar at T=2500 K (left pane) and T=2700 K (right pane)